Stat 6390.001: High Dimensional Probability and Statistics, Fall 2026 **Topic 1**

High Dimensional Probability and Statistics

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Chapter 1

Basic tail and concentration bounds

1.1 Introduction

A central challenge in high-dimensional statistics is the curse of dimensionality. As the dimension d grows, the sample space expands rapidly, so a fixed number of observations becomes increasingly sparse relative to the size of the space. As a result, methods and intuition that work well in low dimensions may become much less informative in high-dimensional settings.

High-dimensional problems also exhibit geometric and probabilistic behavior that can differ substantially from low-dimensional intuition. For example, if

$$X \sim N(0, I_d),$$

then the density is maximized at the origin, but a typical sample is not concentrated near the origin. Instead,

$$\|X\|_2 \approx \sqrt{d},$$

so most samples lie near a thin shell of radius approximately $\sqrt{d}$.

These phenomena motivate the study of probabilistic tools that remain useful when the dimension is large. In particular, this lecture develops tail and concentration bounds that quantify how likely a random variable is to deviate from its mean, with an emphasis on non-asymptotic guarantees.

1.1.1 One-dimensional and multivariate CLT

Let $X_1, \dots, X_n$ be i.i.d. random variables with mean μ and finite variance σ^2. The one-dimensional CLT states that

$$\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right) \xrightarrow{d} N(0, \sigma^2) \qquad \text{as } n \to \infty. \tag{1.1}$$

The multivariate version has the same form. Suppose now that $X_1, \dots, X_n$ are i.i.d. random vectors in $\mathbb{R}^d$ with

$$\mathbb{E}[X_i] = \mu \in \mathbb{R}^d, \qquad \mathrm{Cov}(X_i) = \Sigma.$$

Then

$$\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right) \xrightarrow{d} N(0, \Sigma) \qquad \text{as } n \to \infty. \tag{1.2}$$

For a d-dimensional Gaussian vector $X \sim N(\mu, \Sigma)$ with positive definite covariance matrix Σ, the density is

$$f(x) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left\{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right\}. \tag{1.3}$$

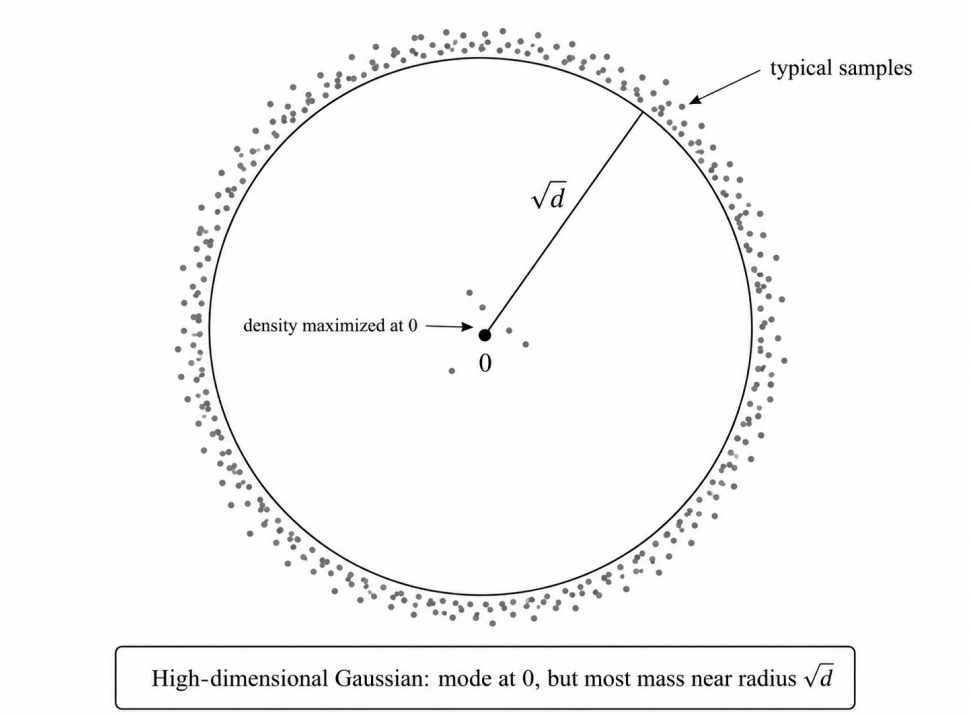


Figure 1.1: A schematic illustration of a high-dimensional standard Gaussian distribution.

Example 1.1.1 (A First High-Dimensional Phenomenon)**.** Consider the standard Gaussian case $X \sim N(0, I_d)$. The density is maximized at the origin, so the mode is $x = 0$. However, when d is large, a typical draw is not close to the origin. Indeed,

$$\mathbb{E}\|X\|_2^2 = \mathbb{E}\left[\sum_{j=1}^{d} X_j^2\right] = \sum_{j=1}^{d} \mathbb{E}[X_j^2] = d. \tag{1.4}$$

Thus the typical Euclidean radius is of order $\sqrt{d}$; more precisely, $\|X\|_2^2$ is a chi-square random variable with d degrees of freedom and is concentrated around d. Hence most of the Gaussian probability mass lies in a thin shell with radius approximately $\sqrt{d}$, even though the density itself is largest at the origin, shown in Figure 1. This illustrates a basic feature of high-dimensional geometry: the location of the mode need not describe where a typical sample lies.

1.2 Classical Tail Bounds

The purpose of a tail bound is to control probabilities such as $\mathbb{P}(X \geq t)$ or $\mathbb{P}(|X - \mathbb{E}X| \geq t)$. The amount of information available about the distribution determines the quality of the bound. Markov's inequality uses only a first moment; Chebyshev's inequality uses a second moment; and the Chernoff method exploits the moment generating function (MGF), thereby incorporating information from all moments when the MGF exists [1].

1.2.1 Markov and Chebyshev inequalities

Proposition 1.2.1 (Markov's inequality)**.** *If random variable X is nonnegative and $\mathbb{E}[X] < \infty$, then for every $t > 0$,*

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}[X]}{t}. \tag{1.5}$$

Proposition 1.2.2 (Chebyshev's inequality)**.** *If random variable X has finite variance and $\mu = \mathbb{E}[X]$, then for every $t > 0$,*

$$\mathbb{P}\big(|X - \mu| \geq t\big) \leq \frac{\mathrm{Var}(X)}{t^2}. \tag{1.6}$$

Chebyshev's inequality is an immediate consequence of Markov's inequality applied to the nonnegative random variable $(X - \mu)^2$. More generally, if the kth centered absolute moment exists, then

$$\mathbb{P}\big(|X - \mu| \geq t\big) = \mathbb{P}\big(|X - \mu|^k \geq t^k\big) \leq \frac{\mathbb{E}|X - \mu|^k}{t^k}. \tag{1.7}$$

1.2.2 From Markov to the Chernoff bound

The exponential function is particularly effective because it converts sums into products and leads naturally to the moment generating function.

Lemma 1.2.3 (Chernoff bound)**.** *Let $\mu = \mathbb{E}[X]$, and suppose*

$$\phi(\lambda) = \mathbb{E}\big[e^{\lambda(X-\mu)}\big]$$

exists for $\lambda \in [0, b]$. Then for every $t > 0$,

$$\log \mathbb{P}(X - \mu \geq t) \leq \inf_{\lambda \in [0,b]} \left\{\log \phi(\lambda) - \lambda t\right\}. \tag{1.8}$$

Proof. Fox $\lambda > 0$. Since the exponential function is increasing,

$$\mathbb{P}(X - \mu \geq t) = \mathbb{P}\big(\lambda(X - \mu) \geq \lambda t\big) \tag{1.9}$$

$$= \mathbb{P}\big(e^{\lambda(X-\mu)} \geq e^{\lambda t}\big). \tag{1.10}$$

Applying Markov's inequality to the nonnegative random variable $e^{\lambda(X-\mu)}$ gives

$$\mathbb{P}(X - \mu \geq t) \leq \frac{\mathbb{E}[e^{\lambda(X-\mu)}]}{e^{\lambda t}} = \phi(\lambda) e^{-\lambda t}. \tag{1.11}$$

Taking logarithms yields

$$\log \mathbb{P}(X - \mu \geq t) \leq \log \phi(\lambda) - \lambda t.$$

Because the inequality holds for every admissible λ, minimizing the right-hand side over $\lambda \in [0, b]$ gives the result. □

Remark 1.2.4. The optimization over λ is essential. Different values of λ produce different valid bounds, and the infimum selects the sharpest one obtained from this argument. In applications, the main work is therefore to control the MGF and then solve a one-dimensional optimization problem.

1.3 Sub-Gaussian Random Variables

1.3.1 Gaussian Tail Bound

As a first example, consider a Gaussian random variable

$$X \sim N(\mu, \sigma^2).$$

Its moment generating function is

$$\mathbb{E}[e^{\lambda X}] = \exp\left(\mu\lambda + \frac{\sigma^2\lambda^2}{2}\right), \qquad \lambda \in \mathbb{R}. \tag{1.12}$$

Applying the preceding lemma, the upper-tail bound is determined by

$$\inf_{\lambda>0}\left\{\log \mathbb{E}[e^{\lambda(X-\mu)}] - \lambda t\right\} = \inf_{\lambda>0}\left\{\frac{\sigma^2\lambda^2}{2} - \lambda t\right\}. \tag{1.13}$$

Here we used the fact that

$$X - \mu \sim N(0, \sigma^2).$$

The quadratic expression is minimized at

$$\lambda^* = \frac{t}{\sigma^2},$$

which gives

$$\inf_{\lambda>0}\left\{\frac{\sigma^2\lambda^2}{2} - \lambda t\right\} = -\frac{t^2}{2\sigma^2}. \tag{1.14}$$

Therefore, for every $t > 0$,

$$\mathbb{P}(X - \mu \geq t) \leq \exp\left(-\frac{t^2}{2\sigma^2}\right). \tag{1.15}$$

(Normal distribution is light tail)

1.3.2 Definition and basic tail bounds

Definition 1.3.1 (Sub-Gaussian random variable)**.** A random variable X with mean $\mu = \mathbb{E}[X]$ is sub-Gaussian with parameter $\sigma > 0$ if

$$\mathbb{E}\left[e^{\lambda(X-\mu)}\right] \leq \exp\left(\frac{\sigma^2\lambda^2}{2}\right) \qquad \text{for all } \lambda \in \mathbb{R}. \tag{1.16}$$

Remark 1.3.2. If X is σ-sub-Gaussian for some $\sigma > 0$, then X is also σ'-sub-Gaussian for some $\sigma' \geq \sigma$.

The definition says that the centered MGF of X grows no faster than the centered MGF of a Gaussian $N(0, \sigma^2)$. Therefore, the same Chernoff calculation gives the Gaussian-type upper-tail bound

$$\mathbb{P}(X - \mu \geq t) \leq \exp\left(-\frac{t^2}{2\sigma^2}\right), \qquad t > 0. \tag{1.17}$$

Lemma 1.3.3 (Symmetry of sub-Gaussianity)**.** *A random variable X is sub-Gaussian with parameter σ if and only if $-X$ is sub-Gaussian with the same parameter.*

Proof. Let $\mu = \mathbb{E}[X]$. Recall that X is sub-Gaussian with parameter σ if

$$\mathbb{E}\left[e^{\lambda(X-\mu)}\right] \leq \exp\left(\frac{\sigma^2\lambda^2}{2}\right), \qquad \lambda \in \mathbb{R}. \tag{1.18}$$

Suppose first that X is sub-Gaussian. Since $\mathbb{E}[-X] = -\mu$,

$$\mathbb{E}\left[e^{\lambda((-X)-\mathbb{E}[-X])}\right] = \mathbb{E}\left[e^{-\lambda(X-\mu)}\right] \tag{1.19}$$

$$\leq \exp\left(\frac{\sigma^2(-\lambda)^2}{2}\right) \tag{1.20}$$

$$= \exp\left(\frac{\sigma^2\lambda^2}{2}\right). \tag{1.21}$$

Hence $-X$ is sub-Gaussian with parameter σ.

Conversely, suppose that $-X$ is sub-Gaussian. Then

$$\mathbb{E}\left[e^{\lambda(X-\mu)}\right] = \mathbb{E}\left[e^{-\lambda((-X)-\mathbb{E}[-X])}\right] \tag{1.22}$$

$$\leq \exp\left(\frac{\sigma^2(-\lambda)^2}{2}\right) \tag{1.23}$$

$$= \exp\left(\frac{\sigma^2\lambda^2}{2}\right). \tag{1.24}$$

Therefore X is sub-Gaussian with parameter σ. □

Lemma 1.3.4 (Lower Tail)**.** *If X is sub-Gaussian with parameter σ, then for every $t > 0$,*

$$\mathbb{P}(X - \mathbb{E}[X] \leq -t) \leq \exp\left(-\frac{t^2}{2\sigma^2}\right). \tag{1.25}$$

Proof. By the symmetry lemma, $-X$ is also sub-Gaussian with parameter σ. Moreover,

$$\mathbb{P}(X - \mathbb{E}[X] \leq -t) = \mathbb{P}(-X - \mathbb{E}[-X] \geq t) \tag{1.26}$$

$$\leq \exp\left(-\frac{t^2}{2\sigma^2}\right), \tag{1.27}$$

where the last step follows from the upper-tail bound applied to $-X$. □

Now if we combine upper and lower tails with the union bound yields the two-sided concentration inequality we will get the following

Lemma 1.3.5. *If X is sub-Gaussian with parameter $\sigma > 0$,*

$$\mathbb{P}\big(|X - \mathbb{E}X| \geq t\big) \leq 2\exp\left(-\frac{t^2}{2\sigma^2}\right), \qquad t > 0. \tag{1.28}$$

Note: This much faster decay is one reason sub-Gaussian assumptions are central in non-asymptotic high-dimensional statistics [1].

Example 1.3.6 (Rademacher random variable)**.** A Rademacher random variable ε takes values in $\{-1, +1\}$ equiprobably:

In particular, $\mathbb{E}[\varepsilon] = 0$. Its MGF is

$$\mathbb{E}[e^{\lambda\varepsilon}] = \frac{1}{2}e^{\lambda} + \frac{1}{2}e^{-\lambda} \tag{1.29}$$

$$= \frac{1}{2}\left(\sum_{k=0}^{\infty}\frac{\lambda^k}{k!} + \sum_{k=0}^{\infty}\frac{(-\lambda)^k}{k!}\right) \tag{1.30}$$

$$= \sum_{m=0}^{\infty}\frac{\lambda^{2m}}{(2m)!}, \tag{1.31}$$

where the odd-order terms cancel.

Since $(2m)! \geq 2^m m!$,

$$\mathbb{E}[e^{\lambda\varepsilon}] \leq \sum_{m=0}^{\infty}\frac{\lambda^{2m}}{2^m m!} \tag{1.32}$$

$$= e^{\lambda^2/2}. \tag{1.33}$$

Since $\mathbb{E}[\varepsilon] = 0$, it follows that

$$\mathbb{E}[e^{\lambda(\varepsilon - \mathbb{E}[\varepsilon])}] \leq e^{\lambda^2/2}.$$

Therefore, ε is sub-Gaussian with parameter $\sigma = 1$.

This example shows that sub-Gaussianity is not the same as Gaussianity. A Rademacher variable is discrete and bounded, yet its MGF is dominated by that of a standard Gaussian. The sub-Gaussian class is therefore broad enough to include many non-Gaussian random variables while retaining Gaussian-like tail behavior.

Note:"equiprobably" meaning

$$\mathbb{P}(\varepsilon = 1) = \mathbb{P}(\varepsilon = -1) = \frac{1}{2}. \tag{1.34}$$

Therefore,

$$\mathbb{E}[\varepsilon] = \frac{1}{2}(1) + \frac{1}{2}(-1) = 0. \tag{1.35}$$

1.3.3 Example of Sub-Gaussian Variables

Example 1.3.7 (Bounded random variables)**.** Let X be zero-mean, and supported on some interval $[a, b]$. We will show that X is sub-Gaussian with parameter at most $\sigma = b - a$.

Letting X' be an independent copy of X, so that $X \overset{d}{=} X'$ and $X \perp\!\!\!\perp X'$. For any $\lambda \in \mathbb{R}$, we have

$$\mathbb{E}_X\left[e^{\lambda X}\right] = \mathbb{E}_X\left[e^{\lambda\left(X - \mathbb{E}_{X'}[X']\right)}\right] \overset{\text{(i)}}{\leq} \mathbb{E}_X\left[\mathbb{E}_{X'}e^{\lambda\left(X - X'\right)}\right] = \mathbb{E}_{X,X'}\left[e^{\lambda\left(X - X'\right)}\right]$$

The first equality is because $\mathbb{E}[X'] = 0$, and (i) follows from the the convexity of the exponential, and Jensen's inequality. Letting ε be an independent Rademacher variable, note that since (X, X') are i.i.d, $X - X' \overset{d}{=} X' - X$, so $(X - X')$ is symmetric about 0. Thus, the distribution of $(X - X')$ is the same as that of $\varepsilon(X - X')$, so that we have

$$\mathbb{E}_{X,X'}\left[e^{\lambda\left(X - X'\right)}\right] = \mathbb{E}_{X,X'}\left[\mathbb{E}_{\varepsilon}\left[e^{\lambda\varepsilon\left(X - X'\right)}\right]\right] \overset{\text{(ii)}}{\leq} \mathbb{E}_{X,X'}\left[e^{\frac{\lambda^2}{2}\left(X - X'\right)^2}\right] \leq e^{\frac{\lambda^2}{2}(b-a)^2}$$

The first equality follows from Tonelli's theorem and (ii) follows from the sub-Gaussianity of Radamacher variable and last inequality is because $|X - X'| \leq b - a$. Putting everything together, we have shown that X is sub-Gaussian with parameter at most $\sigma = b - a$.

Remark 1.3.8. This result is useful, but it can be sharpened. It can be shown that X is sub-Gaussian with parameter at most $\sigma = \frac{b-a}{2}$ (Exercise 2.4). The technique used in this example is a simple example of a *symmetrization argument*, in which we first introduce an independent copy X', and then symmetrize the problem with a Rademacher variable. Such symmetrization arguments are useful in a variety of contexts.

1.3.4 Hoeffding's Inequality

Another property of sub-Gaussianity is preserved by linear operations.

Lemma 1.3.9. *Let X_1 and X_2 be independent sub-Gaussian variables with parameters σ_1 and σ_2, respectively, then $X_1 + X_2$ is sub-Gaussian with parameter $\sqrt{\sigma_1^2 + \sigma_2^2}$.*

Proof. By independence we have

$$\mathbb{E}\left[e^{\lambda(X_1+X_2-\mathbb{E}X_1-\mathbb{E}X_2)}\right] = \mathbb{E}\left[e^{\lambda(X_1-\mathbb{E}X_1)}\right]\mathbb{E}\left[e^{\lambda(X_2-\mathbb{E}X_2)}\right] \leq e^{\frac{\lambda^2}{2}\sigma_1^2} e^{\frac{\lambda^2}{2}\sigma_2^2} = e^{\frac{\lambda^2}{2}\left(\sigma_1^2+\sigma_2^2\right)}$$

□

As a consequence of this fact, we obtain an important result known as the *Hoeffding's inequality*:

Proposition 1.3.10 (Hoeffding's inequality)**.** *Suppose that the variables X_i, $i = 1, \ldots, n$, are independent, and X_i has mean μ_i and sub-Gaussian parameter σ_i. Then for all $t \geq 0$, we have*

$$\mathbb{P}\left[\sum_{i=1}^n (X_i - \mu_i) \geq t\right] \leq \exp\left\{-\frac{t^2}{2\sum_{i=1}^n \sigma_i^2}\right\}$$

and

$$\mathbb{P}\left[\left|\sum_{i=1}^n X_i - \sum_{i=1}^n \mathbb{E}X_i\right| \geq t\right] \leq 2\exp\left\{-\frac{t^2}{2\sum_{i=1}^n \sigma_i^2}\right\}$$

Proof. By previous lemma, $\sum_{i=1}^n X_i$ is $\sqrt{\sigma_1^2 + \cdots + \sigma_n^2}$ sub-Gaussian. So, by basic sub-Gaussian tail bound we have the results. □

Corollary 1.3.11. *If X_i, $i = 1, \ldots, n$, are i.i.d, with mean μ and sub-Gaussian parameter σ. Then for all $t \geq 0$, we have*

$$\mathbb{P}\left[\left|\frac{1}{n}\sum_{i=1}^n X_i - \mu\right| \geq t\right] \leq 2\exp\left\{-\frac{nt^2}{2\sigma^2}\right\}$$

and

$$\mathbb{P}\left[\sqrt{n}\left|\frac{1}{n}\sum_{i=1}^n X_i - \mu\right| \geq t\right] \leq 2\exp\left\{-\frac{t^2}{2\sigma^2}\right\}$$

Remark 1.3.12. The above can be viewed as a finite sample Central Limit Theorem. CLT states that when $n \to \infty$,

$$\sqrt{n}\left|\frac{1}{n}\sum_{i=1}^n X_i - \mu\right| \to \mathcal{N}(0, \sigma^2).$$

The second statement of the above corollary is a finite sample version of CLT, by stating that the tail of $\sqrt{n}\left|\frac{1}{n}\sum_{i=1}^n X_i - \mu\right|$ is upper bounded by the tail of $\mathcal{N}(0, \sigma^2)$.

Proof. By Hoeffding,

$$\mathbb{P}\left[\left|\frac{1}{n}\sum_{i=1}^n X_i - \mu\right| \geq t\right] = \mathbb{P}\left[\left|\sum_{i=1}^n X_i - n\mu\right| \geq nt\right] \leq 2\exp\left\{-\frac{(nt)^2}{2n\sigma^2}\right\} = 2\exp\left\{-\frac{nt^2}{2\sigma^2}\right\}$$

Second statement follows analogously. □

One application of Hoeffding's bound is to derive finite sample guarantees for the error of the sample mean.

Example 1.3.13 (Finite-sample error guarantee)**.** Let $\overline{X}_n := \frac{1}{n}\sum_{i=1}^n X_i$ and fix a confidence level $\delta \in (0,1)$. By the preceding corollary,

$$\mathbb{P}\left[\left|\overline{X}_n - \mu\right| \geq t\right] \leq 2\exp\left\{-\frac{nt^2}{2\sigma^2}\right\}$$

Choose $t = \sqrt{\frac{2\sigma^2}{n}\log\left(\frac{2}{\delta}\right)}$, so that $\delta = 2\exp\left\{-\frac{nt^2}{2\sigma^2}\right\}$. In other words, with probability at least $1-\delta$,

$$\left|\overline{X}_n - \mu\right| \leq \sqrt{\frac{2\sigma^2}{n}\log\left(\frac{2}{\delta}\right)}$$

Example 1.3.14 (Expected absolute error)**.** We can also control the expected absolute error $\mathbb{E}\left|\overline{X}_n - \mu\right|$ directly from the tail bound. Recall that for any nonnegative random variable Y, Tonelli's theorem gives

$$\mathbb{E}[Y] = \int_0^\infty \mathbb{P}(Y \geq t)\, dt$$

Taking $Y = \left|\overline{X}_n - \mu\right|$, the preceding corollary and the trivial bound $\mathbb{P}(Y \geq t) \leq 1$ imply

$$\mathbb{P}\left[\left|\overline{X}_n - \mu\right| \geq t\right] \leq \min\left\{1,\ 2\exp\left\{-\frac{nt^2}{2\sigma^2}\right\}\right\}$$

Let t_0 be the point at which the two upper bounds agree. Thus $2e^{-\frac{nt_0^2}{2\sigma^2}} = 1$, which gives $t_0 = \sigma\sqrt{\frac{2\log 2}{n}}$. Hence

$$\begin{aligned}
\mathbb{E}\left|\overline{X}_n - \mu\right| &= \int_0^\infty \mathbb{P}\left[\left|\overline{X}_n - \mu\right| \geq t\right]\, dt \\
&\leq \int_0^{t_0} 1\, dt + \underbrace{\int_{t_0}^\infty 2\exp\left\{-\frac{nt^2}{2\sigma^2}\right\} dt}_{z = \frac{\sqrt{n}}{\sigma}t,\quad dt = \frac{\sigma}{\sqrt{n}}\, dz} \\
&= t_0 + \frac{2\sigma}{\sqrt{n}}\int_{\frac{\sqrt{n}}{\sigma}t_0}^\infty e^{-z^2/2}\, dz \\
&= \frac{\sigma}{\sqrt{n}}\left(\sqrt{2\log 2} + 2\int_{\sqrt{2\log 2}}^\infty e^{-z^2/2}\, dz\right) \\
&\leq 2\frac{\sigma}{\sqrt{n}}.
\end{aligned}$$

We conclude our discussion of sub-Gaussianity with a theorem that provides different characterizations of sub-Gaussian variables.

Theorem 1.3.15 (Equivalent characterizations of sub-Gaussian variables)**.** *Given any zero-mean random variable X, the following properties are equivalent:*

1. *There is a constant $\sigma \geq 0$ such that*

$$\mathbb{E}\left[e^{\lambda X}\right] \leq e^{\frac{\lambda^2\sigma^2}{2}} \qquad \text{for all } \lambda \in \mathbb{R}.$$

2. *There is a constant $c \geq 0$ and Gaussian random variable $Z \sim \mathcal{N}(0,\tau^2)$ such that*

$$\mathbb{P}\left[|X| \geq s\right] \leq c\,\mathbb{P}\left[|Z| \geq s\right] \qquad \text{for all } s \geq 0.$$

3. *There is a constant $\theta \geq 0$ such that*

$$\mathbb{E}\left[X^{2k}\right] \leq \frac{(2k)!}{2^k k!}\theta^{2k} \qquad \text{for all } k = 1, 2, \ldots$$

4. *There is a constant $\sigma \geq 0$ such that*

$$\mathbb{E}\left[e^{\frac{\lambda X^2}{2\sigma^2}}\right] \leq \frac{1}{\sqrt{1-\lambda}} \qquad \text{for all } \lambda \in [0, 1).$$

Proof. See Appendix A (Section 2.4) in [1]. □

Exercise (Subgaussianity under Lipschitz transformations). Let X be a real-valued random variable such that $X - \mathbb{E}X$ is σ-subgaussian, i.e.,

$$\mathbb{E}\left[e^{\lambda(X-\mathbb{E}X)}\right] \leq e^{\lambda^2\sigma^2/2}, \qquad \forall \lambda \in \mathbb{R}.$$

(a) Let $f : \mathbb{R} \to \mathbb{R}$ be a 1-Lipschitz function, meaning that $|f(x) - f(y)| \leq |x - y|$ for all $x, y \in \mathbb{R}$. Show that $f(X) - \mathbb{E}f(X)$ is $\sqrt{2}\sigma$-subgaussian, i.e.,

$$\mathbb{E}\left[e^{\lambda(f(X)-\mathbb{E}f(X))}\right] \leq e^{\lambda^2\sigma^2}, \qquad \forall \lambda \in \mathbb{R}.$$

(b) As a special case, show that $|X| - \mathbb{E}|X|$ is $\sqrt{2}\sigma$-subgaussian.

Hint. Let X' be an independent copy of X. Use Jensen's inequality, the symmetry of $f(X) - f(X')$, and the Lipschitz property of f.

1.4 Symmetrization

Lemma 1.4.1. *Let $Y \perp\!\!\!\perp Z$ with $\mathbb{E}[Z] = 0$. Then for any convex function F,*

$$\mathbb{E}\left[F(Y)\right] \leq \mathbb{E}\left[F(Y + Z)\right]. \tag{1.36}$$

Example 1.4.2. The function $F(x) = e^{\lambda x}$ is convex. Suppose $\mathbb{E}[X] = 0$. Applying Lemma 1.4.1 with $Y = X$ and $Z = -X'$, where $X' \perp\!\!\!\perp X$ and $X' \sim \text{Law}(X)$, we obtain

$$\mathbb{E}_X\left[e^{\lambda X}\right] \leq \mathbb{E}_{X,X'}\left[e^{\lambda(X-X')}\right]. \tag{1.37}$$

Proof of Lemma 1.4.1. For every $y \in \mathbb{R}$, using $\mathbb{E}[Z] = 0$ and Jensen's inequality,

$$F(y) = F\left(y + \mathbb{E}_Z[Z]\right) = F\left(\mathbb{E}_Z\left[y + Z\right]\right) \leq \mathbb{E}_Z\left[F(y + Z)\right].$$

Substituting $y = Y$ gives $F(Y) \leq \mathbb{E}_Z\left[F(Y + Z)\right]$, and taking expectations over Y yields

$$\mathbb{E}_Y\left[F(Y)\right] \leq \mathbb{E}_Y\mathbb{E}_Z\left[F(Y + Z)\right] = \mathbb{E}_{Y,Z}\left[F(Y + Z)\right],$$

where the last equality uses $Y \perp\!\!\!\perp Z$. □

Definition 1.4.3. A random variable X is called symmetric if it has the same distribution as $-X$.

Remark 1.4.4. If X has pdf $f(x)$, then X is symmetric if and only if $f(x) = f(-x)$.

Lemma 1.4.5 (Constructing symmetric distributions)**.** *Let X be a random variable and let ε be an independent Rademacher random variable.*

(a) εX *and* $\varepsilon\,|X|$ *are identically distributed and symmetric.*

(b) If X *is symmetric, then both* εX *and* $\varepsilon\,|X|$ *have the same distribution as* X*.*

(c) If X' *is an independent copy of* X*, then* $X - X'$ *is symmetric.*

Proof. We only prove that εX is symmetric. For every event $A \subseteq \mathbb{R}$, conditioning on the value of ε gives

$$\mathbb{P}\left[\varepsilon X \in A\right] = \tfrac{1}{2}\mathbb{P}\left[X \in A\right] + \tfrac{1}{2}\mathbb{P}\left[-X \in A\right] = \tfrac{1}{2}\mathbb{P}\left[-X \in A\right] + \tfrac{1}{2}\mathbb{P}\left[X \in A\right] = \mathbb{P}\left[-\varepsilon X \in A\right]. \qquad \square$$

Lemma 1.4.6 (Symmetrization)**.** *Let* $X_1, \dots, X_N$ *be independent, mean-zero random vectors in some normed space, and let* $\varepsilon_1, \dots, \varepsilon_N$ *be independent Rademacher random variables. Then*

$$\frac{1}{2}\,\mathbb{E}\left\|\sum_{i=1}^N \varepsilon_i X_i\right\| \;\le\; \mathbb{E}\left\|\sum_{i=1}^N X_i\right\| \;\le\; 2\,\mathbb{E}\left\|\sum_{i=1}^N \varepsilon_i X_i\right\|. \tag{1.38}$$

Proof. Let $(X_1', \dots, X_N')$ be independent copies of $(X_1, \dots, X_N)$. By Lemma 1.4.5(c) the vectors $X_i - X_i'$ are independent and symmetric (this step uses the assumption that X_i are independent), so that

$$(X_1 - X_1', \dots, X_N - X_N') \overset{d}{=} (\varepsilon_1(X_1 - X_1'), \dots, \varepsilon_N(X_N - X_N')). \tag{1.39}$$

Upper bound. Set $p := \mathbb{E}\left\|\sum_{i=1}^N X_i\right\|$. Since $\mathbb{E}\left[\sum_{i=1}^N X_i'\right] = 0$, we may apply Lemma 1.4.1 with $F(x) = \|x\|$, $Y = \sum_{i=1}^N X_i$ and $Z = -\sum_{i=1}^N X_i'$ to get

$$p = \mathbb{E}\left\|\sum_{i=1}^N X_i - \mathbb{E}\left[\sum_{i=1}^N X_i'\right]\right\| \le \mathbb{E}\left\|\sum_{i=1}^N (X_i - X_i')\right\|.$$

Then by (1.39) and the triangle inequality,

$$\begin{aligned}\mathbb{E}\left\|\sum_{i=1}^N (X_i - X_i')\right\| &= \mathbb{E}\left\|\sum_{i=1}^N \varepsilon_i\,(X_i - X_i')\right\| = \mathbb{E}\left\|\sum_{i=1}^N \varepsilon_i X_i - \sum_{i=1}^N \varepsilon_i X_i'\right\| \\ &\le \mathbb{E}\left\|\sum_{i=1}^N \varepsilon_i X_i\right\| + \mathbb{E}\left\|\sum_{i=1}^N \varepsilon_i X_i'\right\| = 2\,\mathbb{E}\left\|\sum_{i=1}^N \varepsilon_i X_i\right\|.\end{aligned}$$

Lower bound. Conditionally on $\varepsilon = (\varepsilon_1, \dots, \varepsilon_N)$, the vectors $\sum_{i=1}^N \varepsilon_i X_i$ and $-\sum_{i=1}^N \varepsilon_i X_i'$ are independent and the latter has mean zero, so Lemma 1.4.1 applies conditionally and gives, after averaging over ε,

$$\mathbb{E}\left\|\sum_{i=1}^N \varepsilon_i X_i\right\| \le \mathbb{E}\left\|\sum_{i=1}^N \varepsilon_i X_i - \sum_{i=1}^N \varepsilon_i X_i'\right\| = \mathbb{E}\left\|\sum_{i=1}^N \varepsilon_i\,(X_i - X_i')\right\|.$$

Applying (1.39) once more and then the triangle inequality,

$$\mathbb{E}\left\|\sum_{i=1}^N \varepsilon_i\,(X_i - X_i')\right\| = \mathbb{E}\left\|\sum_{i=1}^N (X_i - X_i')\right\| \le \mathbb{E}\left\|\sum_{i=1}^N X_i\right\| + \mathbb{E}\left\|\sum_{i=1}^N X_i'\right\| = 2\,\mathbb{E}\left\|\sum_{i=1}^N X_i\right\|. \qquad \square$$

Remark 1.4.7. The zero-mean assumption is needed for both bounds: in the upper bound to subtract $\mathbb{E}\left[\sum_i X_i'\right]$, and in the lower bound to guarantee that $\mathbb{E}\left[\sum_i \varepsilon_i X_i' \mid \varepsilon\right] = 0$.

1.5 Sub-exponential Random Variables

We now turn to the class of sub-exponential variables, which are defined by a slightly milder condition on the moment generating function.

Definition 1.5.1 ([1, Definition 2.7])**.** A random variable X with mean $\mu = \mathbb{E}[X]$ is sub-exponential if there are non-negative parameters (ν, α) such that

$$\mathbb{E}\left[e^{\lambda(X-\mu)}\right] \leq e^{\frac{\nu^2\lambda^2}{2}} \qquad \text{for all } |\lambda| < \frac{1}{\alpha}. \tag{1.40}$$

Remark 1.5.2. A σ-sub-Gaussian variable is $(\sigma, 0)$-sub-exponential, where we interpret $1/0$ as $+\infty$.

Example 1.5.3 (Sub-exponential but not sub-Gaussian)**.** Let $Z \sim N(0,1)$ and consider $X = Z^2$, so that $\mathbb{E}[X] = 1$. For $\lambda < \frac{1}{2}$,

$$\begin{aligned}\mathbb{E}\left[e^{\lambda(X-1)}\right] &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} e^{\lambda(z^2-1)}e^{-\frac{z^2}{2}}\,dz \\ &= \frac{e^{-\lambda}}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} e^{-\frac{(1-2\lambda)z^2}{2}}\,dz = \frac{e^{-\lambda}}{\sqrt{1-2\lambda}},\end{aligned} \tag{1.41}$$

where the last equality uses $\int_{-\infty}^{+\infty} e^{-az^2/2}dz = \sqrt{2\pi/a}$ with $a = 1-2\lambda > 0$. For $\lambda \geq \frac{1}{2}$ the moment generating function is infinite, so X is not sub-Gaussian. On the other hand,

$$\frac{e^{-\lambda}}{\sqrt{1-2\lambda}} \leq e^{2\lambda^2} = e^{\frac{2^2\lambda^2}{2}} \qquad \text{where } |\lambda| < \tfrac{1}{4} \qquad \text{(check this!)}, \tag{1.42}$$

so that $\mathbb{E}\left[e^{\lambda(X-1)}\right] \leq e^{\frac{2^2\lambda^2}{2}}$ for all $|\lambda| < \frac{1}{4}$, i.e. X is $(2,4)$-sub-exponential.

Exercises

1. Verify the inequality (1.42), that is, show that $\frac{e^{-\lambda}}{\sqrt{1-2\lambda}} \leq e^{2\lambda^2}$ for all $|\lambda| < \frac{1}{4}$.

Proposition 1.5.4 (Sub-exponential tail bound)**.** *Suppose X is sub-exponential with parameters (ν, α) and mean μ. Then*

$$\mathbb{P}\left[X - \mu \geq t\right] \leq \begin{cases} e^{-\frac{t^2}{2\nu^2}}, & \text{if } 0 \leq t \leq \frac{\nu^2}{\alpha}, \\ e^{-\frac{t}{2\alpha}}, & \text{for } t > \frac{\nu^2}{\alpha}, \end{cases} \qquad = \qquad e^{-\min\left\{\frac{t^2}{2\nu^2}, \frac{t}{2\alpha}\right\}}. \tag{1.43}$$

Proof. Assume $\mu = 0$ without loss of generality. By the Chernoff bound from Lecture 1 and Definition 1.5.1,

$$\begin{aligned}\log \mathbb{P}\left[X \geq t\right] &\leq \inf_{0 \leq \lambda < \frac{1}{\alpha}} \left\{\log \mathbb{E}\left[e^{\lambda X}\right] - \lambda t\right\} \leq \inf_{0 \leq \lambda < \frac{1}{\alpha}} \left\{\log e^{\frac{\nu^2\lambda^2}{2}} - \lambda t\right\} \\ &= \inf_{0 \leq \lambda < \frac{1}{\alpha}} \underbrace{\left\{\frac{\nu^2\lambda^2}{2} - \lambda t\right\}}_{=:\, g(\lambda)}.\end{aligned} \tag{1.44}$$

The function g is a convex parabola (see Figure 1.2) with

$$g'(\lambda) = \nu^2\lambda - t = 0 \qquad \Longleftrightarrow \qquad \lambda = \frac{t}{\nu^2}.$$

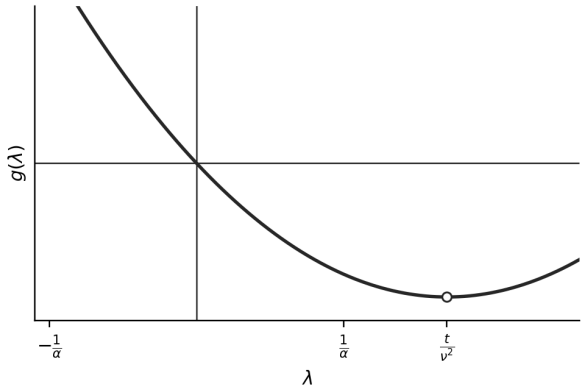


Figure 1.2: The function $g(\lambda) = \frac{\nu^2\lambda^2}{2} - \lambda t$ appearing in (1.44), in the case $\frac{t}{\nu^2} > \frac{1}{\alpha}$. The open circle marks the unconstrained minimizer $\frac{t}{\nu^2}$, which lies outside the feasible region $|\lambda| < \frac{1}{\alpha}$.

① If $\frac{t}{\nu^2} < \frac{1}{\alpha}$ (i.e. $t < \frac{\nu^2}{\alpha}$), the unconstrained minimizer lies in the feasible region, so

$$\inf_{0\le\lambda<\frac{1}{\alpha}} g(\lambda) = g\left(\frac{t}{\nu^2}\right) = \frac{\nu^2}{2}\cdot\frac{t^2}{\nu^4} - \frac{t}{\nu^2}\cdot t = -\frac{t^2}{2\nu^2}.$$

② If $\frac{t}{\nu^2} \ge \frac{1}{\alpha}$ (i.e. $t \ge \frac{\nu^2}{\alpha}$), then g is decreasing on $[0, \frac{t}{\nu^2})$ and the infimum is approached at $\lambda = \frac{1}{\alpha}$, giving

$$\inf_{0\le\lambda<\frac{1}{\alpha}} g(\lambda) = g\left(\frac{1}{\alpha}\right) = \frac{\nu^2}{2}\cdot\frac{1}{\alpha^2} - \frac{1}{\alpha}t = \frac{\nu^2}{2\alpha^2} - \frac{t}{\alpha} \le -\frac{t}{2\alpha},$$

where the last step uses $\nu^2 < \alpha t$. □

1.5.1 Bernstein's Condition

We say that X satisfies Bernstein's condition with parameter b if

$$\left|\mathbb{E}\left[(X-\mu)^k\right]\right| \le \frac{1}{2}\,k!\,\mathrm{Var}(X)\,b^{k-2}, \qquad \text{for any } k = 2,3,\ldots, \tag{1.45}$$

where $\mu = \mathbb{E}[X]$.

Example 1.5.5. If (1.45) is satisfied, then X is sub-exponential. Indeed, writing $\sigma^2 = \mathrm{Var}(X)$ and using the Taylor expansion of the exponential together with $\mathbb{E}[X-\mu] = 0$,

$$\begin{aligned}
\mathbb{E}\left[e^{\lambda(X-\mu)}\right] &= \mathbb{E}\left[\sum_{k=0}^{\infty}\frac{(\lambda(X-\mu))^k}{k!}\right] = 1 + 0 + \mathbb{E}\left[\sum_{k=2}^{\infty}\frac{\lambda^k(X-\mu)^k}{k!}\right] \\
&= 1 + \sum_{k=2}^{\infty}\frac{\lambda^k\,\mathbb{E}\left[(X-\mu)^k\right]}{k!} = 1 + \frac{\lambda^2\sigma^2}{2} + \sum_{k=3}^{\infty}\frac{\lambda^k\,\mathbb{E}\left[(X-\mu)^k\right]}{k!} \\
&\le 1 + \frac{\lambda^2\sigma^2}{2} + \sum_{k=3}^{\infty}\frac{|\lambda|^k\cdot\frac{1}{2}k!\,\sigma^2 b^{k-2}}{k!} = 1 + \frac{\lambda^2\sigma^2}{2} + \frac{\lambda^2\sigma^2}{2}\sum_{k=3}^{\infty}(|\lambda|\,b)^{k-2} = 1 + \frac{\lambda^2\sigma^2}{2}\sum_{s=0}^{\infty}(|\lambda|\,b)^s\,.
\end{aligned} \tag{1.46}$$

For $|\lambda| < \frac{1}{b}$ the geometric series sums to $\frac{1}{1-|\lambda| b}$, so that

$$\mathbb{E}\left[e^{\lambda(X-\mu)}\right] \leq 1 + \frac{\lambda^2\sigma^2/2}{1 - b\,|\lambda|} \leq e^{\frac{\lambda^2\sigma^2/2}{1-b|\lambda|}}, \tag{1.47}$$

using $1 + t \leq e^t$. Restricting further to $|\lambda| < \frac{1}{2b}$ gives $1 - b\,|\lambda| > \frac{1}{2}$, and therefore

$$\mathbb{E}\left[e^{\lambda(X-\mu)}\right] \leq e^{\lambda^2\sigma^2} = e^{\frac{\lambda^2(\sqrt{2}\sigma)^2}{2}}, \qquad \text{for all } |\lambda| < \frac{1}{2b},$$

i.e. X is sub-exponential with parameters $(\nu, \alpha) = (\sqrt{2}\sigma, 2b)$.

Proposition 1.5.6 (Bernstein-type bound)**.** *Suppose X satisfies Bernstein's condition (1.45) with parameter b, mean μ, and variance σ^2. Then*

$$\mathbb{E}\left[e^{\lambda(X-\mu)}\right] \leq \exp\left(\frac{\lambda^2\sigma^2/2}{1 - b\,|\lambda|}\right) \qquad \textit{for all } |\lambda| < \frac{1}{b}, \tag{1.48}$$

and, moreover, the concentration inequality

$$\mathbb{P}\left(|X - \mu| > t\right) \leq 2\exp\left(-\frac{t^2}{2(\sigma^2 + bt)}\right) \qquad \textit{for all } t \geq 0. \tag{1.49}$$

Proof. Equation (1.48) directly follows from (1.46).

Remark 1.5.7. Restricting further to $|\lambda| < \frac{1}{2b}$ gives $b\,|\lambda| < \frac{1}{2}$, hence $1 - b\,|\lambda| > \frac{1}{2}$ and

$$\mathbb{E}\left[e^{\lambda X}\right] \leq \exp\left(\frac{\lambda^2\sigma^2/2}{1/2}\right) = e^{\lambda^2\sigma^2} = e^{\frac{\lambda^2(\sqrt{2}\sigma)^2}{2}}, \qquad |\lambda| < \frac{1}{2b},$$

which recovers the Topic 3 conclusion that X is $(\sqrt{2}\sigma, 2b)$-sub-exponential. Bound (1.48) is sharper: it holds on the full range $|\lambda| < 1/b$ instead of only $|\lambda| < 1/(2b)$, and this extra room is exactly what lets us optimize the Chernoff bound below.

Proof of (1.49). By the Chernoff bound and (1.48), for any $t \geq 0$,

$$\log \mathbb{P}\left(X - \mu \geq t\right) \leq \inf_{0 \leq \lambda < 1/b} \left\{\log \mathbb{E}\left[e^{\lambda(X-\mu)}\right] - \lambda t\right\} \leq \inf_{0 \leq \lambda < 1/b} \underbrace{\left\{\frac{\lambda^2\sigma^2}{2(1 - b\lambda)} - \lambda t\right\}}_{=:\,\theta(\lambda)}. \tag{1.50}$$

Rather than optimizing θ exactly, it suffices to evaluate it at the specific point

$$\lambda^* := \frac{t}{\sigma^2 + bt}.$$

Since $t, \sigma^2, b \geq 0$ we have $\lambda^* \geq 0$, and $\lambda^* < 1/b \iff bt < \sigma^2 + bt \iff \sigma^2 > 0$, which holds whenever X is non-degenerate (if $\sigma^2 = 0$ then $X \equiv \mu$ a.s. and (1.49) is trivial). At $\lambda = \lambda^*$,

$$1 - b\lambda^* = 1 - \frac{bt}{\sigma^2 + bt} = \frac{\sigma^2}{\sigma^2 + bt},$$

so that

$$\theta(\lambda^*) = \frac{\sigma^2(\lambda^*)^2(\sigma^2 + bt)}{2\sigma^2} - \lambda^* t = \frac{(\lambda^*)^2(\sigma^2 + bt)}{2} - \lambda^* t = \frac{t^2}{2(\sigma^2 + bt)} - \frac{t^2}{\sigma^2 + bt} = -\frac{t^2}{2(\sigma^2 + bt)}.$$

Combining with (1.50) (using that λ^* is feasible),

$$\mathbb{P}\left(X-\mu \geq t\right) \leq \exp\left(-\frac{t^2}{2(\sigma^2+bt)}\right), \qquad t \geq 0. \tag{1.51}$$

Finally, $-X$ also satisfies Bernstein's condition with the same (σ^2, b), since $\left|\mathbb{E}[(-(X-\mu))^k]\right| = \left|\mathbb{E}[(X-\mu)^k]\right|$ and $\mathrm{Var}(-X) = \sigma^2$. Applying (1.51) to $-X$ gives $\mathbb{P}(X-\mu \leq -t) \leq \exp(-t^2/(2(\sigma^2+bt)))$ as well, and adding the two one-sided bounds yields (1.49). □

Exercise 1.5.8. Compare the bound (1.49) with the bound one would obtain by plugging the (weaker) parameters $(\sqrt{2}\sigma, 2b)$ derived in the Remark above directly into the general sub-exponential tail bound of Topic 3. Show that (1.49) is never worse, and can be substantially tighter when $bt \ll \sigma^2$.

1.5.2 Sums of Independent Sub-exponential Random Variables

Recall that X with mean μ is sub-exponential with parameters (ν, α) if $\mathbb{E}[e^{\lambda(X-\mu)}] \leq e^{\nu^2\lambda^2/2}$ for all $|\lambda| < 1/\alpha$, and that in this case

$$\mathbb{P}\left(|X-\mu| > t\right) \leq \begin{cases} 2e^{-t^2/(2\nu^2)}, & 0 \leq t \leq \nu^2/\alpha, \\ 2e^{-t/(2\alpha)}, & t > \nu^2/\alpha. \end{cases} \tag{1.52}$$

Lemma 1.5.9. *Let $X_1, \ldots, X_n$ be independent random variables, with X_i sub-exponential with parameters (ν_i, d_i). Then $S_n := \sum_{i=1}^n X_i$ is sub-exponential with parameters*

$$\left(\sqrt{\sum_{i=1}^n \nu_i^2}, \ \max_{1\leq i\leq n} d_i\right).$$

and

$$\mathbb{P}\left(|S_n - \mathbb{E}[S_n]| > t\right) \leq \begin{cases} 2e^{-t^2/(2V^2)}, & 0 \leq t \leq V^2/d, \\ 2e^{-t/(2d)}, & t > V^2/d, \end{cases} \qquad V^2 = \sum_{i=1}^n \nu_i^2, \quad d = \max_{1\leq i\leq n} d_i. \tag{1.53}$$

Proof. Write $d := \max_{1\leq i\leq n} d_i$ and $V^2 := \sum_{i=1}^n \nu_i^2$. For any $|\lambda| < 1/d = \min_{1\leq i\leq n}(1/d_i)$, we have $|\lambda| < 1/d_i$ for every i, so by independence,

$$\mathbb{E}\left[e^{\lambda(S_n - \mathbb{E}[S_n])}\right] = \mathbb{E}\left[e^{\lambda\sum_{i=1}^n (X_i - \mathbb{E}[X_i])}\right] = \prod_{i=1}^n \mathbb{E}\left[e^{\lambda(X_i-\mathbb{E}[X_i])}\right] \leq \prod_{i=1}^n e^{\nu_i^2\lambda^2/2} = e^{\lambda^2 V^2/2}.$$

Hence S_n is (V, d)-sub-exponential.

By (1.52) applied with $(\nu, \alpha) = (V, d)$, we obtain the tail bound

$$\mathbb{P}\left(|S_n - \mathbb{E}[S_n]| > t\right) \leq \begin{cases} 2e^{-t^2/(2V^2)}, & 0 \leq t \leq V^2/d, \\ 2e^{-t/(2d)}, & t > V^2/d, \end{cases} \qquad V^2 = \sum_{i=1}^n \nu_i^2, \quad d = \max_{1\leq i\leq n} d_i.$$

□

Example 1.5.10 (Concentration of χ^2 variables)**.** Let $Z_1, \ldots, Z_n$ i.i.d. $N(0,1)$ and let $Y := \sum_{i=1}^n Z_i^2 \sim \chi^2_n$, so that $\mathbb{E}[Y] = n$. Recall from Topic 3 that each Z_i^2 has mean 1 and is $(2,4)$-sub-exponential, i.e. $\mathbb{E}[e^{\lambda(Z_i^2-1)}] \leq e^{2\lambda^2}$ for all $|\lambda| < 1/4$. Applying Lemma 1.5.9 with $\nu_i = 2$, $d_i = 4$ for every i, Y is $(\sqrt{4n}, 4) = (2\sqrt{n}, 4)$-sub-exponential; here $V^2 = 4n$ and $d = 4$, so the threshold in (1.53) is $V^2/d = n$. Thus

$$\mathbb{P}\left(|Y-n| \geq s\right) \leq \begin{cases} 2e^{-s^2/(8n)}, & 0 \leq s \leq n, \\ 2e^{-s/8}, & s > n. \end{cases}$$

Substituting $s = nt$ (so $s \le n \iff t \le 1$) gives

$$\mathbb{P}\left(\left|\frac{Y}{n} - 1\right| \ge t\right) \le \begin{cases} 2e^{-nt^2/8}, & 0 \le t \le 1, \\ 2e^{-nt/8}, & t > 1. \end{cases} \tag{1.54}$$

1.5.3 Application: The Johnson–Lindenstrauss Lemma

Fix distinct points $u^1, \dots, u^N \in \mathbb{R}^d$, where d is large. We would like a map $F : \mathbb{R}^d \to \mathbb{R}^m$ with $m \ll d$, so that we only need to store the compressed points $\{F(u^1), \dots, F(u^N)\}$, while approximately preserving all pairwise (squared) distances: given a distortion parameter $\delta \in (0,1)$, we want

$$1 - \delta \;\le\; \frac{\left\|F(u^i) - F(u^j)\right\|_2^2}{\left\|u^i - u^j\right\|_2^2} \;\le\; 1 + \delta, \qquad \text{for all } i \ne j. \tag{1.55}$$

Construction. Let $X \in \mathbb{R}^{m \times d}$ have i.i.d. $N(0,1)$ entries, and write its rows as $X_1, \dots, X_m \in \mathbb{R}^d$, so that $X_1, \dots, X_m$i.i.d. $N(0, I_d)$. Define the (random) linear map

$$F(u) := \frac{Xu}{\sqrt{m}}, \qquad u \in \mathbb{R}^d.$$

Step 1: distribution of $\|Xu\|_2^2$. Fix $u \in \mathbb{R}^d$, $u \ne 0$. Writing $Xu = (\langle X_1, u\rangle, \dots, \langle X_m, u\rangle)^\top$, for each i and any unit vector $v = u/\|u\|_2$,

$$\langle X_i, v\rangle = \sum_{k=1}^{d} X_{i,k} v_k \sim N\left(0, \sum_{k=1}^{d} v_k^2\right) = N(0,1) \qquad (\text{check: } \|v\|_2 = 1), \tag{1.56}$$

since $X_i \sim N(0, I_d)$ and any linear combination of independent standard normals with coefficient vector of unit ℓ_2-norm is again standard normal. Moreover $\langle X_1, v\rangle, \dots, \langle X_m, v\rangle$ are independent because $X_1, \dots, X_m$ are independent. Hence

$$Y := \frac{\|Xu\|_2^2}{\|u\|_2^2} = \sum_{i=1}^{m} \frac{\langle X_i, u\rangle^2}{\|u\|_2^2} = \sum_{i=1}^{m} \langle X_i, v\rangle^2 \sim \chi^2_m. \tag{1.57}$$

Step 2: concentration for a single vector. Since $\|F(u)\|_2^2 / \|u\|_2^2 = \|Xu\|_2^2 / (m\|u\|_2^2) = Y/m$, applying (1.54) (with n replaced by m) gives, for any $\delta \in (0,1)$,

$$\mathbb{P}\left(\frac{\|F(u)\|_2^2}{\|u\|_2^2} \notin (1-\delta, 1+\delta)\right) = \mathbb{P}\left(\left|\frac{Y}{m} - 1\right| \ge \delta\right) \le 2e^{-m\delta^2/8}. \tag{1.58}$$

Step 3: union bound over all pairs. Apply (1.58) with $u = u^i - u^j$ for each of the $\binom{N}{2}$ unordered pairs $i \ne j$ (assuming $u^i \ne u^j$):

$$\mathbb{P}\left(\frac{\left\|F(u^i) - F(u^j)\right\|_2^2}{\left\|u^i - u^j\right\|_2^2} \notin (1-\delta, 1+\delta)\right) \le 2e^{-m\delta^2/8}.$$

Let

$$I := \bigcup_{i<j} \left\{\frac{\left\|F(u^i) - F(u^j)\right\|_2^2}{\left\|u^i - u^j\right\|_2^2} \notin (1-\delta, 1+\delta)\right\}$$

be the event that (1.55) fails for at least one pair. By the union bound,

$$\mathbb{P}(I) \le \sum_{i<j} \mathbb{P}\left(\frac{\left\|F(u^i) - F(u^j)\right\|_2^2}{\left\|u^i - u^j\right\|_2^2} \notin (1-\delta, 1+\delta)\right) \le \binom{N}{2} \cdot 2e^{-m\delta^2/8} = N(N-1)\, e^{-m\delta^2/8}. \tag{1.59}$$

Step 4: choosing m. Given a target failure probability $\varepsilon \in (0,1)$, we want $\mathbb{P}(I) < \varepsilon$. By (1.59), it suffices that

$$N(N-1)e^{-m\delta^2/8} < \varepsilon \quad \Longleftrightarrow \quad m > \frac{8}{\delta^2}\log\frac{N(N-1)}{\varepsilon}.$$

Since $N(N-1) \le N^2$, a cleaner sufficient condition is

$$m > \frac{16}{\delta^2}\log\frac{N}{\varepsilon}. \tag{1.60}$$

Theorem 1.5.11 (Johnson–Lindenstrauss Lemma)**.** *Let distinct $u^1, \dots, u^N \in \mathbb{R}^d$ and let $\delta, \varepsilon \in (0,1)$. If*

$$m > \frac{16}{\delta^2}\log\frac{N}{\varepsilon},$$

then, with $X \in \mathbb{R}^{m\times d}$ having i.i.d. $N(0,1)$ entries and $F(u) := Xu/\sqrt{m}$, with probability at least $1-\varepsilon$ we have

$$1-\delta \le \frac{\left\|F(u^i) - F(u^j)\right\|_2^2}{\|u^i - u^j\|_2^2} \le 1+\delta \qquad \text{simultaneously for all } i \ne j.$$

Proof. Immediate from (1.59) and (1.60): the choice of m forces $\mathbb{P}(I) < \varepsilon$, i.e. (1.55) holds for every pair on the complementary event, which has probability at least $1-\varepsilon$. □

Theorem 1.5.12 (Equivalent characterizations of sub-exponential random variables)**.** *For a mean-zero random variable X, the following conditions are equivalent.*

(i) There exist non-negative parameters (ν, α) such that

$$\mathbb{E}\left[e^{\lambda X}\right] \le e^{\nu^2\lambda^2/2} \qquad \text{for all } |\lambda| \le \frac{1}{\alpha}.$$

(ii) There exist constants $c_0, c_0 > 0$ such that

$$\mathbb{E}\left[e^{\lambda X}\right] \le c_0 \qquad \text{for all } |\lambda| \le c_0.$$

(iii) There exist constants $c_1, c_2 > 0$ such that

$$\mathbb{P}\left(|X| \ge t\right) \le c_1 e^{-c_2 t} \qquad \text{for all } t \ge 0.$$

(iv) The quantity

$$\gamma := \sup_{k\ge 2}\left(\frac{\mathbb{E}\left[|X|^k\right]}{k!}\right)^{1/k}$$

is finite.

Remark 1.5.13. A sub-exponential tail bound decays more slowly than a sub-Gaussian tail bound for large t, so the sub-exponential class permits heavier tails.

1.6 Martingale-Based Methods

Definition 1.6.1 (Martingale difference sequence)**.** An adapted sequence $\{(D_k, \mathcal{F}_k)\}_{k=0}^{\infty}$ is called a martingale difference sequence if, for every $k \ge 1$,

$$\mathbb{E}\left[|D_k|\right] < \infty \qquad \text{and} \qquad \mathbb{E}\left[D_k \mid \mathcal{F}_{k-1}\right] = 0.$$

Equivalently, after shifting the index, $\mathbb{E}\left[D_{k+1} \mid \mathcal{F}_k\right] = 0$.

Remark 1.6.2. Here, D_k denotes the difference and $\mathcal{F}_k$ denotes the associated sigma-algebra.

Example 1.6.3. Let $\{(Y_k, \mathcal{F}_k)\}_{k\geq 0}$ be a martingale and define

$$D_k := Y_k - Y_{k-1}, \qquad k \geq 1.$$

Then D_k is $\mathcal{F}_k$-measurable and

$$\mathbb{E}\left[D_{k+1} \mid \mathcal{F}_k\right] = \mathbb{E}\left[Y_{k+1} \mid \mathcal{F}_k\right] - Y_k = 0 \qquad \text{a.s.}$$

Moreover,

$$Y_k - Y_0 = \sum_{i=1}^{k} D_i.$$

Theorem 1.6.4. *Let $\{(D_k, \mathcal{F}_k)\}$ be a martingale difference sequence. Suppose that, for every $k = 1, \ldots, n$,*

$$\mathbb{E}\left[e^{\lambda D_k} \mid \mathcal{F}_{k-1}\right] \leq e^{\lambda^2 \nu_k^2/2} \qquad \textit{for all } |\lambda| < \frac{1}{\alpha_k}.$$

Define

$$\alpha_* := \max_{1\leq k\leq n} \alpha_k.$$

Then the following statements hold.

(a) $\sum_{k=1}^n D_k$ *is sub-exponential with parameters* $\left(\sqrt{\sum_{k=1}^n \nu_k^2},\, \alpha_*\right)$.

(b) $\mathbb{P}\left(\left|\sum_{k=1}^n D_k\right| \geq t\right) \leq \begin{cases} 2\exp\left(-\dfrac{t^2}{2\sum_{k=1}^n \nu_k^2}\right), & 0 \leq t \leq \dfrac{\sum_{k=1}^n \nu_k^2}{\alpha_*}, \\ 2\exp\left(-\dfrac{t}{2\alpha_*}\right), & t > \dfrac{\sum_{k=1}^n \nu_k^2}{\alpha_*}. \end{cases}$

Remark 1.6.5. The moment-generating-function assumption in Theorem 1.6.4 is a conditional sub-exponential condition. Part (a) is the martingale-difference analogue of the standard result that a sum of independent sub-exponential random variables is sub-exponential.

Proof. First, the tower property gives

$$\mathbb{E}[D_k] = \mathbb{E}\left[\mathbb{E}\left[D_k \mid \mathcal{F}_{k-1}\right]\right] = 0,$$

and hence $\mathbb{E}[\sum_{k=1}^n D_k] = 0$.

For $|\lambda| < 1/\alpha_*$, condition successively on the filtration:

$$\begin{aligned}
\mathbb{E}\left[e^{\lambda \sum_{k=1}^n D_k}\right] &= \mathbb{E}\left[\mathbb{E}\left[e^{\lambda \sum_{k=1}^n D_k} \mid \mathcal{F}_{n-1}\right]\right] \\
&= \mathbb{E}\left[e^{\lambda \sum_{k=1}^{n-1} D_k} \mathbb{E}\left[e^{\lambda D_n} \mid \mathcal{F}_{n-1}\right]\right] \\
&\leq \mathbb{E}\left[e^{\lambda \sum_{k=1}^{n-1} D_k} e^{\lambda^2 \nu_n^2/2}\right], \quad |\lambda| < \frac{1}{\alpha_n} \\
&= e^{\lambda^2 \nu_n^2/2} \mathbb{E}\left[e^{\lambda \sum_{k=1}^{n-1} D_k}\right], \quad |\lambda| < \frac{1}{\alpha_n} \\
&\leq e^{\lambda^2 \nu_n^2/2} e^{\lambda^2 \nu_{n-1}^2/2} \mathbb{E}\left[e^{\lambda \sum_{k=1}^{n-2} D_k}\right], \quad |\lambda| < \frac{1}{\alpha_n},\ |\lambda| < \frac{1}{\alpha_{n-1}} \\
&\leq \cdots \leq e^{\lambda^2 \nu_n^2/2} \cdots e^{\lambda^2 \nu_1^2/2}, \quad |\lambda| < \frac{1}{\alpha_n}, \ldots,\ |\lambda| < \frac{1}{\alpha_1} \\
&= \exp\left(\frac{\lambda^2}{2} \sum_{k=1}^n \nu_k^2\right), \qquad |\lambda| < \frac{1}{\max_{1\leq k\leq n} \alpha_k}.
\end{aligned}$$

This proves part (a). Part (b) follows from the standard two-regime sub-exponential tail bound. □

Remark 1.6.6. If the D_k were independent, the MGF would factor as $\prod_{k=1}^{n} \mathbb{E}[e^{\lambda D_k}]$. For a martingale difference sequence, conditional expectation replaces this factorization. Given $\mathcal{F}_{n-1}$, the partial sum $\sum_{k=1}^{n-1} D_k$ is $\mathcal{F}_{n-1}$-measurable and therefore acts as a constant inside the conditional expectation.

1.6.1 Azuma–Hoeffding Inequality

Corollary 1.6.7 (Azuma–Hoeffding random interval)**.** *Let $\{(D_k, \mathcal{F}_k)\}_{k=1}^{\infty}$ be a martingale difference sequence and suppose $\mathbb{E}[D_k \mid \mathcal{F}_{k-1}] \in [a_k, b_k]$, where $a_k, b_k \in \mathcal{F}_{k-1}$ for all $k = 1, \dots, n$. Assume that $b_k - a_k \le L_k$ is a constant. Then, for all $t \ge 0$,*

$$\mathbb{P}\left(\left|\sum_{k=1}^{n} D_k\right| \ge t\right) \le 2\exp\left(-\frac{2t^2}{\sum_{k=1}^{n} L_k^2}\right). \tag{1.61}$$

Proof. Conditioned on $\mathcal{F}_{k-1}$, the random variable D_k still lies in $[a_k, b_k]$ almost surely. By the conditional form of Hoeffding's lemma, D_k is conditionally $(b_k - a_k)/2$-sub-Gaussian, and therefore $L_k/2$-sub-Gaussian and is therefore conditionally sub-exponential with parameters

$$\left(\frac{L_k}{2}, 0\right).$$

By Theorem 1.6.4(a),

$$\sum_{k=1}^{n} D_k \text{ is } \left(\sqrt{\sum_{k=1}^{n} \frac{(L_k)^2}{4}}, 0\right) \text{ sub-exponential } \longrightarrow \left(\sqrt{\sum_{k=1}^{n} \frac{(L_k)^2}{4}}\right) \text{ sub-Gaussian.}$$

The stated tail bound follows. □

Corollary 1.6.8 (Azuma–Hoeffding fix interval)**.** *Let $\{(D_k, \mathcal{F}_k)\}$ be a martingale difference sequence. Suppose there are constants $\{(a_k, b_k)\}_{k=1}^{\infty}$ such that*

$$D_k \in [a_k, b_k] \qquad \text{a.s. for every } k.$$

Then, for every $t \ge 0$,

$$\mathbb{P}\left(\left|\sum_{k=1}^{n} D_k\right| \ge t\right) \le 2\exp\left(-\frac{2t^2}{\sum_{k=1}^{n} (b_k - a_k)^2}\right).$$

Proof. It directly follows from the above corollary. □

Remark 1.6.9. Conditional expectation preserves inequalities: if $X \le b$, then $\mathbb{E}[X \mid \mathcal{F}] \le \mathbb{E}[b \mid \mathcal{F}] = b$. Also, every sub-Gaussian random variable is sub-exponential.

1.6.2 Bounded Differences

Definition 1.6.10 (Bounded difference property)**.** We say that a function f satisfies the bounded difference property with parameters $(L_1, \dots, L_n)$ if, for every $k = 1, \dots, n$,

$$|f(x_1, \dots, x_n) - f(x_1, \dots, x_{k-1}, x_k', x_{k+1}, \dots, x_n)| \le L_k$$

for all $x_1, \dots, x_n, x_k'$.

Example 1.6.11. For $x, y \in \mathbb{R}^n$, define the Hamming distance by

$$d_H(x,y) := \sum_{i=1}^{n} \mathbf{1}\{x_i \neq y_i\}.$$

If f is L-Lipschitz with respect to d_H, meaning that

$$|f(x) - f(y)| \leq L\, d_H(x,y),$$

then changing only the kth coordinate gives

$$|f(x_1,\ldots,x_n) - f(x_1,\ldots,x_{k-1},x_k',x_{k+1},\ldots,x_n)| \leq L \cdot 1 = L.$$

Thus, f has the bounded difference property with $L_k = L$ for every k.

Corollary 1.6.12 (Bounded difference inequality)**.** *Suppose f satisfies the bounded difference property with parameters $(L_1,\ldots,L_n)$ and that $X = (X_1,\ldots,X_n)$ has independent components. Then, for every $t \geq 0$,*

$$\mathbb{P}\left(|f(X) - \mathbb{E}[f(X)]| \geq t\right) \leq 2\exp\left(-\frac{2t^2}{\sum_{k=1}^n L_k^2}\right).$$

Proof. Let

$$\mathcal{F}_k := \sigma(X_1,\ldots,X_k), \qquad Y_k := \mathbb{E}\left[f(X) \mid \mathcal{F}_k\right].$$

Since $\mathcal{F}_n$ is generated by all the components of X,

$$Y_n = \mathbb{E}\left[f(X) \mid \mathcal{F}_n\right] = f(X),$$

while

$$Y_0 = \mathbb{E}\left[f(X) \mid \mathcal{F}_0\right] = \mathbb{E}[f(X)].$$

Therefore,

$$f(X) - \mathbb{E}[f(X)] = Y_n - Y_0 = \sum_{k=1}^{n} D_k, \qquad \text{where} D_k := Y_k - Y_{k-1}.$$

Note

$$\begin{aligned} D_k &= \mathbb{E}\left[f(X) \mid X_1,\ldots,X_k\right] - \mathbb{E}\left[f(X) \mid X_1,\ldots,X_{k-1}\right] \\ &= \mathbb{E}_{X_{k+1},\ldots,X_n}\left[f(X)\right] - \mathbb{E}_{X_k,X_{k+1},\ldots,X_n}\left[f(X)\right] \\ &:= \mathbb{E}_{k+1}\left[f(X)\right] - \mathbb{E}_k\left[f(X)\right]. \end{aligned}$$

Then $D_k \in \mathcal{F}_k$, and

$$\begin{aligned} \mathbb{E}\left[D_k \mid \mathcal{F}_{k-1}\right] &= \mathbb{E}\left[\mathbb{E}\left[f(X) \mid \mathcal{F}_k\right] - \mathbb{E}\left[f(X) \mid \mathcal{F}_{k-1}\right] \mid \mathcal{F}_{k-1}\right] \\ &= \mathbb{E}\left[f(X) \mid \mathcal{F}_{k-1}\right] - \mathbb{E}\left[f(X) \mid \mathcal{F}_{k-1}\right] \\ &= 0 \qquad \text{a.s.} \end{aligned}$$

Thus, $(D_k)_{k=1}^n$ is a martingale difference sequence.
Next,

$$\begin{aligned} D_k &= \mathbb{E}_{k+1}\left[f(X_1,\ldots,X_{k-1},X_k,X_{k+1},\ldots,X_n)\right] - \mathbb{E}_k\left[f(X_1,\ldots,X_{k-1},X_k,X_{k+1},\ldots,X_n)\right] \\ &\leq \sup_x \mathbb{E}_{k+1}\left[f(X_1,\ldots,X_{k-1},x,X_{k+1},\ldots,X_n)\right] - \mathbb{E}_k\left[f(X_1,\ldots,X_{k-1},X_k,X_{k+1},\ldots,X_n)\right] \\ &:= B_k, \end{aligned}$$

where $B_k \in \mathcal{F}_{k-1}$. Similarly,

$$\begin{aligned} D_k &\geq \inf_x \mathbb{E}_{k+1}\left[f(X_1,\ldots,X_{k-1},x,X_{k+1},\ldots,X_n)\right] - \mathbb{E}_k\left[f(X_1,\ldots,X_{k-1},X_k,X_{k+1},\ldots,X_n)\right] \\ &:= A_k, \end{aligned}$$

where $A_k \in \mathcal{F}_{k-1}$. Then, we have

$$\begin{aligned} B_k - A_k &= \sup_x \mathbb{E}_{k+1}\left[f(X_1,\ldots,X_{k-1},x,X_{k+1},\ldots,X_n)\right] - \inf_{x'} \mathbb{E}_{k+1}\left[f(X_1,\ldots,X_{k-1},x',X_{k+1},\ldots,X_n)\right] \\ &\leq \sup_{x_1,\ldots,x_{k-1},\,x_k,x',\,x_{k+1},\ldots,x_n} |f(x_1,\ldots,x_k,\ldots,x_n) - f(x_1,\ldots,x_{k-1},x_k',x_{k+1},\ldots,x_n)| \\ &\leq L_k, \end{aligned}$$

using (**??**). Therefore, by Azuma–Hoeffding with random interval,

$$\mathbb{P}\left(\left|\sum_{k=1}^n D_k\right| \geq t\right) = \mathbb{P}(|f(X) - \mathbb{E}f(X)| \geq t) \leq 2\exp\left(-\frac{2t^2}{\sum_{k=1}^n L_k^2}\right).$$

□

Example 1.6.13 (U-statistics)**.** Let $g : \mathbb{R}^2 \to \mathbb{R}$ be a symmetric function of its arguments. Given i.i.d. sequence X_k, $k \geq 1$ of random variables, the quantity

$$U := \frac{1}{\binom{n}{2}} \sum_{1\leq j<k\leq n} g(X_j, X_k) \tag{1.62}$$

is known as a pairwise U-statistic.

For example, suppose we want to estimate the mean absolute pairwise deviation $\mathbb{E}[|X_1 - X_2|]$. Then U with $g(x,y) = |x-y|$ is one estimate for it.
Since the X_i are identically distributed,

$$\mathbb{E}U = \frac{1}{\binom{n}{2}} \sum_{1\leq j<k\leq n} \mathbb{E}g(X_j, X_k) = \frac{1}{\binom{n}{2}} \sum_{1\leq j<k\leq n} \mathbb{E}g(X_1, X_2) = \mathbb{E}g(X_1, X_2).$$

Suppose $\|g\|_\infty \leq b$. Consider $U = f(X_1,\ldots,X_n)$. Then

$$\begin{aligned} &|f(X_1,\ldots,X_n) - f(X_1,\ldots,X_{k-1},X_k',X_{k+1},\ldots,X_n)| \\ &= \left|\frac{1}{\binom{n}{2}} \sum_{1\leq j<s\leq n} g(X_j,X_s) - \frac{1}{\binom{n}{2}} \sum_{1\leq j<s\leq n} g(X_j',X_s')\right| \qquad [\text{where } X_j' = X_j,\ j = 1,\ldots,k-1,k+1,\ldots,n], \\ &= \frac{1}{\binom{n}{2}} \left|\sum_{1\leq j<s\leq n} g(X_j,X_s) - \sum_{1\leq j<s\leq n} g(X_j',X_s')\right| \\ &\leq \frac{1}{\binom{n}{2}} \sum_{1\leq j<s\leq n} |g(X_j,X_s) - g(X_j,X_s')| \\ &\leq \frac{1}{\binom{n}{2}}\ |2b(n-k+k-1)| \\ &= \frac{4b}{n}. \end{aligned}$$

So the bounded differences property holds with parameter $L_k = \frac{4b}{n}$ in each coordinate. Therefore, the bounded difference inequality gives

$$\mathbb{P}\left(|U - \mathbb{E}U| \geq t\right) \leq 2\exp\left(-\frac{2t^2}{\sum_{k=1}^n (4b/n)^2}\right) = 2\exp\left(-\frac{nt^2}{8b^2}\right). \tag{1.63}$$

Example 1.6.14 (Rademacher Complexity)**.** Let $\{\varepsilon_k\}_{k=1}^n$ be an i.i.d. sequence of Rademacher variables. Given a collection of vectors $\mathcal{A} \subset \mathbb{R}^n$, define the random variable

$$Z(\mathcal{A}) = Z := \sup_{a\in\mathcal{A}} \langle a, \varepsilon\rangle = \sup_{a\in\mathcal{A}} \sum_{i=1}^n a_i\varepsilon_i. \tag{1.64}$$

Then $R(\mathcal{A}) := \mathbb{E}Z(\mathcal{A})$ is the Rademacher complexity of $\mathcal{A}$.

Now let us establish $Z(\mathcal{A}) = f(\varepsilon_1, \ldots, \varepsilon_n)$ is sub-Gaussian. Given two Rademacher vectors $\varepsilon, \varepsilon' \in \{-1, +1\}^n$, and $\varepsilon_i' = \varepsilon_i$ for $i \neq k$, we have

$$\begin{aligned}
|f(\varepsilon_1, \ldots, \varepsilon_n) - f(\varepsilon_1', \ldots, \varepsilon_n')| &= \left|\sup_{a\in\mathcal{A}} \langle a, \varepsilon\rangle - \sup_{a\in\mathcal{A}} \langle a, \varepsilon'\rangle\right| \\
&\leq \sup_{a\in\mathcal{A}} |\langle a, \varepsilon\rangle - \langle a, \varepsilon'\rangle| \qquad \text{(by Lemma 1.6.15)} \\
&= \sup_{a\in\mathcal{A}} |a_k\varepsilon_k - a_k\varepsilon_k'| \\
&= \sup_{a\in\mathcal{A}} |a_k|\,|\varepsilon_k - \varepsilon_k'| \\
&\leq 2\sup_{a\in\mathcal{A}} |a_k|\,.
\end{aligned}$$

Consequently, the bounded difference inequality yields

$$\mathbb{P}(|Z(\mathcal{A}) - R(\mathcal{A})| \geq t) \leq 2\exp\left(-\frac{2t^2}{\sum_{k=1}^n 4\sup_{a\in\mathcal{A}} a_k^2}\right) = 2\exp\left(-\frac{t^2}{2\sum_{k=1}^n \sup_{a\in\mathcal{A}} a_k^2}\right).$$

Thus, $Z(\mathcal{A})$ is sub-Gaussian with parameter $\sqrt{2\sum_{k=1}^n \sup_{a\in\mathcal{A}} a_k^2}$.

Lemma 1.6.15. *The absolute difference of the suprema of two functions is bounded by the supremum of their pointwise absolute difference*

$$\left|\sup_{x\in B} f(x) - \sup_{x\in B} g(x)\right| \leq \sup_{x\in B} |f(x) - g(x)|.$$

1.7 Lipschitz Functions of Gaussian Variables

We first show that Lipschitz functions of a standard Gaussian vector are sub-Gaussian.

Definition 1.7.1. A function $f : \mathbb{R}^n \to \mathbb{R}$ is L-Lipschitz with respect to the Euclidean norm $\|\cdot\|_2$ if

$$|f(x) - f(y)| \leq L\|x - y\|_2 \qquad \text{for all } x, y \in \mathbb{R}^n. \tag{1.65}$$

Theorem 1.7.2 ([1, Theorem 2.26])**.** *Let $X = (X_1, \ldots, X_n) \sim N(0, I_n)$, and let $f : \mathbb{R}^n \to \mathbb{R}$ be L-Lipschitz with respect to $\|\cdot\|_2$. Then $f(X) - \mathbb{E}[f(X)]$ is sub-Gaussian with parameter at most L, and hence*

$$\mathbb{P}\left(|f(X) - \mathbb{E}[f(X)]| \geq t\right) \leq 2e^{-\frac{t^2}{2L^2}} \qquad \text{for all } t \geq 0. \tag{1.66}$$

Remark 1.7.3. The sub-Gaussian parameter in Theorem 1.7.2 depends on f only through L and does not depend on the dimension n. Thus an L-Lipschitz function of an n-dimensional standard Gaussian vector obeys the same tail bound as a scalar $N(0, L^2)$ variable, however large n is. The tail bound (1.66) follows from the sub-Gaussian property by applying the Chernoff bound from Lecture 1 to $f(X)$ and to $-f(X)$.

We do not prove Theorem 1.7.2 here; a proof of the weaker statement with sub-Gaussian parameter $\pi L/2$ can be found in [1, Section 2.3]. We now consider three applications.

Example 1.7.4 (Order statistics, [1, Example 2.29])**.** Given a random vector $(X_1, \dots, X_n)$, its order statistics are obtained by reordering its entries in non-decreasing order,

$$X_{(1)} \le X_{(2)} \le \cdots \le X_{(n)}.$$

In particular, $X_{(n)} = \max_{1\le i\le n} X_i$ and $X_{(1)} = \min_{1\le i\le n} X_i$. For $k \in \{1, \dots, n\}$, let $f_k : \mathbb{R}^n \to \mathbb{R}$ denote the map $x \mapsto x_{(k)}$, so that $X_{(k)} = f_k(X_1, \dots, X_n)$. The Lipschitz constant of f_k is controlled by the following lemma.

Lemma 1.7.5. *For all $x, y \in \mathbb{R}^n$ and $k \in \{1, \dots, n\}$,*

$$\left|x_{(k)} - y_{(k)}\right| \le \|x - y\|_\infty .$$

The proof of Lemma 1.7.5 is left as exercise. Since $\|x - y\|_\infty = \max_{1\le i\le n} |x_i - y_i| \le \|x - y\|_2$, the lemma gives

$$|f_k(x) - f_k(y)| \le \|x - y\|_2 \qquad \text{for all } x, y \in \mathbb{R}^n,$$

so each f_k is 1-Lipschitz. Consequently, if $(X_1, \dots, X_n) \sim N(0, I_n)$, Theorem 1.7.2 with $L = 1$ yields, for every $k = 1, \dots, n$,

$$\mathbb{P}\left(\left|X_{(k)} - \mathbb{E}[X_{(k)}]\right| \ge t\right) \le 2e^{-\frac{t^2}{2}} \qquad \text{for all } t \ge 0. \tag{1.67}$$

In particular, the fluctuations of the maximum $X_{(n)}$ around its mean are of order one, uniformly in n.

Example 1.7.6 (Gaussian complexity, [1, Example 2.30])**.** Let $A \subseteq \mathbb{R}^n$ be a bounded set and $W \sim N(0, I_n)$, and define

$$Z(A) := \sup_{a\in A} \langle a, W\rangle .$$

The expectation $\mathbb{E}[Z(A)]$ is called the Gaussian complexity of A. It is the Gaussian counterpart of the Rademacher complexity from Lecture 6, with the Rademacher vector replaced by W.

To show that $Z(A)$ concentrates, write $Z(A) = f(W)$ with $f(w) := \sup_{a\in A} \langle a, w\rangle$ for $w \in \mathbb{R}^n$, and let $D(A) := \sup_{a\in A} \|a\|_2 < \infty$. For any $w, w' \in \mathbb{R}^n$,

$$\begin{aligned} |f(w) - f(w')| &= \left|\sup_{a\in A} \langle a, w\rangle - \sup_{a\in A} \langle a, w'\rangle\right| \le \sup_{a\in A} |\langle a, w\rangle - \langle a, w'\rangle| \\ &= \sup_{a\in A} |\langle a, w - w'\rangle| \le \sup_{a\in A} \|a\|_2 \, \|w - w'\|_2 = D(A) \, \|w - w'\|_2 . \end{aligned}$$

The second inequality is the Cauchy–Schwarz inequality. Hence f is $D(A)$-Lipschitz, and Theorem 1.7.2 yields

$$\mathbb{P}\left(|Z(A) - \mathbb{E}[Z(A)]| \ge t\right) \le 2\exp\left(-\frac{t^2}{2D(A)^2}\right) \qquad \text{for all } t \ge 0. \tag{1.68}$$

Example 1.7.7 (Singular values of Gaussian random matrices, [1, Example 2.32])**.** Let $n > d$ and let $X \in \mathbb{R}^{n\times d}$ be a random matrix with i.i.d. $N(0, 1)$ entries and ordered singular values $\sigma_1(X) \ge \sigma_2(X) \ge \cdots \ge \sigma_d(X) \ge 0$. Writing $x_1^\top, \dots, x_n^\top$ for the rows of X, we view the matrix as the vector

$$\mathrm{vec}(X) := \left(x_1^\top, x_2^\top, \dots, x_n^\top\right)^\top \in \mathbb{R}^{nd},$$

so that $\mathrm{vec}(X) \sim N(0, I_{nd})$ and each $\sigma_k(X)$ is a function of a standard Gaussian vector in $\mathbb{R}^{nd}$. For $X, Y \in \mathbb{R}^{n\times d}$ we have $\|\mathrm{vec}(X) - \mathrm{vec}(Y)\|_2 = \|X - Y\|_F$, so it suffices to bound $|\sigma_k(X) - \sigma_k(Y)|$ by $\|X - Y\|_F$.

Recall that for $M \in \mathbb{R}^{n\times d}$ the spectral norm and the Frobenius norm are

$$\|M\|_2 := \sigma_1(M) = \sup_{a\in\mathbb{R}^d:\, \|a\|_2=1} \|Ma\|_2\,, \qquad \|M\|_F := \Big(\sum_{i=1}^{n}\sum_{j=1}^{d} M_{ij}^2\Big)^{1/2} = \big(\sigma_1^2(M) + \cdots + \sigma_d^2(M)\big)^{1/2}\,,$$

so that $\|M\|_2 \le \|M\|_F$. By Weyl's inequality (see [1, Exercise 8.3]),

$$\max_{1\le k\le d} |\sigma_k(X) - \sigma_k(Y)| \le \|X - Y\|_2 \le \|X - Y\|_F\,. \tag{1.69}$$

Thus each σ_k is 1-Lipschitz as a function of $\mathrm{vec}(X)$, and Theorem 1.7.2 gives, for each $k = 1, \ldots, d$,

$$\mathbb{P}\left(|\sigma_k(X) - \mathbb{E}[\sigma_k(X)]| \ge t\right) \le 2e^{-\frac{t^2}{2}} \qquad \text{for all } t \ge 0. \tag{1.70}$$

We have not computed $\mathbb{E}[\sigma_k(X)]$, but (1.70) shows that each singular value stays within $O(1)$ of its expectation with high probability, regardless of n and d.

Chapter 2

Concentration of measure

2.1 Concentration by Entropic Techniques

We now introduce the entropy method for deriving concentration inequalities [1, Section 3.1].

Definition 2.1.1 (ϕ-entropy)**.** Let $X \sim P$, and let ϕ be a convex function on an interval containing the support of X such that X and $\phi(X)$ are integrable. The ϕ-entropy of X is

$$H_\phi(X) := \mathbb{E}[\phi(X)] - \phi(\mathbb{E}[X]). \tag{2.1}$$

Remark 2.1.2. By Jensen's inequality, $H_\phi(X) \geq 0$. Moreover, $H_\phi(X) = 0$ if X is almost surely constant or if ϕ is affine. Although the notation suggests a function of X, $H_\phi(X)$ is a number that depends only on the distribution P of X; for instance, $H_\phi(e^{\lambda X})$ below is the ϕ-entropy of the distribution of $e^{\lambda X}$.

Example 2.1.3. Two familiar quantities are ϕ-entropies.

(a) If $\phi(u) = u^2$, then $H_\phi(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \mathrm{Var}(X)$.

(b) If $\phi(u) = -\log u$ for $u > 0$, then applying ϕ to the positive random variable $e^{\lambda X}$ gives

$$\begin{aligned} H_\phi(e^{\lambda X}) &= \mathbb{E}\left[-\log e^{\lambda X}\right] - \left(-\log \mathbb{E}\left[e^{\lambda X}\right]\right) = -\lambda \mathbb{E}[X] + \log \mathbb{E}\left[e^{\lambda X}\right] \\ &= \log e^{-\lambda \mathbb{E}[X]} + \log \mathbb{E}\left[e^{\lambda X}\right] = \log \mathbb{E}\left[e^{\lambda (X - \mathbb{E}[X])}\right], \end{aligned}$$

which is the cumulant generating function of the centered variable $X - \mathbb{E}[X]$.

From now on we work with the convex function $\phi : [0, \infty) \to \mathbb{R}$ defined by

$$\phi(u) := u \log u \quad \text{for } u > 0, \qquad \phi(0) := 0, \tag{2.2}$$

where the value at 0 makes ϕ continuous, since $u \log u \to 0$ as $u \to 0^+$. For a non-negative random variable Z we drop the subscript and write

$$H(Z) := H_\phi(Z) = \mathbb{E}[Z \log Z] - \mathbb{E}[Z] \log \mathbb{E}[Z]. \tag{2.3}$$

Let $\varphi_X(\lambda) := \mathbb{E}\left[e^{\lambda X}\right]$ be the moment generating function of X (not to be confused with the convex function ϕ), and assume that it is finite in a neighborhood of λ, so that $\varphi_X'(\lambda) = \mathbb{E}\left[X e^{\lambda X}\right]$. Taking $Z = e^{\lambda X}$ in (2.3) and using $\log e^{\lambda X} = \lambda X$, we obtain

$$H(e^{\lambda X}) = \mathbb{E}\left[e^{\lambda X} \cdot \lambda X\right] - \mathbb{E}\left[e^{\lambda X}\right] \log \mathbb{E}\left[e^{\lambda X}\right] = \lambda \varphi_X'(\lambda) - \varphi_X(\lambda) \log \varphi_X(\lambda). \tag{2.4}$$

Hence $H(e^{\lambda X})$ is determined by the moment generating function of X and its derivative.

Example 2.1.4 (Entropy of a Gaussian variable, [1, Example 3.1])**.** Let $X \sim N(0, \sigma^2)$. Then $\varphi_X(\lambda) = e^{\lambda^2\sigma^2/2}$ and $\varphi_X'(\lambda) = \lambda\sigma^2 e^{\lambda^2\sigma^2/2}$ for all $\lambda \in \mathbb{R}$, so by (2.4),

$$H(e^{\lambda X}) = \lambda \cdot \lambda\sigma^2 e^{\frac{\lambda^2\sigma^2}{2}} - e^{\frac{\lambda^2\sigma^2}{2}} \log e^{\frac{\lambda^2\sigma^2}{2}} = \frac{\lambda^2\sigma^2}{2} e^{\frac{\lambda^2\sigma^2}{2}} = \frac{1}{2}\sigma^2\lambda^2\varphi_X(\lambda). \tag{2.5}$$

Since the moment generating function yields tail bounds through the Chernoff bound, (2.4) suggests that an upper bound on $H(e^{\lambda X})$ should also lead to a tail bound. The Herbst argument makes this precise for upper bounds of the Gaussian form (2.5).

Proposition 2.1.5 (Herbst argument)**.** *Let $\sigma > 0$ be a constant, $\varphi_X(\lambda) = \mathbb{E}\left[e^{\lambda X}\right]$, and $I = \mathbb{R}$ or $I = [0, +\infty)$. Suppose the entropy $H(e^{\lambda X})$ satisfies*

$$H(e^{\lambda X}) \le \frac{1}{2}\sigma^2\lambda^2\varphi_X(\lambda) \qquad \text{for all } \lambda \in I. \tag{2.6}$$

Then X satisfies

$$\log \mathbb{E}\left[e^{\lambda(X-\mathbb{E}[X])}\right] \le \frac{1}{2}\lambda^2\sigma^2 \qquad \text{for all } \lambda \in I, \tag{2.7}$$

or equivalently, $\mathbb{E}\left[e^{\lambda(X-\mathbb{E}[X])}\right] \le e^{\frac{\lambda^2\sigma^2}{2}}$ for all $\lambda \in I$.

Proof of Proposition 2.1.5. **Case 1: $I = [0, \infty)$.**

① **A differential inequality for the MGF.** Combining (2.4) with the hypothesis (2.6) gives

$$\lambda\varphi_X'(\lambda) - \varphi_X(\lambda)\log\varphi_X(\lambda) \le \frac{1}{2}\sigma^2\lambda^2\varphi_X(\lambda) \qquad \text{for all } \lambda \in I. \tag{2.8}$$

Since $\varphi_X(\lambda) > 0$, we can divide by $\varphi_X(\lambda)$:

$$\frac{\lambda\varphi_X'(\lambda)}{\varphi_X(\lambda)} - \log\varphi_X(\lambda) \le \frac{1}{2}\sigma^2\lambda^2, \qquad \lambda \ge 0. \tag{2.9}$$

② **The auxiliary function G.** The key step is to rewrite the left side of (2.9) as a derivative. Define

$$G(\lambda) := \frac{\log\varphi_X(\lambda)}{\lambda}, \qquad \lambda > 0.$$

By the product rule,

$$G'(\lambda) = -\frac{1}{\lambda^2}\log\varphi_X(\lambda) + \frac{1}{\lambda}\cdot\frac{\varphi_X'(\lambda)}{\varphi_X(\lambda)} = \frac{1}{\lambda^2}\left[\frac{\lambda\varphi_X'(\lambda)}{\varphi_X(\lambda)} - \log\varphi_X(\lambda)\right].$$

So, up to the factor $1/\lambda^2$, $G'(\lambda)$ is exactly the left side of (2.9). Dividing (2.9) by $\lambda^2 > 0$ gives

$$G'(\lambda) \le \frac{1}{2}\sigma^2 \qquad \text{for all } \lambda > 0. \tag{2.10}$$

③ **Integrate.** Fix $\lambda > 0$. For any $\lambda_0 \in (0, \lambda]$, by the fundamental theorem of calculus and (2.10),

$$G(\lambda) - G(\lambda_0) = \int_{\lambda_0}^{\lambda} G'(s)\,ds \le \int_{\lambda_0}^{\lambda} \frac{1}{2}\sigma^2\,ds = \frac{1}{2}\sigma^2(\lambda - \lambda_0).$$

(We need $\lambda_0 \le \lambda$ here: if $\lambda < \lambda_0$, the integral runs backwards and the inequality flips. This does no harm, because we next send $\lambda_0 \to 0^+$.)

④ **Let $\lambda_0 \to 0^+$.** Taking the limit (which exists, as computed below),

$$G(\lambda) - \lim_{\lambda_0 \to 0^+} G(\lambda_0) \le \frac{1}{2}\sigma^2\lambda.$$

To compute the limit, use $\varphi_X(0) = \mathbb{E}\left[e^0\right] = 1$, so $\log \varphi_X(0) = 0$. The limit is then a right-hand difference quotient:

$$\begin{aligned}\lim_{\lambda_0 \to 0^+} G(\lambda_0) &= \lim_{\lambda_0 \to 0^+} \frac{\log \varphi_X(\lambda_0)}{\lambda_0} = \lim_{\lambda_0 \to 0^+} \frac{\log \varphi_X(\lambda_0) - \log \varphi_X(0)}{\lambda_0 - 0} = \frac{d}{d\lambda} \log \varphi_X(\lambda)\Big|_{\lambda=0} \\ &= \frac{\varphi_X'(0)}{\varphi_X(0)} = \frac{\mathbb{E}[X]}{1} = \mathbb{E}[X].\end{aligned}$$

So G extends continuously to 0 with $G(0) := \mathbb{E}[X]$.

⑤ **Unwind.** We have shown that for all $\lambda > 0$,

$$G(\lambda) - \mathbb{E}[X] \le \frac{1}{2}\sigma^2\lambda \iff \frac{\log \varphi_X(\lambda)}{\lambda} - \mathbb{E}[X] \le \frac{1}{2}\sigma^2\lambda \iff \log \varphi_X(\lambda) - \lambda\mathbb{E}[X] \le \frac{1}{2}\sigma^2\lambda^2.$$

Since $\log \varphi_X(\lambda) - \lambda\mathbb{E}[X] = \log \mathbb{E}\left[e^{\lambda X}\right] - \log e^{\lambda \mathbb{E}[X]} = \log \mathbb{E}\left[e^{\lambda(X - \mathbb{E}[X])}\right]$, this is

$$\log \mathbb{E}\left[e^{\lambda(X-\mathbb{E}[X])}\right] \le \frac{1}{2}\sigma^2\lambda^2 \qquad \text{for all } \lambda > 0.$$

The bound also holds at $\lambda = 0$, where both sides equal 0. This proves the claim for $I = [0, \infty)$.

Case 2: $I = \mathbb{R}$. The text leaves this case as an exercise. For $\lambda < 0$, (2.8) still holds. Dividing by $\lambda^2\varphi_X(\lambda) > 0$ again gives $G'(\lambda) \le \frac{1}{2}\sigma^2$ for all $\lambda < 0$. Fix $\lambda < \lambda_0 < 0$ and integrate:

$$G(\lambda_0) - G(\lambda) \le \frac{1}{2}\sigma^2(\lambda_0 - \lambda).$$

Let $\lambda_0 \to 0^-$. The same difference-quotient argument gives $G(\lambda_0) \to \mathbb{E}[X]$, so $\mathbb{E}[X] - G(\lambda) \le -\frac{1}{2}\sigma^2\lambda$, that is, $G(\lambda) \ge \mathbb{E}[X] + \frac{1}{2}\sigma^2\lambda$. Multiplying by $\lambda < 0$ reverses the inequality:

$$\log \varphi_X(\lambda) = \lambda G(\lambda) \le \lambda\mathbb{E}[X] + \frac{1}{2}\sigma^2\lambda^2,$$

which is (2.7) for $\lambda < 0$. □

Remark 2.1.6 (Why G?)**.** The function $\lambda G(\lambda) = \log \varphi_X(\lambda)$ is the cumulant generating function. The target (2.7) reads $\log \varphi_X(\lambda) \le \lambda\mathbb{E}[X] + \frac{1}{2}\sigma^2\lambda^2$, which is $G(\lambda) \le \mathbb{E}[X] + \frac{1}{2}\sigma^2\lambda$. So it is enough to show that G starts at $\mathbb{E}[X]$ and has slope at most $\sigma^2/2$. This reduces the problem to a one-line integration.

Remark 2.1.7 (Interpretation and consequences)**.** (a) If $I = \mathbb{R}$, then (2.7) says exactly that $X - \mathbb{E}[X]$ is sub-Gaussian with parameter σ.

(b) If $I = [0, \infty)$, the Chernoff method still gives the one-sided bound: for every $t \ge 0$,

$$\mathbb{P}\left[X \ge \mathbb{E}[X] + t\right] \le \inf_{\lambda \ge 0} e^{-\lambda t}\, \mathbb{E}\left[e^{\lambda(X-\mathbb{E}[X])}\right] \le \inf_{\lambda \ge 0} e^{-\lambda t + \frac{\lambda^2\sigma^2}{2}} = e^{-\frac{t^2}{2\sigma^2}},$$

where the infimum is attained at $\lambda^* = t/\sigma^2$.

In short, entropy control ⇒ MGF control ⇒ tail control. The entropy method is useful because entropy tensorizes over independent coordinates, which the MGF does not do as easily.

2.2 Wasserstein Distances

This section follows Section 3.3.1 of [1]. The material leads into optimal transport and transportation cost inequalities.

Definition 2.2.1 (Lipschitz functions)**.** Let $(\mathcal{X}, \rho)$ be a metric space. A function $f : \mathcal{X} \to \mathbb{R}$ is L-Lipschitz with respect to ρ if

$$|f(x) - f(x')| \le L \cdot \rho(x, x') \qquad \text{for all } x, x' \in \mathcal{X}.$$

We write $\|f\|_{\mathrm{Lip}}$ for the smallest L for which this holds:

$$\|f\|_{\mathrm{Lip}} = \sup_{x \neq x'} \frac{|f(x) - f(x')|}{\rho(x, x')}.$$

Definition 2.2.2 (Wasserstein distance induced by ρ)**.** Given two probability distributions Q, P on $\mathcal{X}$, define

$$W_\rho(Q, P) := \sup_{\|f\|_{\mathrm{Lip}} \le 1} \left(\int f \, dQ - \int f \, dP \right). \tag{2.11}$$

This is the Wasserstein distance induced by ρ (also called the W_1 or Kantorovich–Rubinstein distance).

Remark 2.2.3 (Invariance to constant offsets)**.** For any constant c, $\int (f + c)\, dQ - \int (f + c)\, dP = \int f \, dQ - \int f \, dP$, because both measures have total mass 1. So in (2.13) we can shift each f by any constant without changing the value.

2.3 Example: Hamming Metric and Total Variation

Consider the Hamming (discrete) metric

$$\rho(x, x') = \mathbf{1}_{\{x \neq x'\}}.$$

Throughout, all sets and functions are measurable.

Definition 2.3.1 (Three definitions of total variation)**.** For probability distributions Q, P on $\mathcal{X}$:

(a) (Sets) $\|Q - P\|_{\mathrm{TV}} = \sup_{A \subseteq \mathcal{X}} |Q(A) - P(A)|$.

(b) (Test functions) $\|Q - P\|_{\mathrm{TV}} = \sup_{f : \mathcal{X} \to [0,1]} \left(\int f \, dQ - \int f \, dP \right)$.

(c) (Densities) If P and Q have densities p and q with respect to a base (dominating) measure ν,

$$\|Q - P\|_{\mathrm{TV}} = \frac{1}{2} \int |p(x) - q(x)| \, \nu(dx).$$

Remark 2.3.2. A dominating measure always exists: take $\nu = \frac{1}{2}(P + Q)$, so that both $P, Q \ll \nu$. The value in Definition 2.3.1(c) does not depend on which dominating ν is chosen.

Proposition 2.3.3. *For the Hamming metric, W_ρ equals the total variation distance:*

$$W_{\mathrm{Ham}}(Q, P) = \|Q - P\|_{\mathrm{TV}} := \sup_{A \subseteq \mathcal{X}} |Q(A) - P(A)|,$$

where the supremum ranges over all measurable A.

Proof of Proposition 2.3.3. **Key observation.** If $\|f\|_{\mathrm{Lip}} \le 1$ with respect to the Hamming metric, then $|f(x) - f(x')| \le 1 \cdot \rho(x, x') \le 1$ for all x, x'. Conversely, any $f : \mathcal{X} \to [0,1]$ satisfies $|f(x) - f(x')| \le \mathbf{1}_{\{x \neq x'\}}$, so it is 1-Lipschitz.

Reduction to $[0,1]$-valued functions. By Remark 2.2.3, W_ρ does not change if f is shifted by a constant. If $\|f\|_{\mathrm{Lip}} \le 1$, the key observation gives $\sup f - \inf f \le 1$, so $\tilde{f} := f - \inf f$ takes values in $[0,1]$ and gives the same objective. Hence we may restrict the supremum to functions with $f(x) \in [0,1]$ for all $x \in \mathcal{X}$:

$$\begin{aligned} W_{\mathrm{Ham}}(Q,P) &= \sup_{\substack{\|f\|_{\mathrm{Lip}} \le 1 \\ f:\mathcal{X}\to[0,1]}} \left(\int f\, dQ - \int f\, dP \right) \\ &= \sup_{f:\mathcal{X}\to[0,1]} \left(\int f\, dQ - \int f\, dP \right) && \text{(key observation)} \\ &= \|Q - P\|_{\mathrm{TV}} && \text{([1, Exercise 3.13]).} \end{aligned}$$

The second line is exactly Definition 2.3.1(b), and the last equality is the equivalence of Definitions 2.3.1(a) and 2.3.1(b) (homework #2). □

Remark 2.3.4. Putting everything together, for the Hamming metric

$$W_{\mathrm{Ham}}(Q,P) = \sup_{f:\mathcal{X}\to[0,1]} \int f\,(dQ - dP) = \|Q - P\|_{\mathrm{TV}} = \frac{1}{2} \int |p - q|\, d\nu = \inf_M M\left[X \neq X'\right].$$

Here the equality with $\frac{1}{2}\int |p-q|\, d\nu$ is Definition 2.3.1(c), and the coupling form $\inf_M M[X \neq X']$ is also part of [1, Exercise 3.13]. In words, W_{Ham} is one half of the $L^1(\nu)$ distance between the densities. The last equality is a concrete instance of the Kantorovich–Rubinstein duality in the next section.

2.4 Optimal Transport and Couplings

Definition 2.4.1 (Coupling)**.** A distribution M on $\mathcal{X} \otimes \mathcal{X}$ is a coupling of the pair (Q,P) if its marginal distributions in the first and second coordinates are Q and P respectively:

$$M(A \times \mathcal{X}) = Q(A), \qquad M(\mathcal{X} \times B) = P(B) \qquad \text{for all } A, B.$$

Equivalently, $(X, X') \sim M$ with $X \sim Q$ and $X' \sim P$. Couplings always exist; for example, the product $M = Q \otimes P$.

Transport cost. The quantity

$$\inf_{\substack{M \text{ a coupling} \\ \text{of } (Q,P)}} \mathbb{E}_M\left[\rho(X,X')\right], \qquad (X,X') \sim M,$$

is also a distance between Q and P. It is the minimal expected cost of moving mass from Q to P when moving a unit of mass from x to x' costs $\rho(x,x')$.

2.4.1 Weak Duality

Proposition 2.4.2 (Weak duality: inf $\ge$ sup)**.** *Assume Q and P have finite first moments. Then*

$$\inf_{\substack{M \text{ coupling} \\ \text{of } (Q,P)}} \int \rho(x,x')\, dM(x,x') \;\ge\; \sup_{\|f\|_{\mathrm{Lip}} \le 1} \left(\int f\, dQ - \int f\, dP \right) = W_\rho(Q,P). \tag{2.12}$$

Proof. Every 1-Lipschitz f is integrable, since $|f(x)| \le |f(x_0)| + \rho(x, x_0)$. Fix any f with $\|f\|_{\mathrm{Lip}} \le 1$ and any coupling M of (Q, P). Then

$$\begin{aligned}
\int \rho(x,x')\,dM(x,x') &\ge \int (f(x) - f(x'))\,dM(x,x') && (\text{since } \|f\|_{\mathrm{Lip}} \le 1)\\
&= \int f(x)\,dM(x,x') - \int f(x')\,dM(x,x') && (\text{linearity})\\
&= \int f(x)\,dQ(x) - \int f(x')\,dP(x') && (\text{marginals of } M \text{ are } Q, P)\\
&= \int f\,dQ - \int f\,dP.
\end{aligned}$$

The left side does not depend on f and the right side does not depend on M. Taking the infimum over couplings on the left and the supremum over 1-Lipschitz f on the right gives (2.12). Kantorovich–Rubinstein duality says equality always holds □

We begin by completing the discussion of distances and divergences between probability measures. Let $(\mathcal{X}, \rho)$ be a metric space, and let $\mathbb{Q}$ and $\mathbb{P}$ be probability distributions on $\mathcal{X}$.

2.4.2 Optimal Transport Theory

Optimal transport theory studies how to transform one probability distribution $\mathbb{Q}$ into another probability distribution $\mathbb{P}$ while minimizing the cost of transporting probability mass.

A transportation plan, or coupling, of $\mathbb{Q}$ and $\mathbb{P}$ is a joint probability distribution M of a pair (X, X') such that

$$X \sim \mathbb{Q} \qquad \text{and} \qquad X' \sim \mathbb{P}.$$

The collection of all couplings of $\mathbb{Q}$ and $\mathbb{P}$ is denoted by $\Pi(\mathbb{Q}, \mathbb{P})$.

Let $\rho(x, x')$ denote the cost of moving one unit of mass from x to x'. For a coupling $M \in \Pi(\mathbb{Q}, \mathbb{P})$, its expected transportation cost is

$$\mathbb{E}_M\left[\rho(X, X')\right].$$

Definition 2.4.3 (Wasserstein distance)**.** For $s \ge 1$, the order-s Wasserstein distance is

$$W_{\rho,s}(\mathbb{Q}, \mathbb{P}) := \inf_{M \in \Pi(\mathbb{Q},\mathbb{P})} \left(\mathbb{E}_M\left[\rho(X, X')^s\right]\right)^{1/s}. \tag{2.13}$$

When $s = 1$, we write $W_\rho(\mathbb{Q}, \mathbb{P}) := W_{\rho,1}(\mathbb{Q}, \mathbb{P})$. The case $s = 2$ is also frequently used in practice.

Example 2.4.4 (Transporting bread to coffee stores)**.** Suppose that bread is produced at two store locations, represented by -1 and 1. The probability measure

$$\mathbb{Q} = \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1$$

describes the distribution of the bread supply. Thus, one half of the bread is produced at location -1, and the other half is produced at location 1.

The bread must be delivered to coffee stores at the same two locations. The probability measure

$$\mathbb{P} = \frac{1}{3}\delta_{-1} + \frac{2}{3}\delta_1$$

describes the distribution of demand. The coffee store at location -1 requires one third of the total bread, whereas the coffee store at location 1 requires two thirds.

Let $m(x, x')$ denote the amount of bread transported from production location x under $\mathbb{Q}$ to coffee-store location x' under $\mathbb{P}$. The four possible transportation quantities are

$$m(-1,-1), \qquad m(-1,1), \qquad m(1,-1), \qquad m(1,1).$$

They have the following interpretations:

- $m(-1,-1)$ is the amount transported from the bread store at -1 to the coffee store at -1;
- $m(-1,1)$ is the amount transported from the bread store at -1 to the coffee store at 1;
- $m(1,-1)$ is the amount transported from the bread store at 1 to the coffee store at -1;
- $m(1,1)$ is the amount transported from the bread store at 1 to the coffee store at 1.

The supply and demand constraints are

$$\begin{aligned} m(-1,-1) + m(-1,1) &= \frac{1}{2}, \quad m(1,-1) + m(1,1) = \frac{1}{2}, \\ m(-1,-1) + m(1,-1) &= \frac{1}{3}, \quad m(-1,1) + m(1,1) = \frac{2}{3}. \end{aligned}$$

An optimal transportation plan is

$$\begin{aligned} m(-1,-1) &= \frac{1}{3}, \quad m(-1,1) = \frac{1}{6}, \\ m(1,-1) &= 0, \qquad m(1,1) = \frac{1}{2}. \end{aligned}$$

Equivalently, the plan is represented by the table

	Coffee store -1	Coffee store 1	Bread supply
Bread store -1	$m(-1,-1) = \frac{1}{3}$	$m(-1,1) = \frac{1}{6}$	$\frac{1}{2}$
Bread store 1	$m(1,-1) = 0$	$m(1,1) = \frac{1}{2}$	$\frac{1}{2}$
Coffee-store demand	$\frac{1}{3}$	$\frac{2}{3}$	1

Suppose that $\rho(x, x') = |x - x'|$. Transporting bread between stores at the same location has zero cost, whereas transporting bread from one location to the other has cost 2. Therefore, the total cost is

$$\begin{aligned} &\rho(-1,-1)m(-1,-1) + \rho(-1,1)m(-1,1) \\ &\qquad + \rho(1,-1)m(1,-1) + \rho(1,1)m(1,1) \\ &= 0\left(\frac{1}{3}\right) + 2\left(\frac{1}{6}\right) + 2(0) + 0\left(\frac{1}{2}\right) = \frac{1}{3}. \end{aligned}$$

The marginal constraints force at least $1/6$ of the bread produced at -1 to be transported to 1. Hence no coupling can have a smaller cost, and

$$W_\rho(\mathbb{Q}, \mathbb{P}) = \frac{1}{3}.$$

Suppose Q and P have densities q and p on $\mathcal{X}$, and a coupling M has density $m(x, x')$. Think of q as an initial pile of mass and p as the target configuration. Then:

- $\rho(x, x')$ is the cost per unit of moving mass from x to x';
- $m(x, x')\,dx\,dx'$ is the amount of mass sent from near x to near x' (a transport plan);
- the marginal constraints say the plan removes exactly q and delivers exactly p;

- the total cost is $\iint \rho(x,x')\, m(x,x')\, dx\, dx'$.

Minimizing over admissible plans gives the Wasserstein distance. This is why it is called the "earth mover's" or transportation distance.

For any 1-Lipschitz function f and any coupling $M \in \Pi(\mathbb{Q},\mathbb{P})$,

$$\begin{aligned}\int_{\mathcal{X}} f\, d\mathbb{Q} - \int_{\mathcal{X}} f\, d\mathbb{P} &= \mathbb{E}_M\left[f(X) - f(X')\right] \\ &\le \mathbb{E}_M\left[\rho(X,X')\right].\end{aligned}$$

Taking the supremum over all 1-Lipschitz functions and then the infimum over all couplings gives the weak-duality inequality.

Theorem 2.4.5 (Kantorovich–Rubinstein duality)**.** *Under the usual regularity and finite-first-moment assumptions,*

$$\begin{aligned}W_\rho(\mathbb{Q},\mathbb{P}) &= \sup_{\|f\|_{\mathrm{Lip}}\le 1}\left[\int_{\mathcal{X}} f\, d\mathbb{Q} - \int_{\mathcal{X}} f\, d\mathbb{P}\right] \\ &= \inf_{M\in\Pi(\mathbb{Q},\mathbb{P})}\int_{\mathcal{X}\times\mathcal{X}} \rho(x,x')\, dM(x,x') \\ &= \inf_{M\in\Pi(\mathbb{Q},\mathbb{P})}\mathbb{E}_M\left[\rho(X,X')\right].\end{aligned} \tag{2.14}$$

If $\mathbb{Q}$ and $\mathbb{P}$ have densities q and p, and the coupling M has density m, then the transportation cost associated with m is

$$\int_{\mathcal{X}\times\mathcal{X}} \rho(x,x')m(x,x')\, dx\, dx'.$$

Thus, the primal formulation (coupling formulation) minimizes transportation cost, whereas the dual formulation (Lipschitz function formulation) compares expectations over 1-Lipschitz functions.

2.4.3 Kullback–Leibler Divergence

Definition 2.4.6 (Kullback–Leibler divergence)**.** The Kullback–Leibler (KL) divergence of $\mathbb{Q}$ from $\mathbb{P}$ is

$$D(\mathbb{Q}\|\mathbb{P}) := \begin{cases}\mathbb{E}_{\mathbb{Q}}\left[\log\dfrac{d\mathbb{Q}}{d\mathbb{P}}\right], & \mathbb{Q}\ll\mathbb{P}, \\ +\infty, & \text{otherwise.}\end{cases} \tag{2.15}$$

If $\mathbb{Q}$ and $\mathbb{P}$ have densities q and p with respect to a common dominating measure ν, then

$$D(\mathbb{Q}\|\mathbb{P}) = \int_{\mathcal{X}} q(x)\log\frac{q(x)}{p(x)}\,\nu(dx). \tag{2.16}$$

Unlike a metric, KL divergence is generally asymmetric and does not satisfy the triangle inequality. It nevertheless controls total variation distance.

Theorem 2.4.7 (Pinsker–Csiszár–Kullback inequality)**.** *With the convention*

$$\|\mathbb{P}-\mathbb{Q}\|_{\mathrm{TV}} := \sup_{A\subseteq\mathcal{X}}\left|\mathbb{P}(A)-\mathbb{Q}(A)\right|,$$

we have

$$\|\mathbb{P}-\mathbb{Q}\|_{\mathrm{TV}} \le \sqrt{\frac{1}{2}D(\mathbb{Q}\|\mathbb{P})}. \tag{2.17}$$

Thus, small KL divergence implies that the two measures assign similar probabilities to every measurable event.

Chapter 3

Supremum of empirical processes

3.1 Motivation

3.1.1 Uniform Convergence of CDFs

Let X be a real-valued random variable with cumulative distribution function

$$F(t) := \mathbb{P}(X \leq t) = \mathbb{E}\left[\mathbf{1}_{\{X \leq t\}}\right], \qquad t \in \mathbb{R}.$$

Suppose that

$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Law}(X).$$

Definition 3.1.1 (Empirical CDF)**.** The empirical cumulative distribution function is

$$\widehat{F}_n(t) := \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}_{\{X_i \leq t\}}, \qquad t \in \mathbb{R}. \tag{3.1}$$

For every fixed t, the strong law of large numbers gives

$$\widehat{F}_n(t) \longrightarrow F(t) \qquad \text{almost surely.}$$

This is pointwise convergence: the exceptional event may initially depend on t.

Many estimation problems can be formulated through a functional γ that maps a CDF F to a real number $\gamma(F)$. The plug-in estimator is $\gamma(\widehat{F}_n)$, and the basic consistency question is whether

$$\gamma(\widehat{F}_n) \overset{?}{\longrightarrow} \gamma(F).$$

Example 3.1.2 (Expectation functionals)**.** Let g be an integrable function and define

$$\gamma_g(F) := \int_{\mathbb{R}} g(x)\, dF(x).$$

Because $\widehat{F}_n$ assigns mass $1/n$ to each observation,

$$\gamma_g(\widehat{F}_n) = \int_{\mathbb{R}} g(x)\, d\widehat{F}_n(x) = \frac{1}{n} \sum_{i=1}^{n} g(X_i).$$

Therefore, if $\mathbb{E}\,|g(X)| < \infty$, the strong law gives

$$\gamma_g(\widehat{F}_n) \longrightarrow \gamma_g(F) \qquad \text{almost surely.}$$

Example 3.1.3 (Quantile functionals)**.** For $\alpha \in (0,1)$, the lower α-quantile functional is

$$Q_\alpha(F) := \inf\{t \in \mathbb{R} : F(t) \geq \alpha\}.$$

The corresponding empirical quantile is

$$\begin{aligned} Q_\alpha(\widehat{F}_n) &= \inf\left\{t \in \mathbb{R} : \widehat{F}_n(t) \geq \alpha\right\} \\ &= \inf\left\{t \in \mathbb{R} : \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{\{X_i \leq t\}} \geq \alpha\right\}. \end{aligned}$$

It is not immediately clear from pointwise convergence alone whether $Q_\alpha(\widehat{F}_n) \to Q_\alpha(F)$. Uniform convergence and an appropriate condition ensuring that F crosses the level α are needed.

Definition 3.1.4 (Supremum norm and continuity)**.** For two CDFs F and G, their supremum distance is

$$\|F - G\|_\infty := \sup_{t \in \mathbb{R}} |F(t) - G(t)|. \tag{3.2}$$

A functional γ is continuous at F in the supremum norm if, for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$\|G - F\|_\infty \leq \delta \quad \Longrightarrow \quad |\gamma(G) - \gamma(F)| \leq \epsilon.$$

Recall that if f is a continuous function on $\mathbb{R}^d$, then

$$x_n \to x \quad \Longrightarrow \quad f(x_n) \to f(x).$$

Similarly, if $F_n \to F$ in the supremum norm and γ is continuous at F, then

$$\gamma(F_n) \to \gamma(F).$$

Therefore, to establish the consistency of $\gamma(\widehat{F}_n)$, it is sufficient to verify that

1. $\left\|\widehat{F}_n - F\right\|_\infty \to 0$;
2. γ is continuous at F with respect to the supremum norm.

Theorem 3.1.5 (Glivenko–Cantelli)**.** *For every distribution function F, the empirical CDF $\widehat{F}_n$ is a strongly consistent estimator of the population CDF in the supremum norm:*

$$\left\|\widehat{F}_n - F\right\|_\infty \longrightarrow 0 \qquad \textit{almost surely.} \tag{3.3}$$

The proof is deferred.

3.1.2 Uniform Laws for General Function Classes

The empirical-CDF result is a special case of a general problem involving empirical averages over a collection of functions. Let

$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathbb{P}$$

on a measurable space $\mathcal{X}$.

Definition 3.1.6 (Empirical measure)**.** The empirical probability measure is

$$\mathbb{P}_n := \frac{1}{n}\sum_{i=1}^{n} \delta_{X_i}. \tag{3.4}$$

For $a \in \mathcal{X}$, the Dirac probability measure δ_a is defined by

$$\delta_a(B) := \begin{cases} 1, & a \in B, \\ 0, & a \notin B, \end{cases} \qquad B \subseteq \mathcal{X}. \tag{3.5}$$

Example 3.1.7 (Bernoulli distribution)**.** The Bernoulli distribution with parameter p can be written as

$$\mathrm{Ber}(p) = p\delta_1 + (1-p)\delta_0.$$

Its probability mass function is

$$\mathrm{pmf}(x) = \begin{cases} p, & x = 1, \\ 1-p, & x = 0. \end{cases}$$

Thus, if $\mathbb{P} = p\delta_1 + (1-p)\delta_0$, then

$$\mathbb{P}(\{1\}) = p, \qquad \mathbb{P}(\{0\}) = 1-p.$$

Definition 3.1.8 (Population and empirical averages)**.** For a measurable function f, define

$$\mathbb{P}f := \int_{\mathcal{X}} f(x)\, d\mathbb{P}(x) = \mathbb{E}_{\mathbb{P}}\left[f(X)\right]. \tag{3.6}$$

The corresponding empirical average is

$$\mathbb{P}_n f := \int_{\mathcal{X}} f(x)\, d\mathbb{P}_n(x) = \frac{1}{n}\sum_{i=1}^{n} f(X_i). \tag{3.7}$$

For each fixed integrable function f, the strong law of large numbers implies

$$\mathbb{P}_n f \longrightarrow \mathbb{P}f \qquad \text{almost surely.}$$

Definition 3.1.9. Let $\mathcal{F}$ be a class of measurable functions. Define

$$\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} := \sup_{f\in\mathcal{F}} \left|\mathbb{P}_n f - \mathbb{P}f\right|. \tag{3.8}$$

Summary of Notation for Empirical Measures and Empirical Processes

Notation	**Definition**	**Interpretation**
$\mathbb{P}$	Probability measure	
$\mathbb{P}f$	$\mathbb{P}f = \int f(x)\,\mathbb{P}(dx)$	$\mathbb{E}_{\mathbb{P}}[f(X)]$
$\{\mathbb{P}f\}_{f\in\mathcal{F}}$	Functions indexed by $\mathcal{F}$	
$\mathbb{P}_n$	$\mathbb{P}_n = \frac{1}{n}\sum_{i=1}^{n}\delta_{X_i}$ Discrete random measure	$\mathbb{P}_n(A) = \frac{1}{n}\sum_{i=1}^{n}\mathbf{1}_A(X_i)$ $\mathbb{P}_n(X_i) = \frac{1}{n},\ \forall i$
$\mathbb{P}_n f$	$\mathbb{P}_n f = \frac{1}{n}\sum_{i=1}^{n} f(X_i)$	Empirical mean of $f(X)$
$\{\mathbb{P}_n f\}_{f\in\mathcal{F}}$	$\left\{\frac{1}{n}\sum_{i=1}^{n} f(X_i) : f\in\mathcal{F}\right\}$	Random process
$\mathbb{P}_n f - \mathbb{P}f$	$\frac{1}{n}\sum_{i=1}^{n} f(X_i) - \mathbb{E}_{\mathbb{P}}[f(X)]$	Random variable
$\{\mathbb{P}_n f - \mathbb{P}f\}_{f\in\mathcal{F}}$	$\left\{\frac{1}{n}\sum_{i=1}^{n}\left[f(X_i) - \mathbb{E}_{\mathbb{P}}f(X)\right]\right\}_{f\in\mathcal{F}}$	Centered empirical process
$\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}}$	$\sup_{f\in\mathcal{F}}\|\mathbb{P}_n f - \mathbb{P}f\|$	Supremum of the empirical process

3.1.3 Glivenko-Cantelli Class

Definition 3.1.10 (Glivenko–Cantelli Class)**.** We say that $\mathcal{F}$ is a **Glivenko–Cantelli class** for $\mathbb{P}$ if

$$\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} \xrightarrow{P} 0, \qquad n \to \infty.$$

If

$$\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}} \xrightarrow{\text{a.s.}} 0,$$

then $\mathcal{F}$ is called a **strong Glivenko–Cantelli class** for $\mathbb{P}$.

Example 3.1.11 (Empirical CDFs and indicator functions)**.** Define the threshold class

$$\mathcal{F}_{\text{cdf}} := \{f_t(\cdot) : t \in \mathbb{R}\}, \qquad f_t(x) := \mathbf{1}_{\{x \le t\}}.$$

For every $t \in \mathbb{R}$,

$$\mathbb{P}f_t = \int_{\mathcal{X}} \mathbf{1}_{\{x \le t\}}\, d\mathbb{P}(x) = \mathbb{P}(X \le t) = F(t),$$

and

$$\mathbb{P}_n f_t = \frac{1}{n}\sum_{i=1}^{n} f_t(X_i) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{\{X_i \le t\}} = \widehat{F}_n(t).$$

Therefore,

$$\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}_{\text{cdf}}} = \sup_{t \in \mathbb{R}} \left|\widehat{F}_n(t) - F(t)\right| = \left\|\widehat{F}_n - F\right\|_{\infty}.$$

By the classical Glivenko–Cantelli theorem,

$$\|\widehat{F}_n - F\|_{\infty} \xrightarrow{\text{a.s.}} 0$$

for every distribution function F.

Hence,

$$\boxed{\mathcal{F} \text{ is a strong Glivenko–Cantelli class for } \mathbb{P}.}$$

The Glivenko–Cantelli theorem says precisely that the threshold class $\mathcal{F}_{\text{cdf}}$ is a universal Glivenko–Cantelli class.

This reformulation is the bridge from empirical CDFs to empirical-process theory. For a single function, the law of large numbers is enough. For an entire class, the size and structure of the class determine whether the same sample controls all empirical averages simultaneously.

Example 3.1.12 (A Class That Is Not Glivenko–Cantelli)**.** Let $\mathcal{S}$ be the class of all finite subsets A of $[0,1]$.

Define

$$\mathcal{F}_{\mathcal{S}} = \{\mathbf{1}_A(\cdot) : A \in \mathcal{S}\}.$$

Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathbb{P},$$

where $\mathbb{P}$ is a probability measure on $[0,1]$ that has no atoms.

Since $\mathbb{P}$ has no atoms,

$$\mathbb{P}(\{y\}) = 0, \qquad \forall\, y \in [0,1].$$

Let

$$A_0 = \{X_1, \ldots, X_n\} \in \mathcal{S}$$

and

$$f_0 = \mathbf{1}_{A_0}(\cdot) \in \mathcal{F}_{\mathcal{S}}.$$

Then

$$\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}_\mathcal{S}} \geq |\mathbb{P}_n f_0 - \mathbb{P} f_0| \,.$$

Since $\mathbb{P}$ has no atoms,

$$\begin{aligned} \mathbb{P} f_0 &= \mathbb{P} \mathbf{1}_{A_0}(\cdot) \\ &= \mathbb{P}(A_0) \\ &= 0. \end{aligned}$$

On the other hand,

$$\begin{aligned} \mathbb{P}_n f_0 &= \mathbb{P}_n \mathbf{1}_{A_0}(\cdot) \\ &= \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}_{A_0}(X_i) \\ &= 1. \end{aligned}$$

Therefore,

$$\boxed{\|\mathbb{P}_n - \mathbb{P}\|_{\mathcal{F}_\mathcal{S}} \geq 1 > 0.}$$

Hence, $\mathcal{F}_\mathcal{S}$ is not a Glivenko–Cantelli class for any atomless probability measure $\mathbb{P}$.

3.2 Empirical Risk Minimization

Let

$$\{\mathbb{P}_\theta : \theta \in \Omega\}$$

be a parameterized family of distributions. The parameter space Ω may be finite-dimensional or infinite-dimensional.

Suppose the observations have distribution P_{θ^*}, where $\theta^* \in \Omega$ is unknown. Suppose

$$X_i \overset{\text{iid}}{\sim} \mathbb{P}_{\theta^*}.$$

A loss function $L_\theta(X)$ measures the fit between a candidate parameter $\theta \in \Omega$ and an observation $X \in \mathcal{X}$. The population risk and empirical risk are, respectively,

$$R(\theta, \theta^*) := \mathbb{E}_{\theta^*}[L_\theta(X)], \tag{3.9}$$

$$\widehat{R}_n(\theta, \theta^*) := \frac{1}{n} \sum_{i=1}^{n} L_\theta(X_i). \tag{3.10}$$

Given a nonempty candidate set $\Omega_0 \subseteq \Omega$, an empirical risk minimizer is any

$$\widehat{\theta}_n \in \operatorname*{arg\,min}_{\theta \in \Omega_0} \widehat{R}_n(\theta, \theta^*). \tag{3.11}$$

Its excess risk is

$$\mathbf{E}(\widehat{\theta}_n, \theta^*) := R(\widehat{\theta}_n, \theta^*) - \inf_{\theta \in \Omega_0} R(\theta, \theta^*). \tag{3.12}$$

Example 3.2.1 (Maximum Likelihood Estimation)**.** Let

$$\{P_\theta : \theta \in \Omega\}$$

be a family of probability distributions with strictly positive densities p_θ with respect to a common underlying measure.

Define

$$L_\theta(x) = \log\left(\frac{p_{\theta^*}(x)}{p_\theta(x)}\right).$$

Then

$$\begin{aligned} R(\theta, \theta^*) &= \mathbb{E}_{\theta^*}[L_\theta(X)] \\ &= \int p_{\theta^*}(x) \log\left(\frac{p_{\theta^*}(x)}{p_\theta(x)}\right) dx \\ &= D(P_{\theta^*}, P_\theta). \end{aligned}$$

Thus, under identifiability,

$$\theta^* = \arg\min_{\theta \in \Omega_0} R(\theta, \theta^*).$$

The empirical risk is

$$\begin{aligned} \widehat{R}_n(\theta, \theta^*) &= \frac{1}{n}\sum_{i=1}^n L_\theta(X_i) \\ &= \frac{1}{n}\sum_{i=1}^n \log\left(\frac{p_{\theta^*}(X_i)}{p_\theta(X_i)}\right). \end{aligned}$$

Therefore,

$$\begin{aligned} \widehat{\theta}_n &\in \arg\min_{\theta \in \Omega_0} \widehat{R}_n(\theta, \theta^*) \\ &= \arg\min_{\theta \in \Omega_0} \left\{ \frac{1}{n}\sum_{i=1}^n \log p_{\theta^*}(X_i) - \frac{1}{n}\sum_{i=1}^n \log p_\theta(X_i) \right\}. \end{aligned}$$

Since

$$\frac{1}{n}\sum_{i=1}^n \log p_{\theta^*}(X_i)$$

does not depend on θ, we obtain

$$\boxed{\widehat{\theta}_n \in \arg\max_{\theta \in \Omega_0} \frac{1}{n}\sum_{i=1}^n \log p_\theta(X_i).}$$

This is the **maximum likelihood estimator**. If $\theta^* \in \Omega_0$,

$$\begin{aligned} E(\widehat{\theta}_n, \theta^*) &= R(\widehat{\theta}_n, \theta^*) - \inf_{\theta \in \Omega_0} R(\theta, \theta^*) \\ &= D(P_{\theta^*}, P_{\widehat{\theta}_n}) - 0 \\ &= D(P_{\theta^*}, P_{\widehat{\theta}_n}). \end{aligned}$$

Example 3.2.2 (Binary Classification)**.** Suppose

$$(X_i, Y_i) \in \mathbb{R}^d \times \{-1, +1\}, \qquad i = 1, \ldots, n,$$

and

$$(X_i, Y_i) \overset{\text{iid}}{\sim} \mathbb{P}_{X,Y}.$$

In particular,

$$X_i \overset{\text{iid}}{\sim} \mathbb{P}_X, \qquad Y_i \mid X_i \sim \mathbb{P}_{Y|X}.$$

Define

$$\psi(x) = \frac{\mathbb{P}(Y = +1 \mid X = x)}{\mathbb{P}(Y = -1 \mid X = x)}.$$

We want to find a classifier

$$f : \mathbb{R}^d \longrightarrow \{-1, +1\}$$

that minimizes

$$\mathbb{P}_{X,Y}\big(f(X) \neq Y\big).$$

Define the loss function

$$L_f(x, y) := \mathbf{1}\{f(x) \neq y\}.$$

Then the risk of a classifier f is

$$\begin{aligned} R(f, f^*) &= \mathbb{E}_{X,Y}\mathbf{1}\{f(X) \neq Y\} \\ &= \mathbb{P}_{X,Y}\big(f(X) \neq Y\big). \end{aligned}$$

The **Bayes classifier** is

$$\boxed{f^*(x) = \begin{cases} +1, & \text{if } \psi(x) \geq 1, \\ -1, & \text{if } \psi(x) < 1. \end{cases}}$$

The empirical risk is

$$\widehat{R}_n(f, f^*) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{f(X_i) \neq Y_i\}.$$

Also,

$$1 - \mathbb{P}_{X,Y}\big(f(X) \neq Y\big) = \mathbb{P}_{X,Y}\big(f(X) = Y\big).$$

By conditioning on X,

$$\begin{aligned} &\mathbb{P}_{X,Y}\big(f(X) = Y\big) \\ &\quad = \mathbb{E}_X\left[\mathbb{P}_{Y|X}\big(f(X) = Y \mid X\big)\right] \\ &\quad = \mathbb{E}_X\Big[\mathbb{P}(Y = +1 \mid X)\mathbf{1}\{f(X) = +1\} \\ &\qquad\qquad + \mathbb{P}(Y = -1 \mid X)\mathbf{1}\{f(X) = -1\}\Big]. \end{aligned}$$

Thus the Bayes classifier f^* maximizes $\mathbb{P}_{X,Y}\big(f(X) = Y\big)$.

Remark 3.2.3 (Correction to the binary classification example)**.** In Example 4.9 of [1], the assumption $P(Y = +1) = P(Y = -1) = 1/2$ is unnecessary for the stated Bayes rule and should be removed.

For simplicity, assume the infimum in (**??**) is attained at a fixed parameter $\theta_0 \in \Omega_0$, so that

$$R(\theta_0, \theta^*) = \inf_{\theta \in \Omega_0} R(\theta, \theta^*).$$

Add and subtract the two empirical risks to obtain

$$\begin{aligned} \mathbf{E}(\widehat{\theta}_n, \theta^*) &= R(\widehat{\theta}_n, \theta^*) - R(\theta_0, \theta^*) \\ &= \underbrace{R(\widehat{\theta}_n, \theta^*) - \widehat{R}_n(\widehat{\theta}_n, \theta^*)}_{T_1} \\ &\quad + \underbrace{\widehat{R}_n(\widehat{\theta}_n, \theta^*) - \widehat{R}_n(\theta_0, \theta^*)}_{T_2 \leq 0} \\ &\quad + \underbrace{\widehat{R}_n(\theta_0, \theta^*) - R(\theta_0, \theta^*)}_{T_3}. \end{aligned}$$

The term T_2 is nonpositive by the minimizing property of $\widehat{\theta}_n$. The first term is

$$T_1 = \mathbb{E}_{\theta^*}[L_{\widehat{\theta}_n}(X)] - \frac{1}{n}\sum_{i=1}^{n} L_{\widehat{\theta}_n}(X_i).$$

To bound this term uniformly over the candidate parameters, define the loss class

$$\mathcal{L}(\Omega_0) := \{x \mapsto L_\theta(x) \mid \theta \in \Omega_0\}.$$

For any distribution P and a P-integrable function f, write

$$Pf := \mathbb{E}[f(X)], \qquad P_n f := \frac{1}{n}\sum_{i=1}^{n} f(X_i),$$

where $X_1, \ldots, X_n$ are i.i.d. from P. For a nonempty function class $\mathcal{F}$, the uniform deviation is

$$\|P_n - P\|_{\mathcal{F}} := \sup_{f\in\mathcal{F}} |P_n f - Pf| = \sup_{f\in\mathcal{F}} \left| \frac{1}{n}\sum_{i=1}^{n} f(X_i) - \mathbb{E}[f(X)] \right|. \tag{3.13}$$

The displayed suprema and expectations are assumed to be well defined. With $P = P_{\theta^*}$, the two remaining terms satisfy

$$\begin{aligned} T_1 &= P_{\theta^*} L_{\widehat{\theta}_n} - P_n L_{\widehat{\theta}_n} \\ &\le \sup_{\theta\in\Omega_0} |P_{\theta^*} L_\theta - P_n L_\theta| = \|P_n - P_{\theta^*}\|_{\mathcal{L}(\Omega_0)}, \\ T_3 &= P_n L_{\theta_0} - P_{\theta^*} L_{\theta_0} \le \|P_n - P_{\theta^*}\|_{\mathcal{L}(\Omega_0)}. \end{aligned}$$

Combining these inequalities gives the following result.

Proposition 3.2.4 (Excess risk bound)**.** *An empirical risk minimizer satisfies*

$$0 \le \mathbf{E}(\widehat{\theta}_n, \theta^*) \le 2\,\|P_n - P_{\theta^*}\|_{\mathcal{L}(\Omega_0)}. \tag{3.14}$$

This motivates the study of $\|P_n - P\|_{\mathcal{F}}$. Recall:

Definition 3.2.5 (Glivenko–Cantelli class)**.** A function class $\mathcal{F}$ is Glivenko–Cantelli for P if

$$\|P_n - P\|_{\mathcal{F}} \longrightarrow 0 \qquad \text{in probability as } n \to \infty.$$

It is strong Glivenko–Cantelli for P if this convergence holds almost surely.

3.3 A Uniform Law via Rademacher Complexity

In the remainder of the lecture, $X_1, \ldots, X_n$ are i.i.d. from P, and $\mathcal{F}$ is a nonempty class of P-integrable functions.

Let $\varepsilon_1, \ldots, \varepsilon_n$ be independent Rademacher variables, with

$$P(\varepsilon_i = +1) = P(\varepsilon_i = -1) = \frac{1}{2}.$$

They are independent of all observations. Recall that for a nonempty set $A \subseteq \mathbb{R}^n$, its Rademacher complexity is

$$\mathcal{R}(A) := \mathbb{E}_\varepsilon\left[\sup_{a\in A} |\langle a, \varepsilon\rangle|\right] = \mathbb{E}_\varepsilon\left[\sup_{a\in A} \left|\sum_{i=1}^{n} a_i\varepsilon_i\right|\right]. \tag{3.15}$$

For a fixed collection $x_1^n = (x_1, \ldots, x_n)$, define

$$\mathcal{F}(x_1^n) := \{(f(x_1), \ldots, f(x_n)) \mid f \in \mathcal{F}\} \subseteq \mathbb{R}^n.$$

These are all the vectors that can be obtained by evaluating functions in $\mathcal{F}$ on the same collection of points.

Definition 3.3.1 (Empirical and population Rademacher complexity)**.** The empirical Rademacher complexity on x_1^n is

$$\mathcal{R}\left(\mathcal{F}(x_1^n)/n\right) := \mathbb{E}_\varepsilon\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^n \varepsilon_i f(x_i)\right|\right]. \tag{3.16}$$

For the random sample $X_1^n = (X_1, \ldots, X_n)$, the Rademacher complexity of the function class is

$$\begin{aligned}\mathcal{R}_n(\mathcal{F}) &:= \mathbb{E}_X\left[\mathcal{R}\left(\mathcal{F}(X_1^n)/n\right)\right] \\ &= \mathbb{E}_{X,\varepsilon}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^n \varepsilon_i f(X_i)\right|\right].\end{aligned} \tag{3.17}$$

We first recall the symmetrization inequality for random vectors.

Theorem 3.3.2 (Symmetrization for mean-zero random vectors)**.** *Let $X_1, \ldots, X_n$ be independent, mean-zero random vectors in a normed space. Let $\varepsilon_1, \ldots, \varepsilon_n$ be independent Rademacher variables, independent of the vectors. Then*

$$\frac{1}{2}\mathbb{E}_{X,\varepsilon}\left\|\sum_{i=1}^n \varepsilon_i X_i\right\| \le \mathbb{E}_X\left\|\sum_{i=1}^n X_i\right\| \le 2\mathbb{E}_{X,\varepsilon}\left\|\sum_{i=1}^n \varepsilon_i X_i\right\|. \tag{3.18}$$

Homework 1: Let $X_1, \ldots, X_n$ be independent random vectors in a normed space, and let $\varepsilon_1, \ldots, \varepsilon_n$ be independent Rademacher variables independent of these vectors. Then

$$\mathbb{E}_X\left\|\sum_{i=1}^n (X_i - \mathbb{E}X_i)\right\| \le 2\mathbb{E}_{X,\varepsilon}\left\|\sum_{i=1}^n \varepsilon_i X_i\right\|.$$

Combining the above two immediately yields:

Theorem 3.3.3 (Symmetrization for general independent random vectors)**.** *Let $X_1, \ldots, X_n$ be independent, random vectors in a normed space. Let $\varepsilon_1, \ldots, \varepsilon_n$ be independent Rademacher variables, independent of these vectors. Then*

$$\begin{aligned}\frac{1}{2}\mathbb{E}_{X,\varepsilon}\left\|\sum_{i=1}^n \varepsilon_i (X_i - \mathbb{E}X_i)\right\| &\le \mathbb{E}_X\left\|\sum_{i=1}^n (X_i - \mathbb{E}X_i)\right\| \\ &\le 2\mathbb{E}_{X,\varepsilon}\left\|\sum_{i=1}^n \varepsilon_i X_i\right\|.\end{aligned} \tag{3.19}$$

Lemma 3.3.4 (Bounds on the mean uniform deviation)**.** *For a class $\mathcal{F}$ of P-integrable functions,*

$$\textit{(a)}\quad \mathbb{E}\left[\|P_n - P\|_{\mathcal{F}}\right] \le 2\mathcal{R}_n(\mathcal{F}). \tag{3.20}$$

$$\textit{(b)}\quad \mathbb{E}\left[\|P_n - P\|_{\mathcal{F}}\right] \ge \frac{1}{2}\mathcal{R}_n(\overline{\mathcal{F}}), \tag{3.21}$$

where

$$\overline{\mathcal{F}} := \{f - Pf \mid f \in \mathcal{F}\}$$

is the recentered function class.

Proof. **(a)**

$$\begin{aligned}\|P_n - P\|_{\mathcal{F}} &= \sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^n f(X_i) - \mathbb{E}[f(X)]\right| \\ &= \sup_{f\in\mathcal{F}}\left|\sum_{i=1}^n \frac{1}{n}\left(f(X_i) - \mathbb{E}[f(X_i)]\right)\right|.\end{aligned}$$

Apply (3.19) to the vectors $(f(X_i)/n)_{f\in\mathcal{F}}$ with the supremum norm over $\mathcal{F}$. The upper bound gives

$$\begin{aligned}\mathbb{E}\left[\|P_n - P\|_{\mathcal{F}}\right] &\le 2\mathbb{E}_{X,\varepsilon}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i f(X_i)\right|\right]\\ &= 2\mathcal{R}_n(\mathcal{F}).\end{aligned}$$

(b) The lower bound in (3.19) gives

$$\begin{aligned}\mathbb{E}\left[\|P_n - P\|_{\mathcal{F}}\right] &\ge \frac{1}{2}\mathbb{E}_{X,\varepsilon}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i\left(f(X_i) - \mathbb{E}[f(X_i)]\right)\right|\right]\\ &= \frac{1}{2}\mathcal{R}_n(\overline{\mathcal{F}}).\end{aligned}$$

□

Remark 3.3.5 (Convergence in mean and in probability)**.** If $\mathcal{R}_n(\mathcal{F}) = o(1)$, then (3.20) implies

$$\mathbb{E}\left[\|P_n - P\|_{\mathcal{F}}\right] \longrightarrow 0.$$

Consequently,

$$\|P_n - P\|_{\mathcal{F}} \longrightarrow 0 \qquad \text{in } L^1 \text{ and in probability.}$$

Thus $\mathcal{F}$ is Glivenko–Cantelli for P.

Definition 3.3.6 (Uniformly bounded function class)**.** A class $\mathcal{F}$ is b-uniformly bounded if

$$\|f\|_\infty := \sup_{x\in\mathcal{X}}|f(x)| \le b \qquad \text{for every } f\in\mathcal{F}.$$

Theorem 3.3.7 (Uniform law via Rademacher complexity)**.** *For a b-uniformly bounded class $\mathcal{F}$ with $b>0$, every $n\ge 1$ and $\delta\ge 0$ satisfy*

$$\|P_n - P\|_{\mathcal{F}} \le 2\mathcal{R}_n(\mathcal{F}) + \delta \tag{3.22}$$

with probability at least

$$1 - \exp\left(-\frac{n\delta^2}{2b^2}\right).$$

Recall the following consequence of the first Borel–Cantelli lemma.

Lemma 3.3.8 (A criterion for almost sure convergence)**.** *If random variables X_n and X satisfy*

$$\sum_{n=1}^{\infty} P\left(|X_n - X| > \epsilon\right) < \infty \qquad \text{for every } \epsilon > 0,$$

then $X_n \to X$ almost surely.

Lemma 3.3.9. *If random variables X_n satisfy*

$$\sum_{n=1}^{\infty} P\left(X_n - > \epsilon\right) < \infty \qquad \text{for every } \epsilon > 0,$$

then $\limsup_n X_n \le 0$ almost surely.

Corollary 3.3.10 (Strong Glivenko–Cantelli property)**.** *If $\mathcal{F}$ is b-uniformly bounded and $\mathcal{R}_n(\mathcal{F}) = o(1)$, then*

$$\|P_n - P\|_{\mathcal{F}} \longrightarrow 0 \qquad \text{almost surely.}$$

Proof. The case $b = 0$ is immediate. Suppose $b > 0$ and fix $\delta > 0$. Theorem 3.3.7 gives

$$\sum_{n=1}^{\infty} P\left(\|P_n - P\|_{\mathcal{F}} - 2\mathcal{R}_n(\mathcal{F}) > \delta\right) \le \sum_{n=1}^{\infty} \exp\left(-\frac{n\delta^2}{2b^2}\right) = \sum_{n=1}^{\infty} \left(e^{-\delta^2/(2b^2)}\right)^n < \infty.$$

By Lemma 3.3.9,

$$\limsup_{n\to\infty} \left[\|P_n - P\|_{\mathcal{F}} - 2\mathcal{R}_n(\mathcal{F})\right] \le 0 \qquad \text{almost surely.}$$

Since $\mathcal{R}_n(\mathcal{F}) \to 0$ and $\|P_n - P\|_{\mathcal{F}} \ge 0$, it follows that $\|P_n - P\|_{\mathcal{F}} \to 0$ almost surely. □

Proof of Theorem 3.3.7. For each $f \in \mathcal{F}$, write $\bar{f}(x) := f(x) - Pf$. Then

$$\|P_n - P\|_{\mathcal{F}} = \sup_{f\in\mathcal{F}} \left|\frac{1}{n}\sum_{i=1}^{n} (f(X_i) - Pf)\right| = \sup_{f\in\mathcal{F}} \left|\frac{1}{n}\sum_{i=1}^{n} \bar{f}(X_i)\right|.$$

Define

$$G(x_1, \ldots, x_n) := \sup_{f\in\mathcal{F}} \left|\frac{1}{n}\sum_{i=1}^{n} \bar{f}(x_i)\right|.$$

Thus $G(X_1, \ldots, X_n) = \|P_n - P\|_{\mathcal{F}}$. Suppose $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$ differ only in coordinate k. Then

$$\begin{aligned} |G(x) - G(y)| &\le \sup_{f\in\mathcal{F}} \left|\frac{1}{n}\sum_{i=1}^{n} \bar{f}(x_i) - \frac{1}{n}\sum_{i=1}^{n} \bar{f}(y_i)\right| \\ &= \frac{1}{n} \sup_{f\in\mathcal{F}} \left|\bar{f}(x_k) - \bar{f}(y_k)\right| \\ &= \frac{1}{n} \sup_{f\in\mathcal{F}} |f(x_k) - f(y_k)| \le \frac{2b}{n}. \end{aligned}$$

Thus G has bounded difference constants $L_k = 2b/n$ for every k.

The one-sided bounded differences inequality yields

$$P\left(G(X_1^n) - \mathbb{E}G(X_1^n) > \delta\right) \le \exp\left(-\frac{2\delta^2}{\sum_{k=1}^{n}(2b/n)^2}\right) = \exp\left(-\frac{n\delta^2}{2b^2}\right).$$

By (3.20), $\mathbb{E}G(X_1^n) \le 2\mathcal{R}_n(\mathcal{F})$. Combining the mean bound with the concentration inequality proves (3.22). □

Our next goal is to obtain a bound on $R_n(\mathcal{F})$ that tends to zero as n increases.

3.4 Classes with Polynomial Discrimination

We follow Section 4.3.1. If $\mathcal{F}$ is a class of binary-valued functions, the number of distinct vectors

$$\mathcal{F}(x_1^n) := \{(f(x_1), \ldots, f(x_n)) : f \in \mathcal{F}\}$$

is at most 2^n, because there are two choices at each of the n coordinates. For example, n sample points allow at most 2^n binary combinations. Polynomial discrimination requires this number to grow no faster than a polynomial in n.

Definition 3.4.1 (Polynomial discrimination)**.** A class $\mathcal{F}$ of functions with domain $\mathcal{X}$ has polynomial discrimination of order $\nu \geq 1$ with parameter n_0 if, for any $n \geq n_0$ and any $x_1^n = (x_1, \ldots, x_n)$ of n points in $\mathcal{X}$,

$$\operatorname{card}(\mathcal{F}(x_1^n)) \leq (n+1)^\nu. \tag{3.23}$$

Lemma 3.4.2. *Suppose that $\mathcal{F}$ has polynomial discrimination of order $\nu \geq 1$ and parameter n_0. Then, for every $n \geq n_0$ and for any $x_1^n = (x_1, \ldots, x_n)$,*

$$R\big(\mathcal{F}(x_1^n)/n\big) \leq 2D(x_1^n)\sqrt{\frac{\nu \log(n+1)}{n}}, \tag{3.24}$$

where

$$D(x_1^n) := \sup_{f \in \mathcal{F}} \sqrt{\frac{1}{n}\sum_{i=1}^n f(x_i)^2} \tag{3.25}$$

is the ℓ_2-radius of the set $\mathcal{F}(x_1^n)/\sqrt{n}$.

Homework 3. The proof is assigned as Homework 3. □

Lemma 3.4.3. *If $\mathcal{F}$ is b-uniformly bounded and has polynomial discrimination of order $\nu \geq 1$ with parameter n_0, then*

$$R_n(\mathcal{F}) \leq 2b\sqrt{\frac{\nu \log(n+1)}{n}}, \qquad \forall n \geq n_0. \tag{3.26}$$

Proof. By the definition of Rademacher complexity and Lemma 3.4.2,

$$\begin{aligned} R_n(\mathcal{F}) &= \mathbb{E}_X\left[R\big(\mathcal{F}(X_1^n)/n\big)\right] \\ &\leq \mathbb{E}_X\left[2D(X_1^n)\sqrt{\frac{\nu \log(n+1)}{n}}\right] \\ &= 2\mathbb{E}_X[D(X_1^n)]\sqrt{\frac{\nu \log(n+1)}{n}}. \end{aligned}$$

Since $|f(x)| \leq b$ for every $f \in \mathcal{F}$ and x,

$$\mathbb{E}_X[D(X_1^n)] = \mathbb{E}_X\left[\sup_{f \in \mathcal{F}} \sqrt{\frac{1}{n}\sum_{i=1}^n f(X_i)^2}\right] \leq b.$$

Therefore,

$$R_n(\mathcal{F}) \leq 2b\sqrt{\frac{\nu \log(n+1)}{n}}.$$

□

3.4.1 The proof of the Classical Glivenko–Cantelli Theorem

Corollary 3.4.4 (Classical Glivenko–Cantelli)**.** *Let $F(t) = P(X \le t)$ be the cumulative distribution function of X, and let $\widehat{F}_n(t) = n^{-1}\sum_{i=1}^n I_{(-\infty,t]}[X_i]$ be the empirical cumulative distribution function. Then, for all $\delta > 0$,*

$$P\left(\|\widehat{F}_n - F\|_\infty \ge 4\sqrt{\frac{\log(n+1)}{n}} + \delta\right) \le \exp\left(-\frac{n\delta^2}{2}\right) \quad \forall \delta \ge 0 \tag{3.27}$$

Consequently,

$$\|\widehat{F}_n - F\|_\infty \longrightarrow 0 \qquad \textit{almost surely.} \tag{3.28}$$

Proof. Consider the class of indicator functions

$$\mathcal{F} := \{f_t : t \in \mathbb{R}\}, \qquad f_t(x) = I_{(-\infty,t]}[x].$$

For this class,

$$\|\widehat{F}_n - F\|_\infty = \|P_n - P\|_{\mathcal{F}}.$$

Since each f_t takes values in $\{0,1\}$, the class is 1-uniformly bounded. Theorem **??** therefore gives

$$P\left(\|\widehat{F}_n - F\|_\infty \ge 2R_n(\mathcal{F}) + \delta\right) \le \exp\left(-\frac{n\delta^2}{2}\right), \qquad \delta \ge 0. \tag{3.29}$$

To verify polynomial discrimination, take any sample points and order them as $x_{(1)} \le \cdots \le x_{(n)}$. As t moves from $-\infty$ to $+\infty$, the vector $(f_t(x_1), \ldots, f_t(x_n))$ can change only when t passes one of the sample points. Thus,

$$\operatorname{card}(\mathcal{F}(x_1^n)) \le n + 1.$$

In particular, $\mathcal{F}$ has polynomial discrimination of order $\nu = 1$ and parameter $n_0 = 1$. Lemma 3.4.3 then gives

$$R_n(\mathcal{F}) \le 2\sqrt{\frac{\log(n+1)}{n}}. \tag{3.30}$$

Substituting (3.30) into (3.29) proves (3.27). The almost-sure convergence follows from the preceding bound, as can be seen by Borel-Cantelli lemma. □

3.5 Vapnik–Chervonenkis Dimension

We now turn to Section 4.3.2 of the textbook: Vapnik–Chervonenkis (VC) dimension.

Definition 3.5.1 (Shattering and VC dimension)**.** Given a class $\mathcal{F}$ of binary-valued functions, we say that a set of points $x_1^n = (x_1, \ldots, x_n)$ is shattered by $\mathcal{F}$ if

$$\operatorname{card}(\mathcal{F}(x_1^n)) = 2^n. \tag{3.31}$$

The VC dimension $\nu(\mathcal{F})$ is the largest integer n for which there exists some x_1^n that is shattered by $\mathcal{F}$. If $\nu(\mathcal{F}) < \infty$, then $\mathcal{F}$ is called a VC class.

For example, a binary-valued function class indexed by a collection $\mathcal{S}$ of subsets can be written as

$$\mathcal{F} = \{\mathbb{I}_S[\cdot] : S \in \mathcal{S}\}.$$

For any $x \in \mathcal{X}$, the coordinate $\mathbb{I}_S[x]$ equals 1 when $x \in S$, and equals 0 otherwise. Thus, selecting a set $S \in \mathcal{S}$ selects which sample points receive label 1. A sample is shattered precisely when all 2^n such labelings can be realized.

Definition 3.5.2 (VC class)**.** A binary-valued function class $\mathcal{F}$ is a VC class if $\nu(\mathcal{F}) < \infty$. In particular, for a set class $\mathcal{S}$, we consider

$$\mathcal{F} = \{\mathbf{1}_S : S \in \mathcal{S}\},$$

where $\mathbf{1}_S(x) = 1$ means that $x \in S$, and it is zero otherwise.

The corresponding evaluation vectors are

$$\mathcal{F}(x_1^n) = \{(\mathbf{1}_S(x_1), \ldots, \mathbf{1}_S(x_n)) : S \in \mathcal{S}\}. \tag{3.32}$$

Each distinct intersection of a set S with the sample determines one such labeling. Thus

$$\operatorname{card}(\mathcal{F}(x_1^n)) = \operatorname{card}\big(\{S \cap \{x_1, \ldots, x_n\} : S \in \mathcal{S}\}\big). \tag{3.33}$$

Identifying each intersection with its indicator vector, we sometimes use the notation

$$\mathcal{S}(x_1^n) = \mathcal{F}(x_1^n), \qquad \nu(\mathcal{S}) = \nu(\mathcal{F}). \tag{3.34}$$

The definition of VC dimension gives the two cases recorded in the notes:

(i) If $n \le \nu(\mathcal{F})$, then

$$\exists\, x_1^n \quad \text{such that} \quad \operatorname{card}(\mathcal{F}(x_1^n)) = 2^n.$$

(ii) If $n > \nu(\mathcal{F})$, then

$$\forall\, x_1^n, \qquad \operatorname{card}(\mathcal{F}(x_1^n)) < 2^n.$$

The bound in (ii) alone would allow $\operatorname{card}(\mathcal{F}(x_1^n)) = 2^n - 1$. The proposition below rules out such exponential growth for a class with fixed finite VC dimension.

Proposition 3.5.3 (Vapnik–Chervonenkis, Sauer and Shelah)**.** *Let $\mathcal{S}$ be a set class with $\nu(\mathcal{S}) < \infty$. For ~~$n > \nu(\mathcal{S})$~~ $n \geq 1$ and every collection x_1^n,*

$$\operatorname{card}(\mathcal{S}(x_1^n)) \leq \sum_{j=0}^{\nu(\mathcal{S})} \binom{n}{j} \leq (n+1)^{\nu(\mathcal{S})}. \tag{3.35}$$

The proof is referred to the textbook [1, Section 4.3.2]. We use the proposition directly in the examples below.

3.5.1 Examples on the Real Line

Example 3.5.4 (Left half-lines)**.** Consider

$$\mathcal{S}_{\text{left}} = \{(-\infty, a] : a \in \mathbb{R}\}, \qquad \mathcal{F}_{\text{left}} = \{\mathbf{1}_{(-\infty,a]} : a \in \mathbb{R}\}.$$

For $n = 1$, either label 0 or 1 can be obtained, so a one-point set is shattered. For $n = 2$ and $x_1 < x_2$,

$$\mathcal{F}_{\text{left}}(x_1, x_2) = \{(0,0), (1,0), (1,1)\}.$$

There are three labelings, fewer than 2^2; the labeling $(0,1)$ is impossible. Therefore

$$\nu(\mathcal{S}_{\text{left}}) = \nu(\mathcal{F}_{\text{left}}) = 1.$$

Applying Proposition 3.5.3 directly, for $n > 1$, the cardinality is at most $n + 1$.

In fact, for distinct ordered points $x_1 < \cdots < x_n$, writing out the labelings gives

$$\begin{aligned} \operatorname{card}(\mathcal{F}_{\text{left}}(x_1^n)) &= \operatorname{card}\big(\{(0,\ldots,0), (1,0,\ldots,0), \ldots, (1,\ldots,1)\}\big) \\ &= n+1. \end{aligned} \tag{3.36}$$

This shows the upper bound is sharp.

Example 3.5.5 (Bounded intervals)**.** Consider

$$\mathcal{S}_{\text{int}} = \{(b, a] : a, b \in \mathbb{R},\ b < a\}, \qquad \mathcal{F}_{\text{int}} = \{\mathbf{1}_{(b,a]} : a, b \in \mathbb{R},\ b < a\}.$$

A one-point set can be shattered. For two distinct points $x_1 < x_2$,

$$\mathcal{F}_{\text{int}}(x_1, x_2) = \{(1,0), (0,1), (1,1), (0,0)\},$$

so all 2^2 labelings are possible. For three points $x_1 < x_2 < x_3$, the labeling $(1,0,1)$ is impossible: an interval containing both x_1 and x_3 must also contain x_2. Thus

$$\nu(\mathcal{S}_{\text{int}}) = \nu(\mathcal{F}_{\text{int}}) = 2.$$

Applying Proposition 3.5.3 directly, for $n > 2$,

$$\operatorname{card}(\mathcal{F}_{\text{int}}(x_1^n)) \leq (n+1)^2. \tag{3.37}$$

3.6 Controlling VC Dimension by Vector Space Structure

Let $\mathcal{G}$ be a class of real-valued functions on $\mathcal{X}$. For each $g \in \mathcal{G}$, define

$$S_g = \{x \in \mathcal{X} : g(x) \leq 0\}, \qquad \mathcal{S}(\mathcal{G}) = \{S_g : g \in \mathcal{G}\}. \tag{3.38}$$

Following the lecture's terminology, $\mathcal{S}(\mathcal{G})$ is the subgraph class at level zero. Here each S_g is a zero-sublevel set in $\mathcal{X}$; the threshold is fixed at zero.

Proposition 3.6.1 (Finite-dimensional vector spaces)**.** *Let $\mathcal{G}$ be a real vector space of functions $g : \mathbb{R}^d \to \mathbb{R}$ with finite dimension m. Then*

$$\nu(\mathcal{S}(\mathcal{G})) \leq m. \tag{3.39}$$

Proof. Set $n = m + 1$, and fix any n distinct points $x_1, \ldots, x_n$. It suffices to show that these points cannot be shattered by $\mathcal{S}(\mathcal{G})$. Define the linear evaluation map

$$L : \mathcal{G} \longrightarrow \mathbb{R}^n, \qquad L(g) = (g(x_1), \ldots, g(x_n)).$$

Its image is a linear subspace satisfying

$$\dim L(\mathcal{G}) \leq \dim \mathcal{G} = m < n. \tag{3.40}$$

To see the dimension inequality explicitly, let $g_1, \ldots, g_m$ be a basis of $\mathcal{G}$. If $g = \sum_{j=1}^m a_j g_j$, then $L(g) = \sum_{j=1}^m a_j L(g_j)$. Thus the m vectors $L(g_j)$ span the image. They need not be linearly independent.

Since the image is a proper subspace of $\mathbb{R}^n$, there is a nonzero vector $\gamma = (\gamma_1, \ldots, \gamma_n)$ orthogonal to it:

$$\langle \gamma, L(g) \rangle = \sum_{i=1}^n \gamma_i g(x_i) = 0 \qquad \text{for every } g \in \mathcal{G}. \tag{3.41}$$

Replacing γ by $-\gamma$ if necessary, we may assume that at least one γ_i is positive.

Rearranging (3.41) yields

$$\sum_{\{i:\gamma_i \leq 0\}} (-\gamma_i) g(x_i) = \sum_{\{i:\gamma_i > 0\}} \gamma_i g(x_i) \qquad \text{for every } g \in \mathcal{G}. \tag{3.42}$$

Suppose, for a contradiction, that the points are shattered. There must then exist $g \in \mathcal{G}$ whose set S_g contains exactly those sample points with $\gamma_i \leq 0$. In other words,

$$g(x_i) \leq 0 \quad \text{if } \gamma_i \leq 0, \qquad g(x_i) > 0 \quad \text{if } \gamma_i > 0.$$

Every term on the left-hand side of (3.42) is nonpositive. Every term on the right-hand side is strictly positive, and there is at least one such term. Hence (3.42) would imply

$$0 \;\geq\; \sum_{\{i:\gamma_i \leq 0\}} (-\gamma_i) g(x_i) = \sum_{\{i:\gamma_i > 0\}} \gamma_i g(x_i) \;>\; 0,$$

a contradiction. No $m + 1$ points can be shattered, proving (3.39). □

To illustrate this proposition, consider the following examples:

Example 3.6.2 (Linear Functions in $\mathbb{R}^d$)**.** Consider the function

$$f_{a,b}(x) = \langle a, x \rangle + b, \qquad a \in \mathbb{R}^d, \quad b \in \mathbb{R}.$$

Define the family of all such linear functions:

$$\mathcal{L}^d = \{f_{a,b}(\cdot) | (a, b) \in \mathbb{R}^d \times \mathbb{R}\}.$$

The class $\mathcal{L}^d$ forms a vector space since it is closed under addition and scalar multiplication.
Let e_i denote the ith standard basis vector in $\mathbb{R}^d$. Then

$$f_{e_i,0}(x) = x_i, \qquad f_{0,1}(x) = 1.$$

Every function in $\mathcal{L}^d$ can be expressed as $f_{a,b}(x) = \sum_{i=1}^{d} a_i x_i + b$.
The functions $\{f_{e_1,0}, \ldots, f_{e_d,0}, f_{0,1}\}$ form a basis for $\mathcal{L}^d$, giving $\dim(\mathcal{L}^d) = d + 1$.
For each $f_{a,b}$, the corresponding subgraph set is

$$H_{a,b} = S_{f_{a,b}} = \{x \in \mathbb{R}^d | \langle a, x \rangle + b \leq 0\}.$$

These sets describe half-spaces in $\mathbb{R}^d$ when $a \neq 0$.
The associated subgraph class is

$$\mathcal{S}(\mathcal{L}^d) = \{H_{a,b} | (a, b) \in \mathbb{R}^d \times \mathbb{R}\}.$$

Since $\mathcal{L}^d$ is a vector space of dimension $d + 1$, the proposition gives

$$\boxed{\nu(\mathcal{S}(\mathcal{L}^d)) \leq d + 1.}$$

To illustrate this result, **consider the special case $d = 1$.**
Here,

$$f_{a,b}(x) = ax + b,$$

and

$$H_{a,b} = \{x \in \mathbb{R} | ax + b \leq 0\}.$$

For $a > 0$,

$$H_{a,b} = \{x \in \mathbb{R} | x \leq \frac{-b}{a}\} = \left(-\infty, -\frac{b}{a}\right],$$

and for $a < 0$,

$$H_{a,b} = \{x \in \mathbb{R} | x \geq \frac{-b}{a}\} = \left[-\frac{b}{a}, \infty\right).$$

Therefore, apart from the empty set and $\mathbb{R}$, the class consists of left-sided and right-sided intervals:

$$\mathcal{S}(\mathcal{L}^1) = \{(-\infty, t] | t \in \mathbb{R}\} \cup \{[s, \infty) | s \in \mathbb{R}\} \cup \{\varnothing, \mathbb{R}\}.$$

Next, consider two distinct points $x_1 < x_2$. Define the subsets:

1. If $t < x_1$, $(-\infty, t]$, then
$$(-\infty, t] \cap \{x_1, x_2\} = \varnothing.$$
2. If $x_1 < t < x_2$,$(-\infty, t]$, then
$$(-\infty, t] \cap \{x_1, x_2\} = \{x_1\}.$$
3. If $t > x_2$, $(-\infty, t]$, then
$$(-\infty, t] \cap \{x_1, x_2\} = \{x_1, x_2\}.$$
4. If $x_1 < s < x_2$, $[s, \infty)$, then
$$[s, \infty) \cap \{x_1, x_2\} = \{x_2\}.$$

All four subsets $\varnothing, \quad \{x_1\}, \quad \{x_2\}, \quad \{x_1, x_2\}$ can be obtained using sets from $\mathcal{S}(\mathcal{L}^1)$.

Hence, two points can be shattered.

Next, **consider three distinct points**

$$x_1 < x_2 < x_3.$$

Any left-sided interval containing x_2 must also contain x_1. Similarly, any right-sided interval containing x_2 must contain x_3. Thus, the subset $\{x_2\}$ cannot be selected without including another point.

Consequently, three points cannot be shattered, and we conclude that the largest n that can be shattered is $n = 2$:

$$\boxed{\nu(\mathcal{S}(\mathcal{L}^1)) = 2.}$$

This agrees with the upper bound obtained from the proposition.

Generally, the VC dimension of affine half-spaces in $\mathbb{R}^d$ is

$$\boxed{\nu(\mathcal{S}(\mathcal{L}^d)) = d + 1.}$$

Example 3.6.3 (Spheres in $\mathbb{R}^d$)**.** Consider the collection of Euclidean balls

$$B_{a,b} = \{x \in \mathbb{R}^d | \|x - a\|_2 \leq b\},$$

where $a \in \mathbb{R}^d$ is the center and $b \in \mathbb{R}_+$ is the radius.

Define the function

$$f_{a,b}(x) = \|x - a\|_2^2 - b^2.$$

Expanding the squared Euclidean norm gives

$$\begin{aligned} f_{a,b}(x) &= \langle x - a, x - a \rangle - b^2 \\ &= \|x\|_2^2 - 2\langle a, x \rangle + \|a\|_2^2 - b^2 \\ &= \|x\|_2^2 - 2\sum_{i=1}^{d} a_i x_i + \|a\|_2^2 - b^2. \end{aligned}$$

Since $\|x - a\|_2 \leq b$ is equivalent to $f_{a,b}(x) \leq 0$, we have

$$B_{a,b} = S_{f_{a,b}} = \{x \in \mathbb{R}^d | f_{a,b}(x) \leq 0\}.$$

Thus, each Euclidean ball is a subgraph set associated with the function $f_{a,b}$.

However, the function class

$$\mathcal{F} = \{f_{a,b}(\cdot) | (a, b) \in \mathbb{R}^d \times \mathbb{R}_+\}$$

is not a vector space, since it is not closed under addition or scalar multiplication.

To apply the preceding proposition, we construct a larger vector space containing $\mathcal{F}$.

Define the feature map

$$\phi(x) = (1, x_1, \ldots, x_d, \|x\|_2^2) \in \mathbb{R}^{d+2}.$$

For any $c \in \mathbb{R}^{d+2}$, define

$$g_c(x) = \langle c, \phi(x) \rangle.$$

Expanding the inner product,

$$g_c(x) = c_1 + \sum_{i=2}^{d+1} c_i x_{i-1} + c_{d+2} \|x\|_2^2.$$

Now consider the function class

$$\mathcal{G} = \{g_c(\cdot) | c \in \mathbb{R}^{d+2}\}.$$

The class $\mathcal{G}$ is a vector space because it is closed under addition and scalar multiplication:

$$g_{c_1} + g_{c_2} = g_{c_1+c_2},$$

and

$$\lambda g_c = g_{\lambda c}.$$

The functions

$$1, x_1, \ldots, x_d, \|x\|_2^2$$

form a basis for $\mathcal{G}$, so

$$\dim(\mathcal{G}) = d + 2.$$

Moreover, every $f_{a,b}$ can be expressed as g_c by choosing

$$c = (\|a\|_2^2 - b^2, -2a_1, \ldots, -2a_d, 1).$$

Therefore,

$$\mathcal{F} \subseteq \mathcal{G}.$$

Applying the proposition gives

$$\nu(\mathcal{S}(\mathcal{G})) \leq \dim(\mathcal{G}) = d + 2.$$

Let

$$\mathcal{B}^d = \{B_{a,b} | (a, b) \in \mathbb{R}^d \times \mathbb{R}_+\}.$$

Since $\mathcal{F} \subseteq \mathcal{G}$, the corresponding subgraph classes satisfy

$$\mathcal{B}^d = \mathcal{S}(\mathcal{F}) \subseteq \mathcal{S}(\mathcal{G}).$$

Hence,

$$\begin{aligned} \nu(\mathcal{B}^d) &= \nu(\mathcal{S}(\mathcal{F})) \\ &\leq \nu(\mathcal{S}(\mathcal{G})) \\ &\leq d + 2. \end{aligned}$$

Therefore,

$$\boxed{\nu(\mathcal{B}^d) \leq d + 2.}$$

This upper bound is not sharp, but it is sufficient for many applications. A more refined argument shows that the VC dimension of Euclidean balls in $\mathbb{R}^d$ is

$$\boxed{\nu(\mathcal{B}^d) = d + 1.}$$

The case $d = 2$ is considered in Exercise 4.13 of Wainwright (2019) **(part of the next homework)**.

3.7 A Remark on the measurability

Recall that

$$\|P_n - P\|_{\mathcal{F}} = \sup_{f \in \mathcal{F}} |P_n f - Pf|.$$

In general, this supremum need not be measurable when $\mathcal{F}$ is uncountable. If $\mathcal{F}$ is separable, the supremum can be taken over a countable dense subset. Otherwise, one may use outer probability, as discussed by van der Vaart and Wellner (1996).

For any finite class $\mathcal{G} \subseteq \mathcal{F}$, the random variable $\|P_n - P\|_{\mathcal{G}}$ is well defined, so it is sensible to define

$$\mathbb{E}\left[\phi\left(\|P_n - P\|_{\mathcal{F}}\right)\right] := \sup\left\{\mathbb{E}\left[\phi\left(\|P_n - P\|_{\mathcal{G}}\right)\right] \mid \mathcal{G} \subset \mathcal{F},\ \mathcal{G} \text{ has finite cardinality}\right\}.$$

Using this definition, we can work with expectations expressed as suprema over finite sets.

For an arbitrary measurable function ϕ, replacing finite subsets by countable subsets in the convention can change its value. Although a countable supremum of measurable random variables is measurable, applying ϕ need not preserve the limit of finite maxima.

Counterexample. Take $n = 1$, let P be the uniform distribution on $\{-1, 1\}$, and let $X_1 \sim P$. For each $k \geq 2$, define $f_k(x) = (1 - 1/k)x$, and let $\mathcal{F} = \{f_k : k \geq 2\}$. Since $Pf_k = 0$, we have $|(P_1 - P)f_k| = |f_k(X_1)| = 1 - 1/k$ almost surely. Thus, for every nonempty finite $\mathcal{G} \subseteq \mathcal{F}$, the empirical-process norm satisfies $\|P_1 - P\|_{\mathcal{G}} = \max_{\{k: f_k \in \mathcal{G}\}}(1 - 1/k) < 1$. On the other hand, $\|P_1 - P\|_{\mathcal{F}} = \sup_{k \geq 2}(1 - 1/k) = 1$.

Now choose $\phi(t) = \mathbf{1}\{t \geq 1\}$. Then

$$\sup_{\substack{\varnothing \neq \mathcal{G} \subseteq \mathcal{F} \\ \mathcal{G} \text{ finite}}} \mathbb{E}[\phi(\|P_1 - P\|_{\mathcal{G}})] = 0,$$

$$\mathbb{E}[\phi(\|P_1 - P\|_{\mathcal{G}})] = 1.$$

So the convention is not the same as the normal definition.

In this example, ϕ is bounded, nonnegative, and nondecreasing. The failure occurs at its jump at 1: the finite maxima approach 1 from below, so their transformed values remain zero, whereas the transformed limit equals one. Thus, even these three properties of ϕ do not suffice for the two conventions to agree.

For any countable class $\mathcal{G} \subseteq \mathcal{F}$, the random variable $\|P_n - P\|_{\mathcal{G}}$ is well defined, so it is sensible to define

$$\mathbb{E}\left[\phi\left(\|P_n - P\|_{\mathcal{F}}\right)\right] := \sup\left\{\mathbb{E}\left[\phi\left(\|P_n - P\|_{\mathcal{G}}\right)\right] \mid \mathcal{G} \subset \mathcal{F},\ \mathcal{G} \text{ has countable cardinality}\right\}.$$

A sufficient condition for equivalence. Suppose that ϕ is nonnegative, nondecreasing, and continuous from below, meaning that $\phi(x_m) \uparrow \phi(x)$ whenever $x_m \uparrow x$. If the supremum can be infinite, impose the same condition at $+\infty$ by defining $\phi(+\infty) = \lim_{t \to \infty} \phi(t)$.

Fix a nonempty countable class $\mathcal{C} \subseteq \mathcal{F}$, and choose increasing nonempty finite classes $\mathcal{G}_m$ whose union is $\mathcal{C}$. Write $S_m = \|P_n - P\|_{\mathcal{G}_m}$ and $S = \|P_n - P\|_{\mathcal{C}}$. Then $S_m \uparrow S$, and the assumptions on ϕ imply $\phi(S_m) \uparrow \phi(S)$. By the monotone convergence theorem,

$$\begin{aligned}\mathbb{E}[\phi(\|P_n - P\|_{\mathcal{C}})] &= \lim_{m \to \infty} \mathbb{E}[\phi(\|P_n - P\|_{\mathcal{G}_m})] \\ &\leq \sup_{\substack{\varnothing \neq \mathcal{G} \subseteq \mathcal{F} \\ \mathcal{G} \text{ finite}}} \mathbb{E}[\phi(\|P_n - P\|_{\mathcal{G}})].\end{aligned}$$

Taking the supremum over countable $\mathcal{C}$ gives one inequality between the two conventions. The reverse inequality holds because every finite class is at most countable. Consequently, the two conventions agree under these conditions. This includes the choices $\phi(t) = t$ and $\phi(t) = t^p$ for $p > 0$.

Interpretation of the convention. For arbitrary ϕ, the finite-subset convention need not agree with the ordinary expectation of the transformed supremum, even when $\mathcal{F}$ is countable and the full supremum is measurable. The counterexample above demonstrates this distinction: the finite-subset convention gives zero, whereas the ordinary expectation of $\phi(\|P_1 - P\|_{\mathcal{F}})$ equals one.

Chapter 4

Metric entropy and Dudley's integral bound

4.1 Covering and packing

4.1.1 Motivation

Many statistical problems involve collections of random variables indexed by a set. Examples include a finite collection $X_1, \ldots, X_n$, a countably infinite collection $\{X_i : i \in \mathbb{N}\}$, and a stochastic process $\{X_t : t \in [0,1]\}$ with an uncountable index set. Further examples discussed in class are:

- The empirical distribution function $\{F_n(t) : t \in \mathbb{R}\}$.
- Quadratic forms $\{v^\top A v : v \in S^{d-1}\}$, where A is a random matrix and $S^{d-1} = \{v \in \mathbb{R}^d : \|v\|_2 = 1\}$.
- An empirical process $\{P_n f - Pf : f \in \mathcal{F}\}$, where

$$P_n f = \frac{1}{n}\sum_{i=1}^{n} f(X_i), \qquad Pf = \mathbb{E}[f(X)].$$

For a finite index set, cardinality provides a natural measure of its size. For an infinite index set, cardinality alone does not describe its geometric complexity. Covering and packing measure this complexity at a given resolution.

4.1.2 Metric Spaces and Examples

A metric space is a pair $(\mathbb{T}, \rho)$, where $\mathbb{T}$ is a nonempty set and $\rho : \mathbb{T} \times \mathbb{T} \to \mathbb{R}$ satisfies:

1. $\rho(\theta, \theta') \geq 0$, with equality if and only if $\theta = \theta'$.
2. $\rho(\theta, \theta') = \rho(\theta', \theta)$.
3. $\rho(\theta_1, \theta_3) \leq \rho(\theta_1, \theta_2) + \rho(\theta_2, \theta_3)$.

Examples include the following distances, with equation labels matching the textbook [1].

Euclidean distance. On $\mathbb{R}^d$,

$$\rho(\theta, \theta') = \|\theta - \theta'\|_2 = \left(\sum_{j=1}^{d} (\theta_j - \theta'_j)^2\right)^{1/2}. \tag{5.1a}$$

Rescaled Hamming distance. On $\{0,1\}^d$,

$$\rho_H(\theta,\theta') = \frac{1}{d}\sum_{j=1}^{d} \mathbf{1}\{\theta_j \neq \theta'_j\}. \tag{5.1b}$$

This is the fraction of coordinates at which the binary vectors differ.

L^2 distance. On $L^2(\mu,[0,1])$,

$$\rho_2(f,g) = \|f-g\|_2 = \left(\int_0^1 |f(x)-g(x)|^2\, d\mu(x)\right)^{1/2}. \tag{5.1c}$$

Functions agreeing μ-almost everywhere are identified.

Uniform distance. On $C([0,1])$, the continuous functions on $[0,1]$,

$$\rho_\infty(f,g) = \|f-g\|_\infty = \sup_{x\in[0,1]} |f(x)-g(x)|. \tag{5.1d}$$

4.1.3 Covering Numbers and Metric Entropy

For $x \in \mathbb{T}$ and $\delta > 0$, define the closed metric ball

$$B_\rho(x,\delta) = \{y \in \mathbb{T} : \rho(x,y) \leq \delta\}.$$

Definition 4.1.1 (Covering number)**.** A δ-cover of $\mathbb{T}$ with respect to ρ is a set $\{\theta^1,\ldots,\theta^N\} \subseteq \mathbb{T}$ such that for every $\theta \in \mathbb{T}$, there is an $i \in \{1,\ldots,N\}$ satisfying $\rho(\theta,\theta^i) \leq \delta$. Equivalently,

$$\mathbb{T} \subseteq \bigcup_{i=1}^{N} B_\rho(\theta^i,\delta).$$

The δ-covering number $N(\delta;\mathbb{T},\rho)$ is the cardinality of the smallest δ-cover. If no finite cover exists, it is infinite.

Geometrically, covering asks how few balls of radius δ are needed to cover the set. Packing asks how many sufficiently separated points can be placed in the set; its formal definition will be considered later.

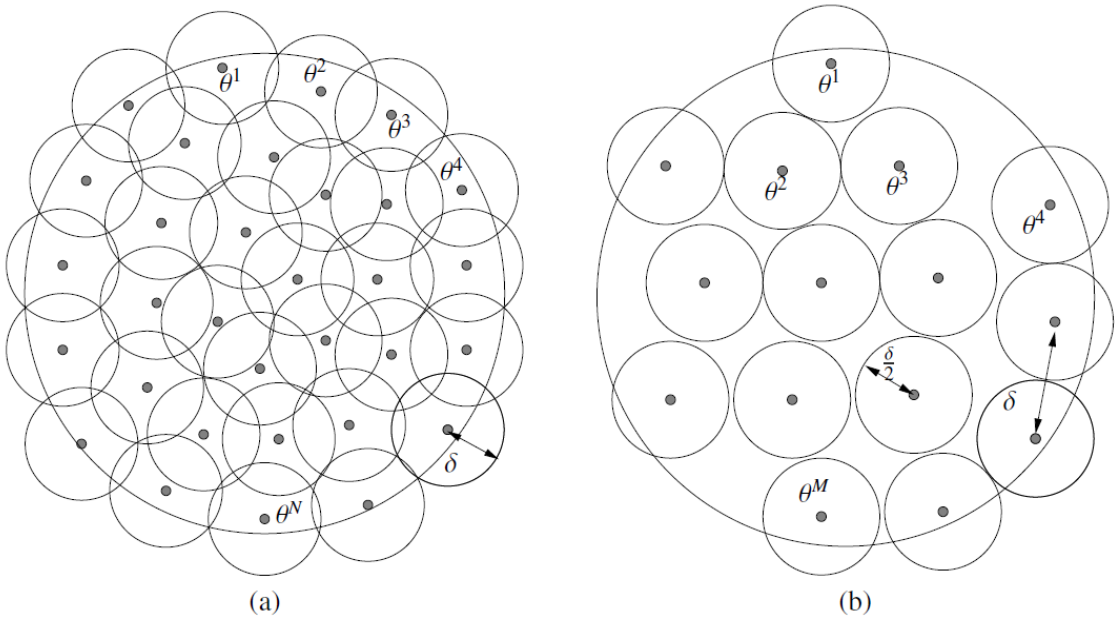


Figure 4.1: Covering and packing (textbook Figure 5.1). Covering balls have radius δ; the disjoint balls illustrating packing have radius $\delta/2$.

Total Boundedness and Metric Entropy

A metric space is **totally bounded** if $N(\delta; \mathbb{T}, \rho) < \infty$ for every $\delta > 0$. Thus, at every positive resolution, finitely many points suffice to approximate the set.

Lemma 4.1.2. *Let T be a subset of a complete metric space. Then T is precompact (that is, its closure is compact) if and only if*

$$N(\delta, T, \rho) < \infty \qquad \text{for every } \delta > 0.$$

The covering number $N(\delta, T, \rho)$ is the smallest number of balls of radius δ needed to cover T. $N(\delta, T, \rho)$ is a quantitative way of measuring "how compact a set is". As the radius $\delta \to 0^+$, the number of balls required to cover T is non-decreasing and typically diverges if the cardinality of T is infinite.

Example (not totally bounded). Consider the continuous ramp functions drawn in class, with $\mathbb{T} = \{f_j : j \geq 1\}$ and $\rho(f, g) = \|f - g\|_\infty$. Although every function takes values in $[0, 1]$, distinct functions satisfy $\|f_i - f_j\|_\infty > 1/2$. Hence $N(\delta; \mathbb{T}, \rho) = \infty$ for $\delta < 1/4$, so this space is not totally bounded.

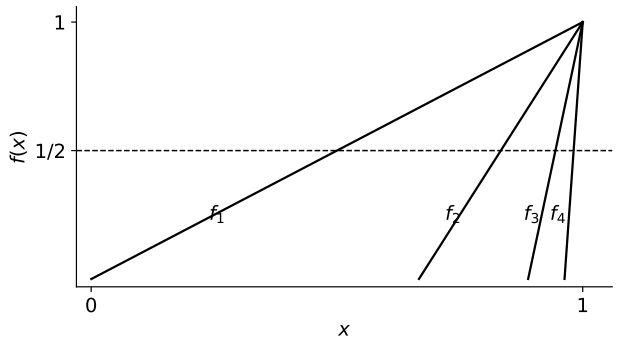


Figure 4.2: The non-totally bounded example drawn in class.

Covering numbers are non-increasing in the radius:

$$0 < \delta_1 \leq \delta_2 \quad \Longrightarrow \quad N(\delta_1; \mathbb{T}, \rho) \geq N(\delta_2; \mathbb{T}, \rho).$$

The quantity $\log N(\delta; \mathbb{T}, \rho)$ is called the **metric entropy**. It measures the size of the set at resolution δ on a logarithmic scale. Unless a base is specified, log denotes the natural logarithm.

Example 4.1.3 (Covering numbers of unit cubes)**.** Consider $\mathbb{T} = [-1, 1]$ with $\rho(x, y) = |x - y|$. For $\delta > 0$, divide the interval into $\lceil 1/\delta \rceil$ equal subintervals, each of length at most 2δ. Their midpoints form a δ-cover. Consequently,

$$N(\delta; [-1, 1], |\cdot|) \leq \left\lceil \frac{1}{\delta} \right\rceil \leq 1 + \frac{1}{\delta}. \tag{5.2}$$

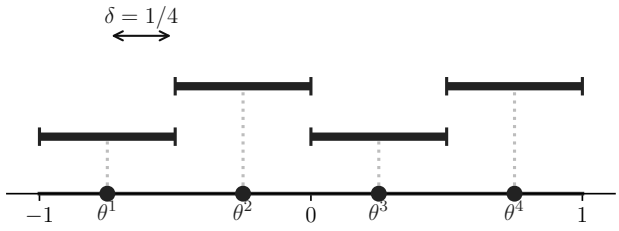
For $[-1, 1]^d$ under the sup-norm, take the Cartesian product of the one-dimensional cover. Each coordinate is approximated within δ, yielding

$$N(\delta; [-1, 1]^d, \|\cdot\|_\infty) \leq \left(1 + \frac{1}{\delta}\right)^d,$$

and hence

$$\log N(\delta; [-1, 1]^d, \|\cdot\|_\infty) \leq d \log \left(1 + \frac{1}{\delta}\right).$$

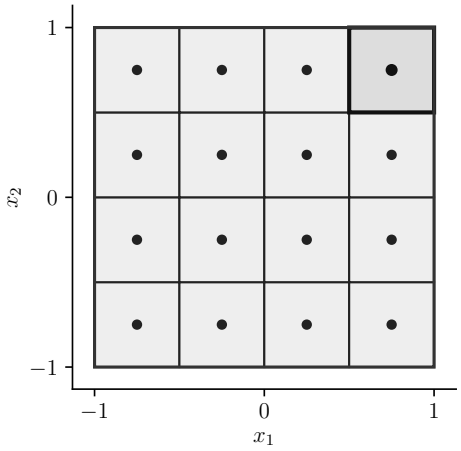
(a) Interval: four balls of radius 1/4



$$[-1,1] \subseteq \bigcup_{i=1}^{4} [\theta^i - \delta, \theta^i + \delta]$$

Dots are centers; segments are the covering balls.

(b) Square: ℓ_∞ balls are squares



$4 \times 4 = 16$ centers, $\delta = 1/4$

Figure 4.3: Example 5.2: covering $[-1,1]$ and $[-1,1]^2$ with $\delta = 1/4$. In two dimensions, balls under the sup-norm are squares.

Remark 4.1.4 (Comparison of metrics)**.** For every $x \in \mathbb{R}^d$,

$$\|x\|_\infty \le \|x\|_2 \le \sqrt{d}\,\|x\|_\infty.$$

Therefore, for the same set $\mathbb{T} \subseteq \mathbb{R}^d$,

$$N(\delta; \mathbb{T}, \|\cdot\|_\infty) \le N(\delta; \mathbb{T}, \|\cdot\|_2) \le N(\delta/\sqrt{d}; \mathbb{T}, \|\cdot\|_\infty).$$

This explains the comparison between square and circular covering regions drawn in class. The comparison involves a change in radius; the two covering numbers need not be equal.

Example 4.1.5 (Covering of the binary hypercube)**.** Consider the binary hypercube $\mathbb{H}^d := \{0,1\}^d$, equipped with the rescaled Hamming metric (5.1b). First, we upper bound its δ-covering number.

For $\delta \in (0,1)$, let

$$S = \{1, 2, \ldots, \lceil (1-\delta)d \rceil\},$$

where $\lceil (1-\delta)d \rceil$ is the smallest integer greater than or equal to $(1-\delta)d$. Consider the set of binary vectors

$$\mathbb{T}(\delta) := \{\theta \in \mathbb{H}^d : \theta_j = 0 \text{ for all } j \notin S\}.$$

For any $\widetilde{\theta} \in \mathbb{H}^d$, construct $\theta \in \mathbb{T}(\delta)$ by matching $\widetilde{\theta}$ on every coordinate in S and setting all coordinates outside S to zero. The vectors agree on S and can differ on at most

$$d - \lceil (1-\delta)d \rceil = \lfloor \delta d \rfloor$$

remaining coordinates. Thus,

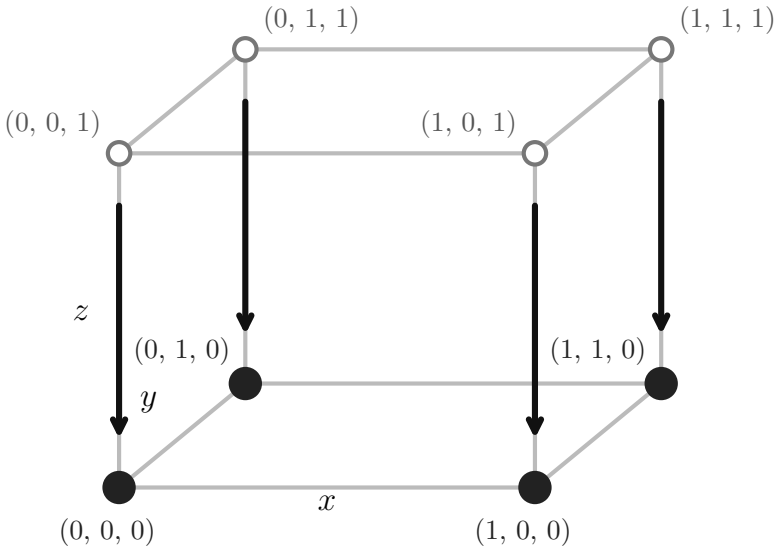
$$\rho_H(\theta, \widetilde{\theta}) \le \frac{\lfloor \delta d \rfloor}{d} \le \delta.$$

This proves that $\mathbb{T}(\delta)$ is a δ-cover. Its coordinates in S can vary freely, so it contains $2^{\lceil (1-\delta)d \rceil}$ vectors. Consequently,

$$N_H(\delta; \mathbb{H}^d) \le 2^{\lceil (1-\delta)d \rceil}, \qquad \text{and hence} \qquad \frac{\log N_H(\delta; \mathbb{H}^d)}{\log 2} \le \lceil d(1-\delta) \rceil.$$

Here $N_H(\delta; \mathbb{H}^d)$ denotes $N(\delta; \mathbb{H}^d, \rho_H)$, as in the textbook.

$$d = 3, \quad \delta = 1/3, \quad m = \lceil (1-\delta)d \rceil = 2$$



$$\mathbb{T}(\delta) = \{(0,0,0), (1,0,0), (0,1,0), (1,1,0)\}$$

Filled dots: cover centers. Arrows: set the third coordinate to zero.

Each mapped vertex differs in one coordinate: $\rho_H = 1/3 = \delta$. Thus $N(\delta) \leq 4$.

Figure 4.4: Example 5.3 with $d = 3$ and $\delta = 1/3$: retain the first two coordinates and set the third to zero. Four centers cover all eight vertices.

4.1.4 Packing number

Definition 4.1.6 (Packing number)**.** A δ-packing of a set T w.r.t a metric ρ is a set $\{\theta^1, \ldots, \theta^M\} \subset T$ s.t. $\rho(\theta^i, \theta^j) > \delta, \forall\, i, j \in [M], i \neq j$. The δ-packing number $M(\delta; T, \rho)$ is the cardinality of the largest δ-packing.

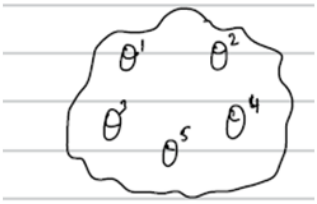


Fig 1: δ-covering of a set

We cannot place infinite points in δ-distance. There should be a maximum number, that is where the δ-packing is coming.

Remark 4.1.7. If N is δ-packing, then the closed $\delta/2$-balls centered in N is disjoint.

Proof. Suppose, $\exists\, i, j$ s.t.

$$A := B(\theta^i, \delta/2) \cap B(\theta^j, \delta/2) \neq \emptyset.$$

Find some $y \in A$.

$$\rho(\theta^i, \theta^j) \leq \rho(\theta^i, y) + \rho(\theta^j, y) \leq \delta. \quad \text{Contradiction.} \qquad \square$$

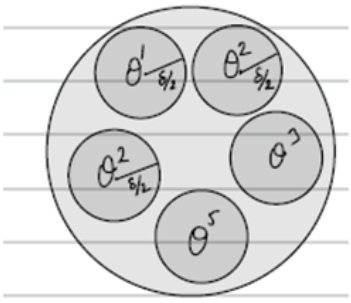


Fig 2: δ-packing of a set

Lemma 4.1.8. *For $\delta > 0$,*

$$M(2\delta; T, \rho) \leq N(\delta; T, \rho) \leq M(\delta; T, \rho)$$

Proof: Homework.

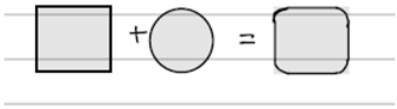
Recall: q-norm. $x \in \mathbb{R}^d$

$$\|x\|_q = \begin{cases} \Big(\sum_{i=1}^{d} |x_i|^q\Big)^{1/q}, & q \geq 1 \\ \max\limits_{1 \leq i \leq d} |x_i|, & q = \infty \end{cases}$$

and $B_q^d(1) := \{x \in \mathbb{R}^d \mid \|x\|_q \leq 1\}$.

Minkowski Sum:

$$A + B := \{a + b \mid a \in A,\ b \in B\}$$


Fig 3: Geometric Illustration of the Minkowski Sum

We take on fixed point in A and connect all the points of B. So the ball can be anywhere.

Lemma 4.1.9. *Consider $K \subset \mathbb{R}^d$. Let $B := \{\theta \in \mathbb{R}^d \mid \|\theta\| \leq 1\}$. Then*

$$\Big(\frac{1}{\delta}\Big)^d \frac{\mathrm{Vol}(K)}{\mathrm{Vol}(B)} \leq N(\delta; K, \|\cdot\|) \leq M(\delta; K, \|\cdot\|) \leq \frac{\mathrm{Vol}(K + \frac{\delta}{2}B)}{\mathrm{Vol}(\frac{\delta}{2}B)} = \frac{\mathrm{Vol}(\frac{2}{\delta}K + B)}{\mathrm{Vol}(B)}$$

Proof. Lower bound: Let $N = N(\delta; K, \|\cdot\|)$. Consider a δ-covering $\{\theta^1, \theta^2, \ldots, \theta^N\}$,

$$K \subset \bigcup_{i=1}^{N} B(\theta^i, \delta).$$

$$\mathrm{Vol}(K) \leq \mathrm{Vol}\Big(\bigcup_{i=1}^{N} B(\theta^i, \delta)\Big) \leq \sum_{i=1}^{N} \mathrm{Vol}\big(B(\theta^i, \delta)\big) = N \cdot \mathrm{Vol}\big(B(0, \delta)\big) = N \cdot \delta^d \,\mathrm{Vol}(B)$$

$$\therefore \Big(\frac{1}{\delta}\Big)^d \frac{\mathrm{Vol}(K)}{\mathrm{Vol}(B)} \leq N \qquad \square$$

Upper bound: Let $M = M(\delta; K, \|\cdot\|)$. Consider $\{\theta^1, \theta^2, \dots, \theta^M\}$ to be δ-packing. By the remark (mentioned above) the closed balls $B(\theta^i, \delta/2)$ are disjoint.

$$\bigcup_{i=1}^{M} B(\theta^i, \delta/2) \subset K + \tfrac{\delta}{2}B = \bigcup_{\theta \in K} B(\theta, \delta/2)$$

$$\mathrm{Vol}\Big(\bigcup_{i=1}^{M} B(\theta^i, \delta/2)\Big) \le \mathrm{Vol}\big(K + \tfrac{\delta}{2}B\big)$$

$$\Rightarrow \sum_{i=1}^{M} \mathrm{Vol}\big(B(\theta^i, \delta/2)\big) \le \mathrm{Vol}\big(K + \tfrac{\delta}{2}B\big)$$

$$\Rightarrow M \,\mathrm{Vol}\big(\tfrac{\delta}{2}B\big) \le \mathrm{Vol}\big(K + \tfrac{\delta}{2}B\big)$$

$$\therefore M \le \frac{\mathrm{Vol}(K + \frac{\delta}{2}B)}{\mathrm{Vol}(\frac{\delta}{2}B)} = \frac{(\delta/2)^d \,\mathrm{Vol}(\frac{2}{\delta}K + B)}{(\delta/2)^d \,\mathrm{Vol}(B)} = \frac{\mathrm{Vol}(\frac{2}{\delta}K + B)}{\mathrm{Vol}(B)} \qquad \square$$

Remark 4.1.10. If $B \subset K$, $\mathrm{Vol}(\frac{2}{\delta}K + B) \le \mathrm{Vol}(\frac{2}{\delta}K + K) = \mathrm{Vol}\big((\frac{2}{\delta} + 1)K\big) = (\frac{2}{\delta} + 1)^d \,\mathrm{Vol}(K)$

$$\Big(\frac{1}{\delta}\Big)^d \boxed{\frac{\mathrm{Vol}(K)}{\mathrm{Vol}(B)}} \le N(\delta; K, \|\cdot\|) \le M(\delta; K, \|\cdot\|) \le \Big(\frac{2}{\delta} + 1\Big)^d \boxed{\frac{\mathrm{Vol}(K)}{\mathrm{Vol}(B)}}.$$

behaves lik δ^{-d} when $\delta \downarrow 0$ since $\boxed{\frac{\mathrm{Vol}(K)}{\mathrm{Vol}(B)}}$ does not depends on δ.

Remark 4.1.11. If $K = B$

$$\Big(\frac{1}{\delta}\Big)^d \le N(\delta; B, \|\cdot\|) \le M(\delta; B, \|\cdot\|) \le \Big(\frac{2}{\delta} + 1\Big)^d$$

Example 4.1.12.

$$\Big(\frac{1}{\delta}\Big)^d \le N\big(\delta, \underbrace{B_\infty^d(1)}_{\ell_\infty\text{-unit ball}}, \|\cdot\|_\infty\big) \le \Big(\frac{2}{\delta} + 1\Big)^d$$

$$\Big(\frac{1}{\delta}\Big) \le N\big(\delta, \underbrace{B_2^d(1)}_{\ell_2\text{-unit ball}}, \|\cdot\|_2\big) \le \Big(\frac{2}{\delta} + 1\Big)^d$$

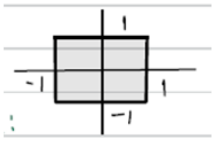


Fig 4: Illustration of the Unit Ball in the l_∞-Norm

Recall: $d = 1$, We show in previous example that:

$$N(\delta, [-1, 1], |\cdot|) \le \Big\lceil \frac{2}{\delta} \Big\rceil \le \frac{2}{\delta} + 1$$

4.2 Gaussian and Rademacher Complexity

$$T \underset{\text{use random process to study } T.\ \text{(Sec: 5.2)}}{\overset{\text{study random process (Sec 5.3, 5.4, } \dots\text{)}}{\rightleftarrows}} \{X_\theta \mid \theta \in T\}$$

Recall: Let $T \subset \mathbb{R}^d$.

Definition 4.2.1 (Rademacher process and Rademacher complexity)**.** Let $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_d)$, where ε_i are iid Rademacher r.v., i.e., $\mathbb{P}(\varepsilon_i = 1) = \mathbb{P}(\varepsilon_i = -1) = 1/2$. For each $\theta \in T$, define

$$R_\theta := \langle \varepsilon, \theta \rangle = \sum_{i=1}^{d} \varepsilon_i \theta_i.$$

The collection $\{R_\theta \mid \theta \in T\}$ is a random process indexed by T, called the canonical Rademacher process. The Rademacher complexity of T is

$$R(T) := \mathbb{E}\Big[\sup_{\theta \in T} R_\theta\Big] = \mathbb{E}\Big[\sup_{\theta \in T} \langle \theta, \varepsilon \rangle\Big].$$

Definition 4.2.2 (Canonical Gaussian process and Gaussian complexity)**.** Let $w = (w_1, \ldots, w_d)$, where w_i are iid $N(0, 1)$. For each $\theta \in T$, define

$$G_\theta := \langle w, \theta \rangle = \sum_{i=1}^{d} w_i \theta_i.$$

The collection $\{G_\theta \mid \theta \in T\}$ is called the canonical Gaussian process indexed by T. The Gaussian complexity of T is

$$\mathcal{G}(T) := \mathbb{E}\Big[\sup_{\theta \in T} G_\theta\Big].$$

Homework:

$$R(T) \leq \sqrt{\pi/2}\, G(T)$$

Example 4.2.3 (Example: the ℓ_2 unit ball)**.** Let $B_2^d = \{\theta \in \mathbb{R}^d : \|\theta\|_2 \leq 1\}$. Using the duality of the Euclidean norm,

$$R(B_2^d) = \mathbb{E} \sup_{\theta \in B_2^d} \langle \varepsilon, \theta \rangle = \mathbb{E}\|\varepsilon\|_2 = \mathbb{E}\Big(\sum_{i=1}^{d} \varepsilon_i^2\Big)^{1/2} = \sqrt{d}, \tag{4.1}$$

$$G(B_2^d) = \mathbb{E} \sup_{\theta \in B_2^d} \langle w, \theta \rangle = \mathbb{E}\|w\|_2 \leq \big(\mathbb{E}\|w\|_2^2\big)^{1/2} = \Big(\sum_{i=1}^{d} \mathbb{E} w_i^2\Big)^{1/2} = \sqrt{d}. \tag{4.2}$$

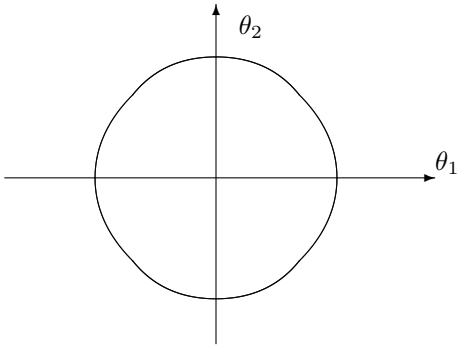
On the other hand,

$$\mathbb{E}\|w\|_2 \geq \sqrt{d}\,(1 - o(1)),$$

which is left as an exercise (hint: use the concentration of the sub-exponential variables $w_i^2 - 1$; see Chapter 2). Hence

$$\frac{G(B_2^d)}{\sqrt{d}} = 1 - o(1).$$

In two dimensions, B_2^2 is the Euclidean disk.



$$B_2^2 = \{(\theta_1, \theta_2) : \theta_1^2 + \theta_2^2 \le 1\}.$$

Example 4.2.4 (Example: the ℓ_1 unit ball)**.** Let $B_1^d = \{\theta \in \mathbb{R}^d : \|\theta\|_1 \le 1\}$.
Recall. For any norm $\|\cdot\|$ and its dual norm $\|\cdot\|_*$,

$$\sup_{\|\theta\| \le 1} \langle y, \theta \rangle = \|y\|_*.$$

In particular, the dual norm of ℓ_1 is ℓ_∞. Consequently,

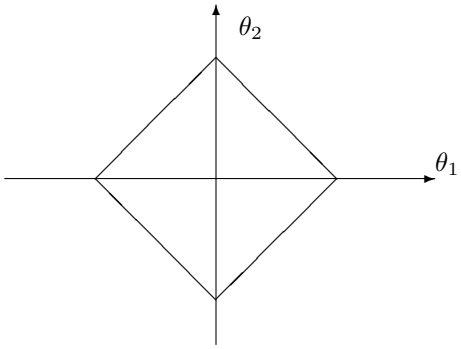
$$R(B_1^d) = \mathbb{E} \sup_{\theta \in B_1^d} \langle \varepsilon, \theta \rangle = \mathbb{E}\|\varepsilon\|_\infty = \mathbb{E} \max_{1 \le i \le d} |\varepsilon_i| = 1, \tag{4.3}$$

$$G(B_1^d) = \mathbb{E} \sup_{\theta \in B_1^d} \langle w, \theta \rangle = \mathbb{E}\|w\|_\infty = \mathbb{E} \max_{1 \le i \le d} |w_i|. \tag{4.4}$$

The Gaussian maximal inequality from Homework 1 gives

$$G(B_1^d) \le \sqrt{2\log(2d)} = \sqrt{2\log d}\,(1 + o(1)) \qquad (d \to \infty). \tag{4.5}$$

In two dimensions, B_1^2 is the diamond $\{(\theta_1, \theta_2) : |\theta_1| + |\theta_2| \le 1\}$.



$$B_1^2 = \{(\theta_1, \theta_2) : |\theta_1| + |\theta_2| \le 1\}.$$

Remark 4.2.5. Homework 5 will ask for the complexity of ℓ_0-balls, where $\|\theta\|_0$ counts nonzero coordinates.

4.3 Metric Entropy and Sub-Gaussian Processes

Definition 4.3.1 (Sub-Gaussian process)**.** A collection of ~~mean-zero~~ random variables $\{X_\theta : \theta \in T\}$ is a sub-Gaussian process with respect to ρ_X if

$$\mathbb{E}\exp\big(\lambda(X_\theta - X_{\widetilde{\theta}})\big) \le \exp\left(\frac{\lambda^2 \rho_X^2(\theta, \widetilde{\theta})}{2}\right) \tag{4.6}$$

for all $\theta, \widetilde{\theta} \in T$ and $\lambda \in \mathbb{R}$. Equivalently, each increment $X_\theta - X_{\widetilde{\theta}}$ is **centered** sub-Gaussian with parameter $\rho_X(\theta, \widetilde{\theta})$.

Example 4.3.2 (Canonical Gaussian process)**.** For $G_\theta = \langle w, \theta \rangle$,

$$G_\theta - G_{\widetilde{\theta}} = \langle w, \theta - \widetilde{\theta} \rangle \sim N\Big(0, \|\theta - \widetilde{\theta}\|_2^2\Big).$$

In particular,

$$\mathbb{E} \exp\big(\lambda(G_\theta - G_{\widetilde{\theta}})\big) = \exp\left(\frac{\lambda^2 \|\theta - \widetilde{\theta}\|_2^2}{2}\right).$$

Thus the canonical Gaussian process is sub-Gaussian with respect to $\rho_X(\theta, \widetilde{\theta}) = \|\theta - \widetilde{\theta}\|_2$.

Example 4.3.3 (Canonical Rademacher process)**.** For $R_\theta = \langle \varepsilon, \theta \rangle$,

$$R_\theta - R_{\widetilde{\theta}} = \langle \varepsilon, \theta - \widetilde{\theta} \rangle = \sum_{i=1}^{d} \varepsilon_i (\theta_i - \widetilde{\theta}_i).$$

Recall:

- ε_i is 1-sub-Gaussian;
- hence $\varepsilon_i(\theta_i - \widetilde{\theta}_i)$ is $|\theta_i - \widetilde{\theta}_i|$-sub-Gaussian;
- since $\varepsilon_1, \ldots, \varepsilon_d$ are independent, $\sum_{i=1}^d \varepsilon_i(\theta_i - \widetilde{\theta}_i)$ is sub-Gaussian with parameter $\big(\sum_{i=1}^d (\theta_i - \widetilde{\theta}_i)^2\big)^{1/2} = \|\theta - \widetilde{\theta}\|_2$.

Hence $\{R_\theta : \theta \in T\}$ is sub-Gaussian with respect to $\rho_X(\theta, \widetilde{\theta}) = \|\theta - \widetilde{\theta}\|_2$.

Our target is to bound $\mathbb{E} \sup_{\theta \in T} X_\theta$.

Lemma 4.3.4. *Let $\{X_\theta : \theta \in T\}$ be a zero-mean random process. Then*

$$\mathbb{E} \sup_{\theta \in T} X_\theta \le \mathbb{E} \sup_{\theta, \widetilde{\theta} \in T} (X_\theta - X_{\widetilde{\theta}}). \tag{4.7}$$

Proof. Fix $\theta_0 \in T$. By the mean-zero assumption,

$$\mathbb{E} \sup_{\theta \in T} X_\theta = \mathbb{E} \sup_{\theta \in T} (X_\theta - X_{\theta_0}) \le \mathbb{E} \sup_{\theta, \widetilde{\theta} \in T} (X_\theta - X_{\widetilde{\theta}}).$$

□

Remark 4.3.5. For every realization of the process,

$$\begin{aligned}
\mathbb{E}\left[\sup_{\theta, \widetilde{\theta} \in T} \big|X_\theta - X_{\widetilde{\theta}}\big|\right] &= \mathbb{E}\left[\sup_{\theta, \widetilde{\theta} \in T} max\{X_\theta - X_{\widetilde{\theta}},\ X_{\widetilde{\theta}} - X_\theta\}\right] \\
&= \mathbb{E}\left[\max\Big\{\sup_{\theta, \widetilde{\theta} \in T} (X_\theta - X_{\widetilde{\theta}}),\ \sup_{\theta, \widetilde{\theta} \in T} (X_{\widetilde{\theta}} - X_\theta)\Big\}\right] \\
&= \mathbb{E}\left[\max\Big\{\sup_{\theta, \widetilde{\theta} \in T} (X_\theta - X_{\widetilde{\theta}}),\ \sup_{\widetilde{\theta}, \theta \in T} (X_{\widetilde{\theta}} - X_\theta)\Big\}\right] \\
&= \mathbb{E}\left[\sup_{\theta, \widetilde{\theta} \in T} \big(X_\theta - X_{\widetilde{\theta}}\big)\right]
\end{aligned}$$

Indeed, interchanging θ and $\widetilde{\theta}$ changes the sign of the increment.

Remark 4.3.6. Fix any $\theta_0 \in T$. By the triangle inequality,

$$\begin{aligned}\mathbb{E}\left[\sup_{\theta\in T}|X_\theta|\right] &= \mathbb{E}\left[\sup_{\theta\in T}|X_\theta - X_{\theta_0} + X_{\theta_0}|\right] \\ &\le \mathbb{E}\left[\sup_{\theta\in T}|X_\theta - X_{\theta_0}|\right] + \mathbb{E}\left[|X_{\theta_0}|\right] \\ &\le \mathbb{E}\left[\sup_{\theta,\widetilde{\theta}\in T}\left|X_\theta - X_{\widetilde{\theta}}\right|\right] + \mathbb{E}\left[|X_{\theta_0}|\right].\end{aligned}$$

Remark 4.3.7. If there exists $\theta_0 \in T$ such that $X_{\theta_0} = 0$ almost surely, then

$$\mathbb{E}\left[\sup_{\theta\in T}|X_\theta|\right] \le \mathbb{E}\left[\sup_{\theta,\widetilde{\theta}\in T}\left|X_\theta - X_{\widetilde{\theta}}\right|\right]$$

4.3.1 Dudley's entropy integral

Let $D = \sup_{\theta,\widetilde{\theta}\in T}\rho_X(\theta,\widetilde{\theta})$ denote the diameter of T. The Dudley entropy integral is

$$J(\delta; D) = \int_\delta^D \sqrt{\log N(u; T, \rho_X)}\, du. \tag{4.8}$$

The term $\log N(u; T, \rho_X)$ is the metric entropy.

Theorem 4.3.8 (Dudley's entropy integral bound)**.** *Let $\{X_\theta : \theta \in T\}$ be a sub-Gaussian process with respect to ρ_X. Then*

$$\mathbb{E}\left[\sup_{\theta,\widetilde{\theta}\in T}(X_\theta - X_{\widetilde{\theta}})\right] \le 8\sqrt{2}J(0; D/2). \tag{4.9}$$

Proof: next class.

4.3.2 Application to Rademacher Complexity

Let $X_1, \ldots, X_n$ be independent with common distribution P, and let $\mathcal{F}$ be a class of functions. Define

$$P_n = \frac{1}{n}\sum_{i=1}^n \delta_{X_i}, \qquad \|P_n - P\|_{\mathcal{F}} = \sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^n f(X_i) - \mathbb{E}_P[f(X)]\right|.$$

Recall the symmetrization inequality:

$$\mathbb{E}\|P_n - P\|_{\mathcal{F}} \le 2R_n(\mathcal{F}) = 2\,\mathbb{E}_X R\big(\mathcal{F}(X_1^n)/n\big), \tag{4.10}$$

where $\mathcal{F}(x_1^n) = \{(f(x_1), \ldots, f(x_n)) : f \in \mathcal{F}\}$ and

$$R\big(\mathcal{F}(x_1^n)/n\big) = \mathbb{E}_\varepsilon \sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^n \varepsilon_i f(x_i)\right|.$$

Example 4.3.9 (Dudley's bound for Rademacher complexity)**.** Fix $x_1^n = (x_1, \ldots, x_n)$ and consider

$$Z_f := \frac{1}{\sqrt{n}}\sum_{i=1}^n \varepsilon_i f(x_i) = \Big\langle \varepsilon, \underbrace{\tfrac{1}{\sqrt{n}}\big(f(x_1), \ldots, f(x_n)\big)}_{=:\,\theta(f)} \Big\rangle. \tag{4.11}$$

By Example 4.3.3, $\{Z_f : f \in \mathcal{F}\}$ is a sub-Gaussian process with respect to

$$\rho_X(f,g) = \|\theta(f) - \theta(g)\|_2 = \left\| \tfrac{1}{\sqrt{n}}\big(f(x_1),\dots,f(x_n)\big) - \tfrac{1}{\sqrt{n}}\big(g(x_1),\dots,g(x_n)\big) \right\|_2,$$

where $\theta(g) = \frac{1}{\sqrt{n}}(g(x_1),\dots,g(x_n))$ for $g \in \mathcal{F}$. In terms of f and g,

$$\rho_X(f,g) = \frac{1}{\sqrt{n}} \left(\sum_{i=1}^n (f(x_i) - g(x_i))^2 \right)^{1/2} = \|f - g\|_{P_n}. \tag{4.12}$$

By Lemma 4.3.4 and Theorem 4.3.8,

$$\begin{aligned} \mathbb{E}_\varepsilon \sup_{f\in\mathcal{F}} Z_f &\le \mathbb{E}_\varepsilon \sup_{f,g\in\mathcal{F}} (Z_f - Z_g) \\ &\le 32 \int_0^\infty \sqrt{\log N(u;\mathcal{F},\|\cdot\|_{P_n})}\, du. \end{aligned} \tag{4.13}$$

Suppose that for any x_1^n, there exists a f_0 such that $f_0(x_1^n) = (0,0,\dots,0)$. Using $|x| = \max\{x,-x\}$, we have

$$\begin{aligned} \mathbb{E}_\epsilon \left[\sup_{f\in\mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \epsilon_i f(x_i) \right| \right] &= \frac{1}{\sqrt{n}} \mathbb{E}_\epsilon \left[\sup_{f\in\mathcal{F}} |Z_f| \right] \\ &= \frac{1}{\sqrt{n}} \mathbb{E}_\epsilon \left[\max \left\{ \sup_{f\in\mathcal{F}} Z_f, \sup_{f\in\mathcal{F}} (-Z_f) \right\} \right] \\ &\le \frac{1}{\sqrt{n}} \left(\mathbb{E}_\epsilon \sup_{f\in\mathcal{F}} Z_f + \mathbb{E}_\epsilon \sup_{f\in\mathcal{F}} (-Z_f) \right) \quad (\text{since } \sup_{f\in\mathcal{F}} Z_f \ge 0) \\ &= \frac{2}{\sqrt{n}} \mathbb{E}_\epsilon \sup_{f\in\mathcal{F}} Z_f \\ &\le \frac{32}{\sqrt{n}} \int_0^D \sqrt{\log N(u;\mathcal{F},\|\cdot\|_{\mathbb{P}_n})}\, du. \end{aligned}$$

Here the equality in the fourth line follows from the symmetry of the Rademacher variables, and the last inequality follows from Dudley's entropy integral bound.

Suppose that $\mathcal{F}$ is b-uniformly bounded. Then

$$\begin{aligned} D &= \sup_{f,g\in\mathcal{F}} \|f - g\|_{\mathbb{P}_n} \\ &= \sup_{f,g\in\mathcal{F}} \left(\frac{1}{n} \sum_{i=1}^n (f(x_i) - g(x_i))^2 \right)^{1/2} \\ &\le \sup_{f,g\in\mathcal{F}} \left(\frac{1}{n} \sum_{i=1}^n 4b^2 \right)^{1/2} = 2b. \end{aligned}$$

Therefore,

$$\mathfrak{R}\left(\frac{\mathcal{F}(x_1^n)}{n} \right) \le \frac{32}{\sqrt{n}} \int_0^{2b} \sqrt{\log N(u;\mathcal{F},\|\cdot\|_{\mathbb{P}_n})}\, du.$$

Since $N(u;\mathcal{F},\|\cdot\|_{\mathbb{P}_n}) = 1$ for $u > 2b$, the integral may equivalently be extended to infinity.

4.3.3 VC Classes

Lemma 4.3.10 (Empirical covering bound for a VC class)**.** *Suppose that $\mathcal{F}$ is VC-class with VC-dimension ν. There is a universal constant $C > 0$ such that, for $0 < u \le b$,*

$$N(u; \mathcal{F}, \|\cdot\|_{P_n}) \le C\nu(16e)^{\nu} \left(\frac{b}{u}\right)^{2\nu}. \tag{4.14}$$

Proof. Homework.

Example 4.3.11 (Rademacher complexity of a bounded VC class)**.** Let $\mathcal{F}$ be b-uniformly bounded with VC-dimension ν. By Example 4.3.9 and the preceding lemma,

$$\begin{aligned}
\mathfrak{R}\left(\frac{\mathcal{F}(x_1^n)}{n}\right) &\le \frac{32}{\sqrt{n}} \int_0^{2b} \sqrt{\log N\left(u; \mathcal{F}, \|\cdot\|_{\mathbb{P}_n}\right)}\, du \\
&\le \frac{32}{\sqrt{n}} \int_0^{2b} \sqrt{\log\left[C\nu(16e)^{\nu}\left(\frac{b}{u}\right)^{2\nu}\right]}\, du \\
&= \frac{32}{\sqrt{n}} \int_0^{2b} \sqrt{\log(C\nu) + \nu\log(16e) + 2\nu\log\left(\frac{b}{u}\right)}\, du \\
&\le \frac{64\sqrt{\nu}}{\sqrt{n}} C_0 \left(1 + \int_0^{2b} \sqrt{\log\left(\frac{b}{u}\right)}\, du\right) \\
&\le C_0' \sqrt{\frac{\nu}{n}},
\end{aligned}$$

where C_0' is independent of n and ν (and may depend on the uniform bound b). The integral is finite, so taking expectation over the sample gives

$$\mathfrak{R}_n(\mathcal{F}) := \mathbb{E}_{X_1^n} \mathfrak{R}\left(\frac{\mathcal{F}(X_1^n)}{n}\right) \le C_0' \sqrt{\frac{\nu}{n}}.$$

Recall from Chapter 4 that polynomial discrimination gave the earlier bound

$$\mathfrak{R}_n(\mathcal{F}) \le C \sqrt{\frac{\nu \log(n+1)}{n}}.$$

Thus, the entropy-integral argument removes the unnecessary factor $\sqrt{\log(n+1)}$.

One may use this result to Classical Glivenko-Cantelli Theorem; see Example 5.24 [1] for more details.

Example 4.3.12. Consider the function class

$$\mathcal{F} := \left\{\mathbf{1}_{(-\infty,t]}(\,\cdot\,) : t \in \mathbb{R}\right\}.$$

The identically zero function is not an element of $\mathcal{F}$. However, for a fixed sample $x_1^n = (x_1, \ldots, x_n)$, choose

$$t_0 < \min_{1 \le i \le n} \{x_i\} \qquad \text{and set} \qquad f_0 := \mathbf{1}_{(-\infty,t_0]}(\,\cdot\,).$$

Then $f_0 \in \mathcal{F}$ and

$$f_0(x_1) = f_0(x_2) = \cdots = f_0(x_n) = 0.$$

4.4 Finite version of Dudley's entropy integral

We finish by proving the finite-set version that motivates Dudley's entropy integral. The central idea is to approximate every point by a sequence of representatives at dyadic resolutions and then telescope along this chain.

Theorem 4.4.1 (Finite-set chaining bound)**.** *Let $\{X_\theta \mid \theta \in T\}$ be a sub-Gaussian process with respect to ρ_X , and define*

$$D := \sup_{\theta,\widetilde{\theta}\in T} \rho_X(\theta,\widetilde{\theta}) < \infty.$$

let $U \in T$ be any subset with $|U| < \infty$, define

$$L := \min\left\{i \geq 1 : N(D2^{-i}; T, \rho_X) \geq |U|\right\}.$$

Then

$$\mathbb{E}\left[\sup_{\theta,\widetilde{\theta}\in U} \left|X_\theta - X_{\widetilde{\theta}}\right|\right] \leq 8\sqrt{2}\int_{D2^{-(L+1)}}^{D/2} \sqrt{\log N(u; T, \rho_X)}\, du. \tag{4.15}$$

Proof. Write $U = \{\theta^1, \ldots, \theta^N\}$, where $N = |U|$, and set $\epsilon_m = D2^{-m}$. If $N = 1$, the result is immediate, so assume $N \geq 2$. In particular, $D > 0$.

Since (T, ρ_X) is a metric space and U is finite, the minimum distance between distinct points of U is positive. A ball of radius less than half this minimum contains at most one point of U. Consequently,

$$L = \min\left\{m \geq 1 : N(\epsilon_m; T, \rho_X) \geq N\right\}$$

is finite.

Construction of the covering sets and chains. For $m = 0, 1, \ldots, L-1$, let U_m be a minimal ϵ_m-cover of T, with centers in T, and set $U_L = U$. Then

$$|U_m| = N(\epsilon_m; T, \rho_X) \quad (0 \leq m < L), \qquad |U_L| = |U| \leq N(\epsilon_L; T, \rho_X).$$

For $1 \leq m < L$, the definition of L gives $|U_m| < N$, so all these sets are finite. Since $\epsilon_0 = D$, we may take $U_0 = \{\theta_0\}$ for some $\theta_0 \in T$. Here covering balls are understood to be closed.

For each $m = 0, \ldots, L-1$, fix a nearest-point map $\pi_m : T \to U_m$, with ties resolved deterministically, such that $\pi_m(\alpha) \in \arg\min_{\beta\in U_m} \rho_X(\alpha, \beta)$. Because U_m covers T, we have $\rho_X(\alpha, \pi_m(\alpha)) \leq \epsilon_m$ for every $\alpha \in T$.

Fix $\theta \in U$, and define its chain backward by

$$\gamma^L = \theta, \qquad \gamma^{m-1} = \pi_{m-1}(\gamma^m), \qquad m = L, L-1, \ldots, 1.$$

Thus $\gamma^m \in U_m$, $\gamma^0 = \theta_0$, and $\rho_X(\gamma^m, \gamma^{m-1}) \leq \epsilon_{m-1}$. The increments telescope:

$$X_\theta - X_{\theta_0} = \sum_{m=1}^{L} \left(X_{\gamma^m} - X_{\gamma^{m-1}}\right).$$

The deterministic chaining inequality. For $m = 1, \ldots, L$ and $\gamma \in U_m$, define

$$Z_{m,\gamma} = X_\gamma - X_{\pi_{m-1}(\gamma)}, \qquad R_m = \max_{\gamma\in U_m} Z_{m,\gamma} - \min_{\gamma\in U_m} Z_{m,\gamma}.$$

The recursive definition of the chain gives $Z_{m,\gamma^m} = X_{\gamma^m} - X_{\gamma^{m-1}}$.

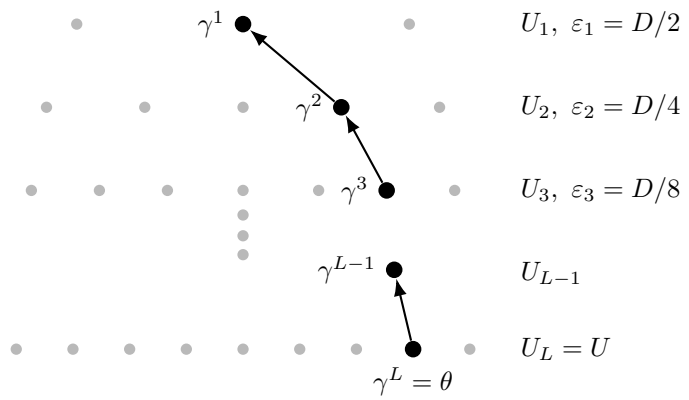


Figure 4.5: The multiscale covering sets and one backward chain from the target $\theta \in U$ to the coarsest displayed level. Each arrow is an increment controlled at the corresponding scale.

For another point $\tilde{\theta} \in U$, construct $\tilde{\gamma}^L, \dots, \tilde{\gamma}^0$ in the same way. Both chains end at θ_0, so subtracting their telescoping identities yields

$$X_\theta - X_{\tilde{\theta}} = \sum_{m=1}^{L} \big(Z_{m,\gamma^m} - Z_{m,\tilde{\gamma}^m} \big).$$

At each level m, both γ^m and $\tilde{\gamma}^m$ belong to U_m. Therefore, for every realization of the process, both increments lie between $\min_{\gamma \in U_m} Z_{m,\gamma}$ and $\max_{\gamma \in U_m} Z_{m,\gamma}$, and hence

$$\big| Z_{m,\gamma^m} - Z_{m,\tilde{\gamma}^m} \big| \le \max_{\gamma \in U_m} Z_{m,\gamma} - \min_{\gamma \in U_m} Z_{m,\gamma} = R_m.$$

By the triangle inequality,

$$\begin{aligned} |X_\theta - X_{\tilde{\theta}}| &\le \sum_{m=1}^{L} \big| Z_{m,\gamma^m} - Z_{m,\tilde{\gamma}^m} \big| \\ &\le \sum_{m=1}^{L} R_m. \end{aligned}$$

The last expression is independent of $\theta, \tilde{\theta}$. Taking the maximum over these two indices therefore gives the pointwise bound

$$\max_{\theta, \tilde{\theta} \in U} |X_\theta - X_{\tilde{\theta}}| \le \sum_{m=1}^{L} \left(\max_{\gamma \in U_m} Z_{m,\gamma} - \min_{\gamma \in U_m} Z_{m,\gamma} \right).$$

Notice that π_0 is constant, so R_1 is precisely the range of $\{X_\gamma : \gamma \in U_1\}$. The covering sets need not be nested, because the parent maps connect consecutive levels.

Application of the maximal inequality. By the sub-Gaussian increment assumption, for every $\gamma \in U_m$, since

$$\rho_X(\gamma, \pi_{m-1}(\gamma)) \le \epsilon_{m-1},$$

$Z_{m,\gamma}$ is sub-Gaussian with parameter ϵ_{m-1}. Apply the known maximal inequality separately to $\{Z_{m,\gamma} : \gamma \in U_m\}$ and $\{-Z_{m,\gamma} : \gamma \in U_m\}$. This gives

$$\begin{aligned} \mathbb{E} R_m &= \mathbb{E} \max_{\gamma \in U_m} Z_{m,\gamma} + \mathbb{E} \max_{\gamma \in U_m} (-Z_{m,\gamma}) \\ &\le 2\epsilon_{m-1} \sqrt{2 \log |U_m|}. \end{aligned}$$

When $|U_m| = 1$, the same inequality holds because $R_m = 0$. Taking expectations in the deterministic chaining inequality yields

$$\mathbb{E} \max_{\theta,\tilde{\theta}\in U} |X_\theta - X_{\tilde{\theta}}| \le 2\sqrt{2} \sum_{m=1}^{L} \epsilon_{m-1} \sqrt{\log |U_m|}.$$

Comparison with the entropy integral. For every $u \in [\epsilon_{m+1}, \epsilon_m]$, monotonicity of covering numbers and the cardinality bounds above imply

$$N(u; T, \rho_X) \ge N(\epsilon_m; T, \rho_X) \ge |U_m|.$$

Since $\epsilon_m - \epsilon_{m+1} = \epsilon_{m-1}/4$, we obtain

$$\begin{aligned} \int_{\epsilon_{m+1}}^{\epsilon_m} \sqrt{\log N(u; T, \rho_X)}\, du &\ge (\epsilon_m - \epsilon_{m+1}) \sqrt{\log |U_m|} \\ &= \frac{\epsilon_{m-1}}{4} \sqrt{\log |U_m|}. \end{aligned}$$

Consequently,

$$\begin{aligned} \mathbb{E} \max_{\theta,\tilde{\theta}\in U} |X_\theta - X_{\tilde{\theta}}| &\le 8\sqrt{2} \sum_{m=1}^{L} \int_{\epsilon_{m+1}}^{\epsilon_m} \sqrt{\log N(u; T, \rho_X)}\, du \\ &= 8\sqrt{2} \int_{\epsilon_{L+1}}^{\epsilon_1} \sqrt{\log N(u; T, \rho_X)}\, du \end{aligned}$$

□

4.5 Dudley's Entropy Integral Bound

Recall that the expected supremum is defined by

$$\mathbb{E} \sup_{t\in T} X_t := \sup \left\{ \mathbb{E} \left[\sup_{t\in U} X_t \right] \;\middle|\; U \subseteq T,\ 0 < |U| < \infty \right\}.$$

Proof. This is Homework Problem.

Lemma 4.5.1 (Expected supremum over finite subsets)**.** *Let T be a nonempty index set, and let $\{X_t : t \in T\}$ be a family of integrable real-valued random variables on a common probability space. Define*

$$\mathbb{E} \left[\sup_{s,t\in T} (X_s - X_t) \right] := \sup_{\substack{F\subseteq T\times T \\ 0<|F|<\infty}} \mathbb{E} \left[\max_{(s,t)\in F} (X_s - X_t) \right].$$

Then

$$\mathbb{E} \left[\sup_{s,t\in T} (X_s - X_t) \right] = \sup_{\substack{U\subseteq T \\ 0<|U|<\infty}} \mathbb{E} \left[\max_{s,t\in U} (X_s - X_t) \right].$$

The equality is understood in the extended-real sense, so the common value may be $+\infty$.

Theorem 4.5.2 (Dudley's entropy integral bound)**.** *Let $\{X_\theta \mid \theta \in T\}$ be a sub-Gaussian process with respect to ρ_X, and suppose that*

$$D := \sup_{\theta,\widetilde{\theta}\in T} \rho_X(\theta, \widetilde{\theta}) < \infty.$$

Then

$$\mathbb{E} \left[\sup_{\theta,\widetilde{\theta}\in T} \left(X_\theta - X_{\widetilde{\theta}} \right) \right] \le 8\sqrt{2} J(0; D).$$

Proof. By Lemma 4.5.1 and Theorem 4.4.1,

$$\begin{aligned}\mathbb{E}\left[\sup_{\theta,\widetilde{\theta}\in T}\left(X_\theta - X_{\widetilde{\theta}}\right)\right] &= \sup_{\substack{U\subseteq T\\ 0<|U|<\infty}} \mathbb{E}\left[\sup_{\theta,\widetilde{\theta}\in U}\left(X_\theta - X_{\widetilde{\theta}}\right)\right] \\ &\leq \sup_{\substack{U\subseteq T\\ 0<|U|<\infty}} 8\sqrt{2}J(0;D) \\ &= 8\sqrt{2}J(0;D).\end{aligned}$$

□

Remark 4.5.3. The conclusion of Theorem 4.5.2 can equivalently be written as

$$\mathbb{E}\left[\sup_{\theta,\widetilde{\theta}\in T}\left|X_\theta - X_{\widetilde{\theta}}\right|\right] \leq 8\sqrt{2}J(0;D).$$

Some other equivalent forms are:

$$\mathbb{E}\left[\sup_{\theta\in T}(X_\theta - X_{\theta_0})\right] \leq 8\sqrt{2}J(0;D).$$

and

$$\mathbb{E}\left[\sup_{\theta\in T}|X_\theta - X_{\theta_0}|\right] \leq 8\sqrt{2}J(0;D).$$

Chapter 5

Random matrices and Covariance estimation

5.1 Introduction

A random matrix has random variables as its entries. Write

$$\mathbf{X} = \begin{pmatrix} x_{11} & \cdots & x_{1d} \\ \vdots & & \vdots \\ x_{n1} & \cdots & x_{nd} \end{pmatrix} = \begin{pmatrix} x_1^{\mathrm{T}} \\ \vdots \\ x_n^{\mathrm{T}} \end{pmatrix} \in \mathbb{R}^{n \times d}, \qquad x_i = (x_{i1}, \ldots, x_{id})^{\mathrm{T}}.$$

For covariance estimation, suppose $x_1, \ldots, x_n$ are i.i.d. with mean zero. Their common covariance matrix is

$$\begin{aligned} \operatorname{cov}(x_1) &= \mathbb{E}\big[(x_1 - \mathbb{E}[x_1])(x_1 - \mathbb{E}[x_1])^{\mathrm{T}}\big] \\ &= \mathbb{E}[x_1 x_1^{\mathrm{T}}] - \mathbb{E}[x_1]\big(\mathbb{E}[x_1]\big)^{\mathrm{T}} = \mathbb{E}[x_1 x_1^{\mathrm{T}}]. \end{aligned}$$

The matrix product gives

$$\begin{aligned} \frac{1}{n}\mathbf{X}^{\mathrm{T}}\mathbf{X} &= \frac{1}{n}\begin{pmatrix} x_1 & \cdots & x_n \end{pmatrix}\begin{pmatrix} x_1^{\mathrm{T}} \\ \vdots \\ x_n^{\mathrm{T}} \end{pmatrix} \\ &= \frac{1}{n}\big(x_1 x_1^{\mathrm{T}} + \cdots + x_n x_n^{\mathrm{T}}\big) = \frac{1}{n}\sum_{i=1}^{n} x_i x_i^{\mathrm{T}}. \end{aligned}$$

This matrix estimates $\operatorname{cov}(x_1)$. We consider the regimes $d < n$, $d \asymp n$, and $d \gg n$.

Asymptotic results

Wigner semicircle law. Suppose $\mathbf{W}_n = (w_{ij})_{i,j=1}^n$ is real symmetric ($w_{ji} = w_{ij}$), with independent upper-triangular entries of mean zero and unit variance. Let $\lambda_i(\mathbf{A}_n)$ be the ith eigenvalue of the normalized matrix $\mathbf{A}_n = \mathbf{W}_n/\sqrt{n}$. The empirical distribution of these eigenvalues is

$$\mu_n = \frac{1}{n}\sum_{i=1}^{n} \delta_{\lambda_i(\mathbf{A}_n)},$$

which places mass $1/n$ at each eigenvalue. Under the hypotheses of the semicircle law, $\mu_n \Rightarrow \mu_{\mathrm{sc}}$ as $n \to \infty$, where μ_{sc} has density

$$f_{\mathrm{sc}}(x) = \frac{\sqrt{4-x^2}}{2\pi}\,\mathbb{I}[|x| < 2].$$

Although there are $n(n+1)/2$ independent entries, the eigenvalues of $\mathbf{W}_n$ are on the scale of $\sqrt{n}$. The trace of $\mathbf{W}_n^2$ gives the intuition.

$$\begin{aligned}\operatorname{tr}(\mathbf{W}_n^2) &= \sum_{i=1}^n \sum_{j=1}^n w_{ij} w_{ji} = \sum_{i=1}^n \sum_{j=1}^n w_{ij}^2 \\ &= \sum_{k=1}^n \lambda_k(\mathbf{W}_n^2) = \sum_{k=1}^n \big(\lambda_k(\mathbf{W}_n)\big)^2.\end{aligned}$$

Since $\mathbb{E}[w_{ij}] = 0$, we have $\operatorname{var}(w_{ij}) = \mathbb{E}[w_{ij}^2] - (\mathbb{E}[w_{ij}])^2 = \mathbb{E}[w_{ij}^2] = 1$, so

$$\mathbb{E}\left[\sum_{i=1}^n \sum_{j=1}^n w_{ij}^2\right] = n^2.$$

This suggests $\sum_{k=1}^n (\lambda_k(\mathbf{W}_n))^2 \approx n^2$, so $|\lambda_k(\mathbf{W}_n)| \approx \sqrt{n}$ if the eigenvalues have comparable magnitudes.

Remark. The eigenvalues are dependent. The eigenvectors are delocalized, meaning that their components are spread across the coordinates.

Marčenko–Pastur law. Suppose $\mathbf{X} \in \mathbb{R}^{n \times d}$ has rows $x_1^{\mathrm{T}}, \ldots, x_n^{\mathrm{T}}$, with $\mathbb{E}[x_{ij}] = 0$ and $\operatorname{var}(x_{ij}) = 1$. The sample covariance matrix is

$$\widehat{\mathbf{\Sigma}} = \frac{1}{n}\mathbf{X}^{\mathrm{T}}\mathbf{X} \in \mathbb{R}^{d \times d}.$$

Assume $d, n \to \infty$ with $d/n \to \gamma \in (0, \infty)$. In the Marčenko–Pastur setting, the empirical distribution of the eigenvalues of $\widehat{\mathbf{\Sigma}}$ converges as follows.

$$\mu_n = \frac{1}{n}\sum_{i=1}^d \delta_{\lambda_i(\widehat{\mathbf{\Sigma}})} \;\Rightarrow\; \mu_{\mathrm{MP},\gamma}.$$

The density of the continuous part of $\mu_{\mathrm{MP},\gamma}$ is

$$f_\gamma(x) = \frac{\sqrt{(b-x)(x-a)}}{2\pi\gamma x}\,\mathbb{I}[a \le x \le b], \qquad a = (1-\sqrt{\gamma})^2, \qquad b = (1+\sqrt{\gamma})^2.$$

If $\gamma > 1$, the distribution also has a point mass at zero of size $1 - 1/\gamma$, and $\int_a^b f_\gamma(x)\,dx = 1/\gamma$.

When $\gamma > 1$, we eventually have $d > n$, so $\operatorname{rank}(\mathbf{X}^{\mathrm{T}}\mathbf{X}) = \operatorname{rank}(\mathbf{X}) \le n < d$. Consequently, $\mathbf{X}^{\mathrm{T}}\mathbf{X}$ has at least $d - n$ zero eigenvalues, and

$$\frac{d-n}{d} = 1 - \frac{n}{d} \;\longrightarrow\; 1 - \frac{1}{\gamma}.$$

For example, when $\gamma = 1$, we have $a = 0$ and $b = 4$. Even when $\mathbf{\Sigma} = \mathbf{I}_d$, the sample eigenvalues can range asymptotically from 0 to 4.

Under suitable moment assumptions, when $0 < \gamma \le 1$, we have

$$\lambda_{\min}(\widehat{\mathbf{\Sigma}}) \longrightarrow (1-\sqrt{\gamma})^2, \qquad \lambda_{\max}(\widehat{\mathbf{\Sigma}}) \longrightarrow (1+\sqrt{\gamma})^2.$$

Thus, in the identity-covariance case, $\|\widehat{\mathbf{\Sigma}} - \mathbf{I}_d\|_2$ fails to converge to zero in probability. In particular, $v^{\mathrm{T}}(\widehat{\mathbf{\Sigma}} - \mathbf{I}_d)v$ fails to converge to zero in probability when v is a unit eigenvector corresponding to $\lambda_{\max}(\widehat{\mathbf{\Sigma}})$.

We seek non-asymptotic probability bounds for finite n and d. We start with Gaussian random matrices, then consider sub-Gaussian ensembles and ensembles with milder tail conditions.

5.2 Some preliminaries

6.1.1 Notation and basic facts

Given a square matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$, an eigenvalue λ and a corresponding nonzero eigenvector v satisfy $\mathbf{A}v = \lambda v$. If $\mathbf{A}$ is real and symmetric, it has the decomposition

$$\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{\mathrm{T}} = \begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix} \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \begin{pmatrix} v_1^{\mathrm{T}} \\ \vdots \\ v_n^{\mathrm{T}} \end{pmatrix}.$$

For $\mathbf{A} \in \mathbb{R}^{n\times m}$ with $n \geq m$, the ith singular value is the square root of the ith eigenvalue of $\mathbf{A}^{\mathrm{T}}\mathbf{A}$. Thus,

$$\sigma_i(\mathbf{A}) = \sqrt{\lambda_i(\mathbf{A}^{\mathrm{T}}\mathbf{A})}.$$

The singular values are nonnegative. The singular value decomposition is $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathrm{T}}$, with

$$\mathbf{A}v_j = \sigma_j(\mathbf{A})u_j, \qquad \mathbf{A}^{\mathrm{T}}u_j = \sigma_j(\mathbf{A})v_j.$$

We order the singular values as

$$\sigma_{\max}(\mathbf{A}) = \sigma_1(\mathbf{A}) \geq \sigma_2(\mathbf{A}) \geq \cdots \geq \sigma_m(\mathbf{A}) = \sigma_{\min}(\mathbf{A}) \geq 0.$$

The largest and smallest singular values have the variational characterizations

$$\sigma_{\max}(\mathbf{A}) = \max_{v\in\mathbb{S}^{d-1}} \|\mathbf{A}v\|_2, \qquad \sigma_{\min}(\mathbf{A}) = \min_{v\in\mathbb{S}^{d-1}} \|\mathbf{A}v\|_2. \tag{6.1}$$

Here $\mathbb{S}^{d-1} \coloneqq \{v \in \mathbb{R}^d \mid \|v\|_2 = 1\}$ is the Euclidean unit sphere in $\mathbb{R}^d$, and $\|\mathbf{A}v\|_2^2 = v^{\mathrm{T}}\mathbf{A}^{\mathrm{T}}\mathbf{A}v$. The matrix ℓ_2-operator norm, also called the spectral norm, is

$$|||\mathbf{A}|||_2 \coloneqq \sup_{\|v\|_2=1} \|\mathbf{A}v\|_2 = \sigma_{\max}(\mathbf{A}).$$

Other texts also write this norm as $\|\mathbf{A}\|_{\mathrm{op}}$ or $\|\mathbf{A}\|_2$.

Since covariance matrices are symmetric, we focus on the set $\mathcal{S}^{d\times d} \coloneqq \{\mathbf{Q} \in \mathbb{R}^{d\times d} \mid \mathbf{Q} = \mathbf{Q}^{\mathrm{T}}\}$ of symmetric matrices and the subset of positive semidefinite matrices

$$\mathcal{S}_+^{d\times d} \coloneqq \{\mathbf{Q} \in \mathcal{S}^{d\times d} \mid \mathbf{Q} \succeq 0\}. \tag{6.2}$$

Any $\mathbf{Q} \in \mathcal{S}^{d\times d}$ is diagonalizable by an orthogonal transformation. In particular, $\mathbf{Q} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^{\mathrm{T}}$ with $\mathbf{P}^{\mathrm{T}}\mathbf{P} = \mathbf{P}\mathbf{P}^{\mathrm{T}} = \mathbf{I}_d$. We denote its vector of eigenvalues by $\gamma(\mathbf{Q}) \in \mathbb{R}^d$, ordered as

$$\gamma_{\max}(\mathbf{Q}) = \gamma_1(\mathbf{Q}) \geq \gamma_2(\mathbf{Q}) \geq \cdots \geq \gamma_d(\mathbf{Q}) = \gamma_{\min}(\mathbf{Q}).$$

The matrix $\mathbf{Q}$ is positive semidefinite, written $\mathbf{Q} \succeq 0$, if and only if $\gamma_{\min}(\mathbf{Q}) \geq 0$.

The Rayleigh–Ritz variational characterization gives

$$\gamma_{\max}(\mathbf{Q}) = \max_{v\in\mathbb{S}^{d-1}} v^{\mathrm{T}}\mathbf{Q}v, \qquad \gamma_{\min}(\mathbf{Q}) = \min_{v\in\mathbb{S}^{d-1}} v^{\mathrm{T}}\mathbf{Q}v. \tag{6.3}$$

For symmetric $\mathbf{Q}$, $\mathbf{Q}^{\mathrm{T}}\mathbf{Q} = \mathbf{Q}^2$ and

$$\mathbf{Q}^2 = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^{\mathrm{T}}\mathbf{P}\mathbf{\Lambda}\mathbf{P}^{\mathrm{T}} = \mathbf{P}\mathbf{\Lambda}^2\mathbf{P}^{\mathrm{T}}.$$

Consequently, the operator norm can be written as

$$\begin{aligned} |||\mathbf{Q}|||_2 &= \max_{v \in \mathbb{S}^{d-1}} \|\mathbf{Q}v\|_2 = \max_{v \in \mathbb{S}^{d-1}} \left(v^{\mathrm{T}} \mathbf{Q}^{\mathrm{T}} \mathbf{Q} v\right)^{1/2} \\ &= \left[\gamma_{\max}(\mathbf{Q}^2)\right]^{1/2} = \max\left\{\gamma_{\max}(\mathbf{Q}), |\gamma_{\min}(\mathbf{Q})|\right\}. \end{aligned} \tag{6.4a}$$

It follows that

$$|||\mathbf{Q}|||_2 = \max_{v \in \mathbb{S}^{d-1}} \left|v^{\mathrm{T}} \mathbf{Q} v\right|. \tag{6.4b}$$

Finally, for $\mathbf{A} \in \mathbb{R}^{n \times m}$ with $n \geq m$, let m- dimensional symmetric $\mathbf{R} := \mathbf{A}^{\mathrm{T}}\mathbf{A}$. Then we have relationship

$$\gamma_j(\mathbf{R}) = \left(\sigma_j(\mathbf{A})\right)^2, \qquad j = 1, \ldots, m.$$

6.1.2 Set-up of covariance estimation

Let $\{x_1, \ldots, x_n\}$ be n i.i.d. samples from a distribution in $\mathbb{R}^d$ with mean zero. We use a lowercase x_i for a random vector and uppercase $\mathbf{X}$ for the random matrix. The covariance is

$$\mathbf{\Sigma} = \mathrm{cov}(x_1) = \mathbb{E}\left[(x_1 - \mathbb{E}[x_1])(x_1 - \mathbb{E}[x_1])^{\mathrm{T}}\right].$$

The sample covariance matrix estimates $\mathbf{\Sigma}$ and is given by

$$\widehat{\mathbf{\Sigma}} := \frac{1}{n} \sum_{i=1}^{n} x_i x_i^{\mathrm{T}}. \tag{6.5}$$

Since each x_i has mean zero, $\mathbb{E}[x_i x_i^{\mathrm{T}}] = \mathbf{\Sigma}$ and $\mathbb{E}[\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma}] = 0$. Thus $\widehat{\mathbf{\Sigma}}$ is unbiased.

Our goal is to bound the estimation error in the ℓ_2-operator norm. The sample quadratic form is

$$v^{\mathrm{T}} \widehat{\mathbf{\Sigma}} v = v^{\mathrm{T}} \left(\frac{1}{n} \sum_{i=1}^{n} x_i x_i^{\mathrm{T}}\right) v = \frac{1}{n} \sum_{i=1}^{n} (v^{\mathrm{T}} x_i)(x_i^{\mathrm{T}} v) = \frac{1}{n} \sum_{i=1}^{n} \langle x_i, v \rangle^2.$$

By (6.4b), the bound $|||\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma}|||_2 \leq \epsilon$ is equivalent to

$$\max_{v \in \mathbb{S}^{d-1}} \left| \frac{1}{n} \sum_{i=1}^{n} \langle x_i, v \rangle^2 - v^{\mathrm{T}} \mathbf{\Sigma} v \right| \leq \epsilon. \tag{6.6}$$

Since $v^{\mathrm{T}} \mathbf{\Sigma} v = \mathbb{E}[\langle x_i, v \rangle^2]$, this is also equivalent to

$$\max_{v \in \mathbb{S}^{d-1}} \left| \frac{1}{n} \sum_{i=1}^{n} \left(\langle x_i, v \rangle^2 - \mathbb{E}[\langle x_i, v \rangle^2]\right) \right| \leq \epsilon.$$

Thus controlling the operator norm error amounts to establishing a uniform law of large numbers for $f_v(x) = \langle x, v \rangle^2$, indexed by $v \in \mathbb{S}^{d-1}$ (see Chapter 4).

Control in operator norm also controls the eigenvalue errors uniformly. By Weyl's theorem,

$$\max_{j=1,\ldots,d} \left|\gamma_j(\widehat{\mathbf{\Sigma}}) - \gamma_j(\mathbf{\Sigma})\right| \leq |||\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma}|||_2. \tag{6.7}$$

An analogous eigenvector guarantee requires control on gaps between adjacent eigenvalues, for example $\gamma_j(\mathbf{\Sigma}) > \gamma_{j+1}(\mathbf{\Sigma}) + \Delta$ for a positive gap Δ (see Chapter 8).

Finally, let $\{\sigma_j(\mathbf{X})\}_{j=1}^{\min\{n,d\}}$ denote the singular values of the random matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ with rows x_i^{T}. Then

$$\widehat{\mathbf{\Sigma}} = \frac{1}{n} \sum_{i=1}^{n} x_i x_i^{\mathrm{T}} = \frac{1}{n} \mathbf{X}^{\mathrm{T}} \mathbf{X} = \left(\frac{\mathbf{X}}{\sqrt{n}}\right)^{\mathrm{T}} \left(\frac{\mathbf{X}}{\sqrt{n}}\right).$$

The eigenvalues of $\widehat{\mathbf{\Sigma}}$ are the squared singular values of $\mathbf{X}/\sqrt{n}$, padded with zeros if $n < d$.

5.3 Wishart matrices and their behavior

Suppose $x_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \mathbf{\Sigma})$ for $i = 1, \dots, n$. We say that $\mathbf{X} \in \mathbb{R}^{n \times d}$, with ith row x_i^{T}, is drawn from the $\mathbf{\Sigma}$-Gaussian ensemble. For the sample covariance $\widehat{\mathbf{\Sigma}} = \mathbf{X}^{\mathrm{T}}\mathbf{X}/n$, the Wishart relation is

$$n\widehat{\mathbf{\Sigma}} \sim W_d(n, \mathbf{\Sigma}).$$

When $d = 1$ and the variance is σ^2, this gives $W_1(n, \sigma^2)$.

Theorem 5.3.1. *Let $\mathbf{X} \in \mathbb{R}^{n \times d}$ be drawn from the $\mathbf{\Sigma}$-Gaussian ensemble, with $\mathbf{\Sigma} \neq 0$. For all $\delta > 0$, the maximum singular value satisfies*

$$\mathbb{P}\left[\frac{\sigma_{\max}(\mathbf{X})}{\sqrt{n}} \geq \gamma_{\max}(\sqrt{\mathbf{\Sigma}})(1+\delta) + \sqrt{\frac{\operatorname{tr}(\mathbf{\Sigma})}{n}}\right] \leq e^{-n\delta^2/2}. \tag{6.8}$$

For $n \geq d$, the minimum singular value satisfies

$$\mathbb{P}\left[\frac{\sigma_{\min}(\mathbf{X})}{\sqrt{n}} \leq \gamma_{\min}(\sqrt{\mathbf{\Sigma}})(1-\delta) - \sqrt{\frac{\operatorname{tr}(\mathbf{\Sigma})}{n}}\right] \leq e^{-n\delta^2/2}. \tag{6.9}$$

For $n < d$, the variational minimum $\min_{v \in \mathbb{S}^{d-1}} \|\mathbf{X}v\|_2$ is zero.

Example 5.3.2 (Operator norm bounds for the standard Gaussian ensemble)**.** Let $\mathbf{W} \in \mathbb{R}^{n \times d}$ have rows w_i^{T}, where $w_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \mathbf{I}_d)$. Suppose $n \geq d$. Let A and B be the events in the brackets of (6.8) and (6.9) for $\mathbf{X} = \mathbf{W}$ and $\mathbf{\Sigma} = \mathbf{I}_d$. On $A^c \cap B^c$ we have

$$\frac{\sigma_{\max}(\mathbf{W})}{\sqrt{n}} \leq 1 + \delta + \sqrt{\frac{d}{n}}, \qquad \frac{\sigma_{\min}(\mathbf{W})}{\sqrt{n}} \geq 1 - \delta - \sqrt{\frac{d}{n}}. \tag{6.10}$$

By Theorem 5.3.1, both $\mathbb{P}(A^c)$ and $\mathbb{P}(B^c)$ are at least $1 - e^{-n\delta^2/2}$. Hence

$$\begin{aligned}\mathbb{P}(A^c \cap B^c) &= \mathbb{P}(A^c) + \mathbb{P}(B^c) - \mathbb{P}(A^c \cup B^c) \\ &\geq \mathbb{P}(A^c) + \mathbb{P}(B^c) - 1 \geq 1 - 2e^{-n\delta^2/2}.\end{aligned}$$

The bounds in (6.10) imply

$$\left\| \frac{1}{n}\mathbf{W}^{\mathrm{T}}\mathbf{W} - \mathbf{I}_d \right\|_2 \leq 2\epsilon + \epsilon^2, \qquad \epsilon = \sqrt{\frac{d}{n}} + \delta, \tag{6.11}$$

with the same probability lower bound.

To see this, write $M = \sigma_{\max}(\mathbf{W})/\sqrt{n}$ and $m = \sigma_{\min}(\mathbf{W})/\sqrt{n}$. Then $-\epsilon \leq m - 1 \leq M - 1 \leq \epsilon$, so each singular value satisfies

$$1 - \epsilon \leq \frac{\sigma_j(\mathbf{W})}{\sqrt{n}} \leq 1 + \epsilon.$$

Using nonnegativity of the singular values, we can square the bounds as

$$\big(\max\{0, 1-\epsilon\}\big)^2 \leq \left(\frac{\sigma_j(\mathbf{W})}{\sqrt{n}}\right)^2 \leq (1+\epsilon)^2. \tag{*}$$

The eigenvalues of $\mathbf{W}^{\mathrm{T}}\mathbf{W}/n$ are precisely these squared singular values. If $\mathbf{W}^{\mathrm{T}}\mathbf{W}/n = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{\mathrm{T}}$, then

$$\frac{1}{n}\mathbf{W}^{\mathrm{T}}\mathbf{W} - \mathbf{I}_d = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{\mathrm{T}} - \mathbf{V}\mathbf{I}_d\mathbf{V}^{\mathrm{T}} = \mathbf{V}(\mathbf{\Lambda} - \mathbf{I}_d)\mathbf{V}^{\mathrm{T}}.$$

Thus subtracting $\mathbf{I}_d$ shifts each eigenvalue by -1, so

$$\begin{aligned}\left\|\frac{1}{n}\mathbf{W}^{\mathrm{T}}\mathbf{W}-\mathbf{I}_d\right\|_2 &= \max_{1\le j\le d}\left|\gamma_j\left(\frac{1}{n}\mathbf{W}^{\mathrm{T}}\mathbf{W}-\mathbf{I}_d\right)\right| \\ &= \max_{1\le j\le d}\left|\gamma_j\left(\frac{1}{n}\mathbf{W}^{\mathrm{T}}\mathbf{W}\right)-1\right| \\ &= \max_{1\le j\le d}\left|\left(\frac{\sigma_j(\mathbf{W})}{\sqrt{n}}\right)^2-1\right| \\ &\le (1+\epsilon)^2-1 = 2\epsilon+\epsilon^2.\end{aligned}$$

The inequality follows from (*), since $1-(\max\{0,1-\epsilon\})^2 \le (1+\epsilon)^2-1$.

If $d/n \to 0$, we can choose $\delta_n \to 0$ with $n\delta_n^2 \to \infty$, for instance $\delta_n = \sqrt{\log(n)/n}$. Then $\epsilon \to 0$ and the failure probability bound $2e^{-n\delta_n^2/2}$ tends to zero. Thus $\widehat{\mathbf{\Sigma}} = \mathbf{W}^{\mathrm{T}}\mathbf{W}/n$ is consistent in operator norm, with

$$\|\widehat{\mathbf{\Sigma}} - \mathbf{I}_d\|_2 \xrightarrow{\mathbb{P}} 0.$$

Example 5.3.3 (Gaussian covariance estimation)**.** Let the rows of $\mathbf{X}$ be x_i^{T}, with $x_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,\mathbf{\Sigma})$. Writing $x_i = \sqrt{\mathbf{\Sigma}}\,w_i$, where $w_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,\mathbf{I}_d)$, gives $\mathbf{X} = \mathbf{W}\sqrt{\mathbf{\Sigma}}$. Hence

$$\begin{aligned}\left\|\frac{1}{n}\mathbf{X}^{\mathrm{T}}\mathbf{X}-\mathbf{\Sigma}\right\|_2 &= \left\|\sqrt{\mathbf{\Sigma}}\Big(\frac{1}{n}\mathbf{W}^{\mathrm{T}}\mathbf{W}-\mathbf{I}_d\Big)\sqrt{\mathbf{\Sigma}}\right\|_2 \\ &\le \|\sqrt{\mathbf{\Sigma}}\|_2^2\left\|\frac{1}{n}\mathbf{W}^{\mathrm{T}}\mathbf{W}-\mathbf{I}_d\right\|_2 \\ &= \|\mathbf{\Sigma}\|_2\left\|\frac{1}{n}\mathbf{W}^{\mathrm{T}}\mathbf{W}-\mathbf{I}_d\right\|_2.\end{aligned}$$

The inequality uses submultiplicativity of the operator norm, which follows from $\|\mathbf{A}\mathbf{B}v\|_2 \le \|\mathbf{A}\|_2\|\mathbf{B}v\|_2$. The last equality uses positive semidefiniteness of $\mathbf{\Sigma}$.

Applying (6.11) for $n \ge d$ gives, for $\mathbf{\Sigma} \ne 0$ and $\delta > 0$,

$$\frac{\|\widehat{\mathbf{\Sigma}}-\mathbf{\Sigma}\|_2}{\|\mathbf{\Sigma}\|_2} \le 2\sqrt{\frac{d}{n}} + 2\delta + \left(\sqrt{\frac{d}{n}}+\delta\right)^2. \tag{6.12}$$

This bound holds with probability at least $1-2e^{-n\delta^2/2}$. The relative error converges to zero in probability as $d/n \to 0$.

We now consider additional structure through control of the eigenvalues of $\mathbf{\Sigma}$.

Example 5.3.4 (Faster rates under trace constraints)**.** Let $\{\gamma_j(\mathbf{\Sigma})\}_{j=1}^d$ be the ordered eigenvalues of a nonzero covariance matrix $\mathbf{\Sigma}$, with $\gamma_1(\mathbf{\Sigma})$ its largest eigenvalue. Consider

$$\frac{\operatorname{tr}(\mathbf{\Sigma})}{|||\mathbf{\Sigma}|||_2} = \frac{\sum_{j=1}^d \gamma_j(\mathbf{\Sigma})}{\gamma_1(\mathbf{\Sigma})} \leq C, \tag{6.13}$$

where C is independent of dimension. One possible bound is $C = \operatorname{rank}(\mathbf{\Sigma})$.

A matrix can be full rank and still have relatively fast eigenvalue decay. Consider the Schatten q-balls

$$\mathbb{B}_q(R_q) := \left\{ \mathbf{\Sigma} \in \mathcal{S}^{d\times d} \;\middle|\; \sum_{j=1}^d |\gamma_j(\mathbf{\Sigma})|^q \leq R_q \right\}, \tag{6.14}$$

where $q \in [0,1]$ and the radius is $R_q > 0$. At $q = 0$, the sum counts the nonzero eigenvalues.

Restrict attention to matrices with eigenvalues in $[-1,1]$. For $q_1 < q_2$,

$$\sum_{j=1}^d |\gamma_j(\mathbf{\Sigma})|^{q_2} \leq \sum_{j=1}^d |\gamma_j(\mathbf{\Sigma})|^{q_1}.$$

Thus, at a fixed radius R, the restricted sets are nested as $\mathbb{B}_{q_1}(R) \subseteq \mathbb{B}_{q_2}(R)$. The smallest set has $q = 0$ and the largest has $q = 1$. Under this eigenvalue restriction, a nonzero covariance matrix $\mathbf{\Sigma} \in \mathbb{B}_q(R_q)$ satisfies (6.13) with $C = R_q/(\gamma_1(\mathbf{\Sigma}))^q$.

For any matrix class satisfying (6.13), Theorem 5.3.1 gives, with high probability,

$$\frac{\sigma_{\max}(\mathbf{X})}{\sqrt{n}} \leq \gamma_{\max}(\sqrt{\mathbf{\Sigma}})\left(1 + \delta + \sqrt{\frac{C}{n}}\right). \tag{6.15}$$

Comparing this with (6.10) for $\mathbf{\Sigma} = \mathbf{I}_d$ shows that C plays the role of the effective dimension.

5.4 Sub-Gaussian Random Vectors and covariance matrix estimation

Let $X_1, \ldots, X_n \in \mathbb{R}^d$ be independent and identically distributed random vectors with $\mathbb{E}X_i = 0$ and covariance matrix $\Sigma \in \mathbb{R}^{d\times d}$. Define

$$S_n = \frac{1}{n}\sum_{i=1}^n X_i X_i^\top. \tag{5.1}$$

and

$$X = \begin{pmatrix} X_1^\top \\ X_2^\top \\ \vdots \\ X_n^\top \end{pmatrix} \in \mathbb{R}^{n\times d}, \qquad \widehat{\Sigma} = S_n = \frac{1}{n}X^\top X.$$

Because the population mean is zero, $\widehat{\Sigma}$ is the uncentered sample covariance estimator, and $\mathbb{E}\widehat{\Sigma} = \Sigma$. Our goal is to bound the spectral-norm estimation error $\left\|\widehat{\Sigma} - \Sigma\right\|_2$.

For a real matrix A, its largest and smallest singular values are

$$s_{\max}(A) = \sup_{\|x\|_2=1} \|Ax\|_2, \qquad s_{\min}(A) = \inf_{\|x\|_2=1} \|Ax\|_2.$$

Definition 5.4.1 (Sub-Gaussian random vector)**.** A zero-mean random vector $X \in \mathbb{R}^d$ is sub-Gaussian with parameter $\sigma > 0$ if, for all $\lambda \in \mathbb{R}$ for each fixed $v \in \mathbb{S}^{d-1}$ unit sphere in $\mathbb{R}^d$,

$$\mathbb{E}\exp\{\lambda\langle v, X\rangle\} \le \exp\left(\frac{\lambda^2\sigma^2}{2}\right). \tag{5.2}$$

That is, $\langle v, X\rangle$ is a one-dimensional zero-mean sub-Gaussian with parameter σ for any $v \in \mathbb{S}^{d-1}$.

Theorem 5.4.2. *Suppose $X_1, \dots, X_n \in \mathbb{R}^d$ are i.i.d., mean-zero, and sub-Gaussian with parameter σ. Let $\Sigma = \mathbb{E}[X_i X_i^\top]$ denote the population covariance matrix. The sample covariance $\widehat{\Sigma} = n^{-1}\sum_{i=1}^n X_i X_i^\top$ satisfies the bounds*

$$\mathbb{E}e^{\lambda\left\|\widehat{\Sigma}-\Sigma\right\|_2} \le e^{c_0\frac{\lambda^2\sigma^4}{n}+4d}, \qquad \forall|\lambda| \le \frac{n}{c_1\sigma^2}. \tag{5.3}$$

Consequently, for all $\delta > 0$,

$$\mathbb{P}\left(\frac{\left\|\widehat{\Sigma}-\Sigma\right\|_2}{\sigma^2} \ge c_2\left(\sqrt{\frac{d}{n}}+\frac{d}{n}\right)+\delta\right) \le c_3 e^{-c_4 n\min(\delta,\delta^2)}, \tag{5.4}$$

and

$$\frac{\mathbb{E}\left\|\widehat{\Sigma}-\Sigma\right\|_2}{\sigma^2} \le c_5\left(\sqrt{\frac{d}{n}}+\frac{d}{n}\right). \tag{5.5}$$

Here $c_0, \dots, c_5$ are positive universal constants.

Proof. Denote $Q = \widehat{\Sigma} - \Sigma$. Recall from Section 6.1 the variational representation

$$\|Q\|_2 = \max_{v\in S^{d-1}} |\langle v, Qv\rangle|.$$

Let $\mathcal{N} = \{v^1, \dots, v^N\} \subset S^{d-1}$ be a $\frac{1}{8}$-covering of the unit sphere S^{d-1} with respect to the Euclidean norm. From Chapter 5, there exists such a covering with

$$N \le \left(1+\frac{2}{1/8}\right)^d = 17^d. \tag{5.6}$$

Given any $v \in S^{d-1}$, there exists j such that

$$\left\|v - v^j\right\|_2 \le \frac{1}{8}.$$

Write $v = v^j + \Delta$, where $\|\Delta\|_2 \le 1/8$.

Since Q is symmetric,

$$\begin{aligned} v^\top Q v &= (v^j+\Delta)^\top Q(v^j+\Delta) \\ &= (v^j)^\top Q v^j + 2\Delta^\top Q v^j + \Delta^\top Q\Delta. \end{aligned}$$

By Cauchy–Schwarz and the definition of the spectral norm,

$$\begin{aligned} |v^\top Q v| &\le |(v^j)^\top Q v^j| + 2\|\Delta\|_2\|Q\|_2\left\|v^j\right\|_2 + \|\Delta\|_2^2\|Q\|_2 \\ &\le |(v^j)^\top Q v^j| + \left(\frac{1}{4}+\frac{1}{64}\right)\|Q\|_2 \\ &\le |(v^j)^\top Q v^j| + \frac{1}{2}\|Q\|_2 \\ &\le \max_{1\le j\le N}|(v^j)^\top Q v^j| + \frac{1}{2}\|Q\|_2. \end{aligned}$$

Taking the supremum over $v \in S^{d-1}$ gives

$$\|Q\|_2 = \sup_{v \in S^{d-1}} |v^\top Q v| \le \max_{1 \le j \le N} |(v^j)^\top Q v^j| + \frac{1}{2}\|Q\|_2 . \tag{5.7}$$

Consequently,

$$\|Q\|_2 \le 2 \max_{1 \le j \le N} |(v^j)^\top Q v^j|. \tag{5.8}$$

For all $\lambda \ge 0$,

$$\begin{aligned} \mathbb{E}e^{\lambda \|Q\|_2} &\le \mathbb{E}e^{2\lambda \max_{1 \le j \le N} |(v^j)^\top Q v^j|} \\ &\le \sum_{j=1}^{N} \left(\mathbb{E}e^{2\lambda (v^j)^\top Q v^j} + \mathbb{E}e^{-2\lambda (v^j)^\top Q v^j} \right). \end{aligned} \tag{5.9}$$

Next, we claim that for any fixed unit vector $u \in S^{d-1}$,

$$\mathbb{E}e^{t\langle u, Qu\rangle} \le e^{512 \frac{t^2}{n} e^4 \sigma^4}, \qquad \forall |t| \le \frac{n}{32 e^2 \sigma^2}. \tag{5.10}$$

We take this bound as given for the moment and use it to complete the proof of (5.3).
For each vector v^j in the covering set, we apply (5.10) twice, once with $t = 2\lambda$ and once with $t = -2\lambda$.
Combining the resulting bounds with (5.9), we obtain

$$\begin{aligned} \mathbb{E}e^{\lambda \|Q\|_2} &\le 2N e^{2048 \frac{\lambda^2}{n} e^4 \sigma^4} \\ &\le e^{c_0 \frac{\lambda^2 \sigma^4}{n} + 4d}, \end{aligned}$$

valid for all $0 \le \lambda$, where the last step uses $2(17^d) \le e^{4d}$.
For $\lambda < 0$, the bound follows from $\mathbb{E}e^{\lambda \|Q\|_2} \le 1 \le e^{4d}$.
Having established the MGF bound (5.3), the tail bound (5.4) follows from Proposition 2.9.

□

Lemma 5.4.3. *$\forall u \in S^{d-1}$, ($\langle u, X\rangle$ is 1-dimensional σ-sub-Gaussian),*

$$\mathbb{E}e^{t u^T Q u} \le e^{C \frac{t^2}{n} \sigma^4}, \qquad \forall |t| < \frac{Cn}{\sigma^2}.$$

Proof. Fix $u \in \mathbb{S}^{d-1}$. By definition of Q,

$$\begin{aligned} u^\top Q u &= u^\top \left(\widehat{\Sigma} - \Sigma \right) u \\ &= u^\top \left(\frac{1}{n} \sum_{i=1}^{n} X_i X_i^\top - \Sigma \right) u \\ &= \frac{1}{n} \sum_{i=1}^{n} (u^\top X_i)^2 - u^\top \Sigma u \\ &= \frac{1}{n} \sum_{i=1}^{n} \left[(u^\top X_i)^2 - \mathbb{E}(u^\top X_i)^2 \right], \end{aligned} \tag{5.11}$$

since

$$\mathbb{E}(u^\top X_i)^2 = u^\top \mathbb{E}(X_i X_i^\top) u = u^\top \Sigma u.$$

Independence of the observations gives

$$\begin{aligned}\mathbb{E}e^{tu^TQu} &= \mathbb{E}e^{t\frac{1}{n}\sum_{i=1}^n\left[(u^Tx_i)^2-u^T\Sigma u\right]}\\ &= \mathbb{E}\prod_{i=1}^n e^{t\frac{1}{n}\left[(u^Tx_i)^2-u^T\Sigma u\right]}\\ &= \left(\mathbb{E}e^{t\frac{1}{n}\left[(u^Tx_1)^2-u^T\Sigma u\right]}\right)^n.\end{aligned}$$

$\langle u^TX_i\rangle$ is 1-dimensional σ-sub-Gaussian.

Lemma 5.4.4. *If Y is zero-mean σ-sub-Gaussian, then*

$$Y^2 - \mathbb{E}Y^2$$

is sub-exponential with parameters $(4\sigma^2, 4\sigma^2)$, i.e.,

$$\mathbb{E}e^{t(Y^2-\mathbb{E}Y^2)} \le e^{\frac{t^2}{2}(4\sigma^2)^2}, \qquad \forall|t| < \frac{1}{4\sigma^2}.$$

Proof. Homework. □

By the lemma, $(u^TX_1)^2 - \mathbb{E}(u^TX_1)^2$ is $(4\sigma^2, 4\sigma^2)$-sub-exponential.
Therefore,

$$e^{\frac{t}{n}\left[(u^TX_1)^2-u^T\Sigma u\right]} \le e^{\frac{t^2}{2n^2}(4\sigma^2)^2}, \qquad \forall|t| < \frac{n}{4\sigma^2}.$$

□

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